Ocean surface drift modelling with uncertainty assessment using fuzzy number theory

by

Hauke Blanken
B.Sc., University of Alberta, 2011
M.Eng., McGill University, 2014

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## ABSTRACT

Predicting the drift of objects floating at the ocean surface is a challenging task, with significant implications on marine emergency operations such as oil spill response and search and rescue efforts. Object drift is governed by a combination of currents, winds, and waves at the ocean surface, and depends strongly on object geometry. This introduces considerable uncertainty into drift prediction, as the model uncertainty from hydrodynamic models for these governing forces is aggregated and combined with systemic uncertainty about their respective influence on the drift of objects with various geometries.

The first portion of this thesis examines the observed drift and dispersion of 206 GPS-tracked drifting buoys deployed in a fjord system on the west coast of Canada, which is subject to proposed expansion of shipping for oil and gas resource extraction. The observed mean drift patterns are found to be best explained by considering a combination of observed near-surface currents and winds, but the trajectories of individual buoys are characterized by significant dispersion around the mean. This dispersion results in buoys grounding against the shoreline on timescales of 12 15 hours, and corresponds with observed changes in the dispersion regime from near-field to far-field. Drift tracks exhibit fractal characteristics, and dispersion is found to be well modelled using fractional Brownian motion, rather than a traditional random-walk. Based on this, a statistical model for surface drift in the region is proposed, and shown to skillfully reproduce historical observations of sheening from an oil spill in the region.

In the second portion of the thesis, uncertainty assessment in drift prediction using fuzzy numbers is introduced and the relationship between forcing from currents, wind, and waves is examined in detail. Forcing data is taken from an observing platform in the northeast Pacific (Ocean Station Papa). Uncertainty in the forcing data is characterized from reported instrument uncertainty and spectral estimates of energy at unresolved time scales. This uncertainty is expressed and aggregated as fuzzy numbers, and propagated through seven day simulations of the trajectories of a set of surface drifting buoys with five different geometries using the transformation method. For generality, the effect of buoy geometry is accounted for deterministically, thereby avoiding traditionally used empirical coefficients. All possible linear combinations of forcing terms are explored, and the optimal combination of forcing terms for each drifter type is identified through the use of a novel skill metric. With the

optimal forcing combination, the model is found to correctly identify the area where a buoy is expected to be found for the duration of the simulation. Comparison with the commonly used nine parameter leeway method shows similar performance, but without the need for object-specific empirical coefficients.

Finally, the fuzzy-based method for propagating uncertainty in drift trajectory models is extended to spatiotemporally variable velocity fields. For this, a 4th-order Runge-Kutta solver for the integration of velocities, with inverse-distance weighting interpolation, is expanded to include possibility as an additional dimension. Performance of the method is demonstrated by tracking particles through one revolution of a monopole vortex, the strength of which varies sinusoidally with distance from the center. Both steady and unsteady maximum amplitudes are considered, and a steady uncertainty is prescribed as a fuzzy number. The model is shown to produce adequate results with reasonable computational effort, and the free parameters of the simulation are optimized through a sensitivity analysis.

# Contents

**Supervisory Committee** **ii**

**Abstract** **iii**

**Table of Contents** **v**

**List of Tables** **viii**

**List of Figures** **ix**

**Acknowledgements** **xix**

**Dedication** **xx**

**1 Introduction** **1**

**2 Background** **5**

2.1 Oceanography . . . . . 5

2.1.1 Ocean Currents . . . . . 5

2.1.2 Fjords and Inland Waters . . . . . 8

2.2 Methods for Drift Prediction . . . . . 10

2.2.1 Governing Equations . . . . . 10

2.2.2 Windage . . . . . 14

2.2.3 Dispersion . . . . . 17

2.2.4 Research Opportunities . . . . . 19

2.3 Fuzzy Numbers . . . . . 21

**3 Surface drift and dispersion in a multiply-connected fjord system** **27**

3.1 Introduction . . . . . 27

3.2 Observations . . . . . 30

3.2.1 Physical Setting . . . . . 30
3.2.2 Drifter Program . . . . . 36
3.3 Methods . . . . . 39
3.3.1 Channel Coordinates . . . . . 39
3.3.2 Drift Velocity Statistics . . . . . 41
3.3.3 Fractional Brownian Motion . . . . . 42
3.4 Results . . . . . 44
3.4.1 Mean Drift Patterns and Driving Forces . . . . . 44
3.4.2 Along-Channel Drift and Dispersion . . . . . 47
3.4.3 Effects of Cross-Channel Dispersion . . . . . 49
3.4.4 Spectra and Fractal Dimension . . . . . 51
3.5 Discussion . . . . . 55
3.6 Application to Oil Spill Risk Analysis . . . . . 60
3.7 Conclusions . . . . . 64

**4 A fuzzy - based framework for assessing uncertainty in drift prediction using observed currents and winds 66**
4.1 Introduction . . . . . 66
4.2 Drift Trajectory Model . . . . . 69
4.2.1 Stokes Drift . . . . . 71
4.2.2 Current Shear . . . . . 73
4.2.3 Wind Shear . . . . . 75
4.3 Uncertainty Propagation . . . . . 76
4.3.1 Overview of Relevant Fuzzy Number Theory . . . . . 76
4.3.2 Fuzzification of Input Data . . . . . 81
4.3.3 Integration of Fuzzy Drift Velocities . . . . . 82
4.3.4 Model Evaluation . . . . . 84
4.4 Forcing and Verification Data . . . . . 86
4.4.1 Drifter Deployment . . . . . 88
4.4.2 Comparison of Current Meter and ADCP at Ocean Station Papa 91
4.5 Results . . . . . 97
4.5.1 Fuzzy Forcing . . . . . 97
4.5.2 Optimal Forcing Combinations . . . . . 103
4.5.3 Trajectory Results . . . . . 107
4.6 Discussion . . . . . 113

4.7 Conclusions . . . . . 115

**5 Propagating particle tracking uncertainty defined by fuzzy numbers in spatially variable velocity fields 117**

5.1 Introduction . . . . . 117
5.2 Materials and Methods . . . . . 119
5.2.1 Forcing Field . . . . . 119
5.2.2 A Quick Primer on Fuzzy Numbers . . . . . 121
5.2.3 Uncertainty & Fractional Brownian Motion . . . . . 122
5.2.4 Fuzzy Particle Tracking . . . . . 124
5.2.5 Skill Assessment . . . . . 126
5.3 Results . . . . . 126
5.3.1 Steady Forcing Field . . . . . 126
5.3.2 Unsteady Forcing Field . . . . . 129
5.4 Discussion . . . . . 131
5.5 Conclusions . . . . . 134

**6 Conclusions 136**

**A Additional Information 138**

A.1 Fuzzy Numbers - A Hypothetical Example . . . . . 138

**Bibliography 140**

# List of Tables

Table 3.1 Seasonally resolved regression coefficients, $\alpha$, with number of samples, $N$, and fraction of variance explained by the model $\vec{u}_{drift_X} = \alpha\vec{u}_{wind_X} + \vec{u}_{current_X}$ . . . . . 46

Table 3.2 Regression fit parameters describing the evolution of the first two moments of observed displacement probabilities. Linear fits describe the mean, and power laws of the form $\sigma^2 = A\tau^B$ describe the two regimes of variance growth (ballistic, subscript b, and diffusive, subscript d). $\tau_T$ is the time of transition between the regimes, determined as the time which minimizes the total error in the regression fits to the observed variance. . . . . 52

Table 3.3 Fractal dimensions derived from log-transformed regression fits in Figure 3.20 ($D$=1 $-$ *slope*), power law exponents ($\sigma^2 \propto t^B$) predicted by fractional Brownian motion ($B_{predicted}$), and those observed during the first 15 hours of dispersion based regression fit to observations of displacement variance growth ($B_{observed}$). . 59

Table 4.1 Leeway coefficients for proxy object types for drifters used in this study (see Equation 2.13). Reproduced from the OpenDrift source code [28]. . . . . 91

Table 4.2 Amplitude of energy in unresolved timescales and associated power law coefficients. . . . . 97

Table 4.3 Percent change in mean trajectory prediction skill for given adjustments to deterministic wind drag factor, $\alpha$. . . . . 107

# List of Figures

Figure 3.1 Map of the Kitimat fjord system showing sub-regions DC, EAST, and SOUTH, and grid system for analysis of along-channel dispersion. Channels in region DC are shown in red, region EAST in yellow, and region SOUTH in green. . . . . 30

Figure 3.2 Mean (arrows) and RMS (ellipses) velocities recorded by acoustic Doppler current profilers deployed in the Kitimat fjord system between July 2013 and July 2016. Statistics are shown for the entire available record (a), as well as seasonal statistics separated into spring (b), summer (c), and fall (d). The record length at each location is indicated by the marker color. Size of the ellipse axes indicates one standard deviation. . . . . 32

Figure 3.3 Cross-section of mean temperature (left column) and salinity (right column) in the upper 20 m of the water column between Kitimat and Hecate Strait, based on CTD casts taken between July 2013 and July 2016. Values are shown for spring (March - May, a-b), summer (June - August, c-d), and fall (September - November, e-f). Isopycnals (lines of constant potential density, $\rho_0$) are contoured as $\rho_0 - 1000$ kg/m$^3$. The location of Wright Sound is indicated by the dashed vertical line. . . . . 34

Figure 3.4 Seasonal wind distributions at Nanakwa Shoal (ECCC Buoy 46181)for spring (Mar - May, left), summer (Jun - Aug, middle), and fall (Sep - Nov, right). . . . . 35

Figure 3.5 Distribution of significant wave height vs. peak wave period observations recorded at Nanakwa Shoal (ECCC Buoy 46181) between Mar 1, 2014 and Sep 1, 2016. . . . . 36

Figure 3.6 The final drifter design. . . . . 37

Figure 3.7 Sketch of the final drifter design. . . . . 38

Figure 3.8 Deviation between 186 positions reported by at-rest SPOT GPS transmitters and the mean position. The component-wise mean is shown by the red marker, and component-wise standard deviations are shown by the dashed ellipse. . . . . 39
Figure 3.9 Illustration of simulated fractional Brownian motion tracing a dispersing particle's position for various fractal dimensions using the method of [61] over 1000 time increments. . . . . 43
Figure 3.10 Distribution of mean (arrows) and RMS (dashed ellipses) drift velocities in the Kitimat system. Results are shown the average over the data set (a) along with seasonal averages (b - d). Shading indicates the number of independent samples of drift velocity in the regions over which statistics were derived. . . . 45
Figure 3.11 Observed along-channel drift velocity (seaward positive) near mid-channel around mooring FOC1 against drift velocity predicted by $u_{drift_X} = \alpha u_{wind_X} + u_{current_X}$ using values of $\alpha$ from Table 3.1 for the complete data set (a), spring (March - May, b), summer (June - August, c), and fall (September - November, d). 46
Figure 3.12 Distribution of drifter displacements, $\Delta X$, over ten days with mean (dash-dotted line, with implied mean drift speed), 95% limits (dashed line), and 95% limits of the corresponding normal distribution (solid line). Distributions are derived by binning observed displacements into 2 km by 0.5 hour bins, and for each half hour interval reporting the percentage of observed displacements in each displacement bin. Results are shown for the complete data set (a), as well as seasonal subsets of data from March - May (b), June - September (c), and October - November (d), and regional subsets for DC (e), EAST (f), and SOUTH (g). . . . . 48
Figure 3.13 Mean and variance of drifter displacement, $\Delta X$, as functions of time for the seasonally - resolved (a, c) and regionally - resolved (b, d) data. Optimal linear (power-law) regression fits are superimposed on the plotted mean (variance). Fit parameters are given in Table 3.2. . . . . 50

Figure 3.14 $3^{rd}$ and $4^{th}$ moments of the observed displacement distributions for the complete data set (left) as well as the seasonal (middle) and regional (right) subsets. For clarity values have been smoothed using a moving average with a 3 hour window and outliers more than 3 standard deviations from the mean are excluded. . . . . 51

Figure 3.15 a) Distribution of times taken for a drifter to move from mid-channel ($\chi > 0.8$) to near the shoreline ($\chi < 0.1$). Mean time is indicated by the dashed black line and dotted lines indicate one standard deviation around the mean. b,c) distributions of drifter along- and cross-channel velocities near mid-channel (shown in green; inside green polygon), near the shore (shown in red; inside blue polygon, outside green polygon), and for all data (shown in blue; inside blue polygon). Along-channel velocity is positive down-channel (approximately southwest) and cross-channel velocity is positive 90° clockwise from down-channel (approximately northwest). . . . . 53

Figure 3.16 Distribution for times required for cross-channel transport from $\chi > 0.8$ to $\chi < 0.1$. Distributions are shown for the complete data set (a), spring (b), summer (c), and fall (d). Mean transit time is denoted by the dashed black line and dotted lines denote one standard deviation. . . . . 54

Figure 3.17 Observations of drift speed, $u_{drift}$ as function of cross-channel position, $\chi$. The linear regression fit line (black), corresponds to the equation $u_{drift}(\chi) = 0.18 + 0.2\chi$. . . . . 55

Figure 3.18 a) Mean energy spectra of drift velocity $0.5(P_{uu}+P_{vv})$ for the complete data set (red), as well as spring (green), summer (yellow), and fall (blue). Binned density of observations from individual drifter tracks are shown by the colorbar. b) Ratio of mean amplitude spectra of velocity in the cross-channel: along-channel directions as a function of frequency for all data (red), spring (green), summer (yellow), and fall (blue). Bounds for approximately isotropic motion are indicated by the dashed lines. c) Energy spectrum of drifter displacement in the same style as a). . . . . 56

Figure 3.19 Ratio of mean amplitude spectra of velocity in the meridional:zonal directions as a function of frequency for all data (red), spring (green), summer (yellow), and fall (blue). A value of 1 (dashed horizontal line) indicates isotropic motion. The dotted lines represent 30% deviations from isotropy which may be acceptable for fractal analysis, following [125]. . . . . 57

Figure 3.20 Yardstick trajectory length estimates for shoreline, and drifter tracks with seasonal subsets. Dashed lines show log-transformed regression fits to $L(\Delta)$, with slopes $1 - D_L$. Regression results are reported in Table 3.3. Dotted vertical lines indicate the range of spatial scales over which fractal behaviour is evident. . 58

Figure 3.21 Horizontal velocity modes 1 - 3 (Mode 1 = green, $R_i$ = 6.5-9.5 km, Mode 2 = blue, $R_i$ = 4-6 km, Mode 3 = red, $R_i$ = 2.5-3.5 km) as a function of depth in Upper Douglas Channel (top, a - c) and Whale Channel (bottom, d - f). Values are shown for spring (left, a & d), summer (center, b & e), and fall (right, c & f). Modes were calculated using the standard vertical structure equation following [3]. All modes are normalized by their surface value. . . . . 60

Figure 3.22 Density of hourly drifter positional fixes up to 48 hours after passing within 2km (a) and 5km (b) of the site of the wreck of the *Queen of the North*. Observations of oil sheening resulting from the wreck after 48 hours are shown in blue. Potential seasonal maximum spill extents predicted by applying a normal distribution with mean and variance specified by the regression fits from Figure 3.13 are shown in cyan (spring), orange (summer) and red (fall). . . . . 61

Figure 4.1 Profiles of temperature (yellow) and salinity (dashed blue) taken between Feb 18 - 19, 2015. . . . . 74

Figure 4.2 Examples of possible shapes of the membership function for a fuzzy number describing possible values of an arbitrary parameter. Functions for a triangular (dash-dotted), transformed Gaussian (solid), transformed Laplace (dashed), and expert knowledge-informed (dotted) fuzzy number are shown. The transformed membership functions are derived using a crisp value $\mu_K = 0.4$, and standard deviation $\sigma = 0.2$. All functions have a support width of $3\sigma$. . . . . 78
Figure 4.3 Illustration of the discretization of a fuzzy number into array form using the transformation method. Adapted from [55]. . . 80
Figure 4.4 Illustration of the scheme to derive all possible crisp time series corresponding to a fuzzy time series of drift velocity. An example of the first four time steps starting from point $m = 3$ with $n_p = 3$ is shown here. . . . . 83
Figure 4.5 Illustration of instantaneous fuzzy skill score for trajectory evaluation. For clarity only a single membership level is shown. . . 85
Figure 4.6 Tracks of drifters for seven days after deployment near Ocean Station Papa (panels A & B), and time series of u,v components of currents (panels C & D) and wind (panel E) during the deployment. . . . . 87
Figure 4.7 The drifters deployed for this study (left), and their idealized geometry, including approximate drag coefficients (right). The SVP 8 (bottom-left photograph) and SVP 18 drifters (no photograph shown) are identical except for the length of the tether. 89
Figure 4.8 Distance of drifters from Ocean Station Papa during the first seven days of the deployment. . . . . 90
Figure 4.9 Magnitude (left) and phase (in degrees, right) of the vector correlation between the ADCP record at all available depths and the current meter record at 15 m. Red lines indicate correlation with the raw current meter record, while green lines indicate correlations with the smoothed current meter record. Dashed lines in the left-hand panel indicate the maximum correlation for each case. . . . . 92

Figure 4.10 Timeseries of currents recorded by the current meter at 15 m depth (red) and the ADCP at 6 m depth (blue). $u$ components (E-W) are shown on top and $v$ components (N-S) below. . . . 93
Figure 4.11 Time-averaged profile of velocity components from the ADCP, and the model for current shear (Equation 4.7) over the course of the simulation. . . . . 94
Figure 4.12 Depth profile of data return for the ADCP after quality control flags are applied. Bins with no good data are marked by red x's on the y-axis. . . . . 94
Figure 4.13 Difference between current meter at 15 m and ADCP measurements at 6 m, in u,v space. Distributions of component differences are shown in the side panels, with a Gaussian distribution superimposed. The solid line indicates a zero-mean distribution while the dotted line indicates a distribution that is shifted to the bias between the two data sources. The time-averaged currents recorded by the current meter and the ADCP are shown by the colored markers. . . . . 95
Figure 4.14 Same as Figure 4.13, but for ADCP bin at 16m. . . . . 96
Figure 4.15 Power spectra of measured wind (left) and current (right). Power law fit to the high-frequency tail is shown in dashed red and the area corresponding to the energy in unresolved timescales is hatched in red. . . . . 98
Figure 4.16 Daily-mean currents in the study area at 15 m depth from the GLORYS12 numerical re-analysis for Feb 19 - 26, 2015 [73]. . . 99
Figure 4.17 Differences in the daily-mean currents from GLORYS12 between OSP and the locations of the drifters. Each row represents values for a drifter type, with $u$-components (E-W, east positive) on the left and $v$ components (N-S, north positive) on the right. The differences between modelled and observed currents are shown for the observations from the current meter (red) and the ADCP (blue). Mean differences are shown by the dashed line while the shading represents one standard deviation. . . . 100
Figure 4.18 Generalized comparison of Gaussian and Laplace distributions with a simulated distribution from a red noise process with $f^{-\frac{3}{4}}$. 101

Figure 4.19 Components of fuzzy wind (panels A & B), current (panels C & D), Stokes drift (panels E & F), current shear (panels G & H), and wind shear (panels I & J) for the first 7 days of the drifter deployment. The crisp value (measurement) is shown in black and the three fuzzy membership levels are indicated in grayscale. For clarity in panels E - J only values for SCT drifters with the velocity profile applied by the 'centroid' scheme are shown. . . 102

Figure 4.20 Summary of optimal forcing terms for each drifter as a function of time. In the left-hand panels (A, C, E, G, & I), timesteps where a forcing term contributed to the solution achieving the highest skill score are shaded in blue for each drifter type. In the right hand panels (B, D, F, H, & J) the percentage of applicable time each forcing term contributed to the optimal solution is summarized. . . . . 104

Figure 4.21 Percentage of time current shear (Panel A) and Stokes drift (Panel B) profiles applied via the centroid and area-weighted average of velocity or force produce the optimal solution for a the trajectories of each of the five drifters. No results are shown for Stokes drift on the SVP drifters, as the analysis indicates that Stokes drift does not contribute to the optimal trajectory prediction at any time. Panel C shows the percentage of time the currents measured by the current meter (red) and the ADCP (blue) contribute to the optimal solution. . . . . 105

Figure 4.22 Daily predicted tracks of the sphere drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Membership levels of the possible displacement, $P_{\vec{x}}^{(j)}$ for $j = 0, 1, 2$, are indicated in grayscale shading, increasing in intensity towards the crisp solution ($j = 3$) which is indicated by the dashed black line. Equivalent results obtained using the leeway method are shown in red shading. Observed drifter tracks are superimposed in orange. Model skill is also shown in grayscale for the optimal forcing combination and in red for the leeway method. Shading indicates the maximum and minimum skill for the ensemble of drifters, while the solid line indicates the mean. Red markers on the x-axis of the skill plot indicate times when at least one observed drifter fell outside of the lowest membership level. . . . . 108

Figure 4.23 Daily predicted tracks of the SCT drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22. . . . . 109

Figure 4.24 Daily predicted tracks of the Davis drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22. . . . . 110

Figure 4.25 Daily predicted tracks of the SVP 8 drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22. . . . . 111

Figure 4.26 Daily predicted tracks of the SVP 18 drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22. . . . . 112

Figure 5.1 Analytical velocity fields used in this study. Both fields consider a counterclockwise rotating monopole vortex, with amplitude $A$ (left panel). In the steady case, $A$ is held constant at 1 (top right panel). In the unsteady case, $A$ varies sinusoidally with time, with a maximum amplitude of 1 (bottom right panel). . . 120
Figure 5.2 Schematic showing the fuzzy particle tracking scheme described in Section 5.2.4 . . . . . 123
Figure 5.3 Membership, $\mu$, function of velocity perturbations. . . . . 125
Figure 5.4 Mean represented uncertainty (green), given by the integral of the membership function of final positions, and computational runtime required (right y-axis, blue) for simulations in the steady velocity field. Shading indicates the range of values for the examined fractal dimensions, D. . . . . 127
Figure 5.5 Performance summary for relevant combinations of free model parameters. Plots show mean cost (or components of cost) as a function of time step for the steady velocity case. Time cost ($T/T_{max}$) is shown in blue, uncertainty cost ($1 - F/F_{max}$) in green, crisp skill ($s$) in orange, and total cost (Equation 5.2.5) in red. $N_g$ increases from first to second row, and $N_l$ increases from left to right. Shading indicates the range of values for the examined fractal dimensions, D. . . . . 129
Figure 5.6 Evolution of particle trajectory for steady velocity case, with $\delta t = \frac{2\pi}{270}$, $N_g$=32, $N_l$=7, and $D$=1.5. The crisp position is shown by the red and white marker. The membership function of positions is shaded, and individual fuzzy positions are superimposed. $t$ (top-right corner of each panel) refers to the time index of the simulation. . . . . 130
Figure 5.7 Mean represented uncertainty (green), given by the integral of the membership function of final positions, and computational runtime required (right y-axis, blue) for simulations in the unsteady velocity field. Shading indicates the range of values for the examined fractal dimensions, D. . . . . 131

Figure 5.8 Performance summary for relevant combinations of free model parameters. Plots show mean cost (or components of cost) as a function of time step for the unsteady velocity case. Time cost ($T/T_{max}$) is shown in blue, uncertainty cost ($1 - F/F_{max}$) in green, crisp skill ($s$) in orange, and total cost (Equation 5.2.5) in red. $N_g$ increases from first to second row, and $N_l$ increases from left to right. Shading indicates the range of values for the examined fractal dimensions, D. . . . . 132

Figure 5.9 Evolution of particle trajectory for unsteady velocity case, with $\delta t = \frac{2\pi}{270}$, $N_g$=32, $N_l$=7, and $D$=1.5. The crisp position is shown by the red and white marker. The membership function of positions is shaded, and individual fuzzy positions are superimposed. $t$ (top-right corner of each panel) refers to the time index of the simulation. . . . . 133

Figure 5.10 Uncertainty content ($F$, round markers) and cost (x markers) of additional simulations with $N_g$=64 and $N_l \in [5, 7, 9, 11, 13, 15]$ as a function of $N_g$ (top) and $N_l$ (bottom). Results are shown for the steady velocity case. . . . . 134

Figure A.1 Membership function derived as hypothetical example in the appendix, for illustrative purposes only. . . . . 139

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# DEDICATION

To the precious and beautiful ecosystems on the west coast of North America. May they long retain their wonders.

# Chapter 1

# Introduction

Humans interact with the ocean on a daily basis, whether directly through commercial, scientific, governmental, and recreational activity or indirectly as the beneficiaries of these activities. The vast majority of direct interactions occur at the ocean surface, and conducting them safely and productively requires a strong understanding of the movement of objects at the ocean surface under the influence of currents, wind, and waves. This is particularly true in the coastal ocean, where the majority of human activity takes place.

The ability to accurately predict the drift of material at the ocean surface is critical to the response to environmental emergencies such as oil spills [cf. 5, 54, 52, 175], and human emergencies like search and rescue operations [cf. 32, 11, 127]. For scientific and regulatory activities, this knowledge is equally critical for ecosystem monitoring [cf. 49, 120, 25] and environmental impact assessments and tracking of marine pollutants [cf. 45, 43, 9].

A recent example of the requirement for accurate prediction of drift trajectories is given in the release of fuel oil from the cargo vessel *M/V Marathassa* in the harbour of Vancouver, BC on April 8, 2015. The environmental response to this incident, which occurred in an area of significant social and economic importance, is detailed in [52]. Regarding the drift trajectory predictions that were provided during the response they state:

> "It was also noted that the spill trajectory models that were provided ... were all in different platforms and did not correctly identify the spill trajectory. "

While the response to this particular incident was hampered by multiple factors, it can

be concluded that inaccurate drift trajectory models have significant negative impacts on urgent emergency response efforts such as this one by contributing to mis-allocation of available resources for response. The trajectory modelling for the fuel spilled in this incident was re-visited by [175], who found that the trajectory could be reproduced but a high-resolution numerical model of the surrounding ocean was required to do so. Development of such operational high-resolution systems has received significant attention recently [cf. 51, 123, 101], however they are not ubiquitously present along the worlds coastlines. Therefore it is necessary to continue strengthening our ability to predict ocean surface drift and account for uncertainties where inputs to such predictions are less than optimal.

Ocean surface drift prediction is a challenging endeavour as the development of a drift trajectory depends on not only ocean surface currents, but also wind and waves. All of these driving forces are related, but their interaction is complex and depends strongly on the geometry of the drifting object [cf. 12, 141]. Imperfect knowledge of the current, wind, and waves generally necessitates empirically fitted coefficients to describe the response of different objects [2]. This approach has shown adequate success in operational response scenarios, but is limited by definition.

In practice, drift prediction is conducted by using a combination of deterministic and stochastic modelling techniques [cf. 173, 36, 127, 28]. Generally information on currents, winds, and waves is taken from numerical circulation models (where available) and used to perform particle tracking in a deterministic sense. A stochastic component representing the effects of turbulence and uncertainty through randomized perturbation of the displacements is added to the deterministic solution [13, 72]. However the appropriate strength of these perturbations is generally unknown, and an active research topic [108, 74, 31].

Uncertainty arises from various sources in drift prediction, a primary one being the differences between modelled current, wind, and waves and their real-world analogues. These differences are due to computational and numerical limitations on models' ability to resolve detail in the governing Navier-Stokes equations [134] and the simple fact that they are an abstraction of reality. Additional uncertainty is introduced by the difficulty in obtaining direct and precise measurements of ocean surface currents [133], and a general sparsity of observations at the small flow scales governing drift except during concerted efforts such as the one described in [31]. Even further uncertainty is introduced through the methodology used to make drift trajectory predictions, as assumptions about the relative strength of forcing parameters are required in the

derivation of the governing equation for deterministic drift [30, 122], and stochastic models require additional assumptions about the homogeneity and self-correlation of turbulence statistics and underlying uncertainty [145, 125].

Fuzzy numbers [170, 171, 172] offer a promising framework to assess and propagate uncertainty in ocean surface drift prediction. They can be constructed directly from data [20] and hence offer a physically-based modelling approach that requires minimal input. A minimum of three data points is required to construct a fuzzy number, though additional data will certainly improve results. This avoids the assumption of a normal distribution, or any distribution, often required in probabilistic uncertainty modelling [148]. Conversions between fuzzy number-based (possibilistic) and probabilistic frameworks have been established, if required [38]. Additionally, a wealth of knowledge on the arithmetic of fuzzy numbers has been developed, which allows for ready aggregation of uncertainty from various sources and propagation of this uncertainty through complex mathematical systems [65, 55, 68].

In this thesis, several contributions to the field of ocean surface drift prediction are presented. Studies were initially developed as standalone works, and are presented in a largely unaltered form in Chapters 3 - 5. Chapter 2 provides an overview of relevant technical background.

In Chapter 3, a detailed statistical analysis of observations of ocean surface drift in a coastal fjord system is presented. This is an area where results from high-resolution ocean and atmospheric numerical models were unavailable at the time of writing, and a data-driven statistical model for surface drift is proposed for consideration in emergency response and marine spatial planning. The model is a novel method to treat dispersion in topographically constrained areas, and is found to evaluate favourably against historical observations of sheening from an oil spill in the area.

Chapter 4 presents a fuzzy-based approach for propagating uncertainty through a drift trajectory model. Uncertainties are aggregated from instrument uncertainty and observation-based estimates of the strength of turbulence in the forcing data. This model is compared to observations from five different types of drifting buoys, with deterministic treatment of the geometric differences between them. It is further used to evaluate the sensitivity of the model to the forcing terms and their various parameterizations. Forcing fields are based on observations taken at a platform in the northeast Pacific, and considered to be spatially constant. The model is found to reproduce results from the commonly used leeway search and rescue model, which makes use of empirically determined parameterizations of wind effects [28], which

gives confidence that drift trajectory modelling with a generalized treatment of object geometry is feasible given appropriate accounting for uncertainty.

The modelling approach developed in Chapter 4 is extended to spatially variable velocity fields in Chapter 5. This involves expanding the commonly used $4^{th}$-order Runge-Kutta integration scheme with inverse-distance weighted interpolation to fuzzy numbers, by including an additional dimension for possibility. The scheme is tested in both steady and unsteady idealized velocity fields, including a sensitivity analysis to optimize free parameter values. It is found to produce adequate results with reasonable computational effort, and therefore offers a strong platform for direct consideration of model - observation differences and the additional identified sources of uncertainty in a drift trajectory model.

A summary set of conclusions is presented in Chapter 6 to complete the thesis.

# Chapter 2

# Background

In this chapter, relevant background knowledge is given for the work presented in this thesis. This includes key concepts on the oceanographic principles of surface currents in open ocean and coastal settings, a review of current methods of predicting the trajectories of drifting objects and challenges associated with these methods, as well as an overview of relevant fuzzy number theory. Given that Chapters 3 - 5 are each a standalone work, some background information is referred to these chapters to minimize repetition.

## 2.1 Oceanography

### 2.1.1 Ocean Currents

Wind, sea level gradients, horizontal density gradients due to variations in temperature and salinity, and tides are the primary forcing mechanisms of ocean currents. The relative importance of these mechanisms varies with setting, from the open ocean to nearshore regions. Various types of waves propagate information on sea level and density differences over large distances [113, 50, 153, 81]. For the purpose of drift prediction, and this thesis, it is assumed that observations and numerical circulation models provide an adequate representation of the currents forced by these mechanisms. However two mechanisms require special consideration: the current shear in the upper few meters of the ocean, which is a primary source of short-term variability in the currents; and the mean Lagrangian circulation set up by the orbital motion of the surface wave field (Stokes drift).

### Wind-driven currents and the Ekman spiral

At the ocean surface, wind exerts a drag force which raises the surface wave field and imparts momentum on the ocean surface. The magnitude of this drag force is generally parameterized through a drag coefficient that is a function of wind speed, measurement height, and atmospheric stability (degree of stratification in the air masses near the ocean surface) [169]. In steady state, the momentum flux from the wind is balanced by turbulent dissipation of the input energy, and the Coriolis force due to the rotation of the earth [81]. In the original mathematical description of this phenomenon by [40] this balance was found to yield surface currents moving at 45 degrees to the right (left) of the wind in the northern (southern) hemisphere.

The exact values of the deflection and mathematical formulation of the momentum balance have since been updated, but the principles remain valid. One such update was made by [23], who revisited the problem using numerical experiments and the turbulence closure scheme of [83] instead of the eddy viscosity closure in the original formulation. This turbulent closure, the author's so-called level 2.5 scheme, solves the full turbulent moment equations, with the simplifying assumption that density fluctuations are negligible along with several other simplifications by scaling detailed in the original paper [83]. This resulted in a description which included a surface layer where turbulence was enhanced within the upper few meters by the action of breaking waves, and a result of this is a reduced deflection angle between the wind and corresponding surface current. The work of [23] is used as a model for wind-driven current shear in the study described in Chapter 4, and is further detailed in Section 4.2.2.

When the magnitude of near surface wind decreases, the inertia of the currents that have been set in motion is balanced by Coriolis force resulting in clockwise (in the northern hemisphere) rotary motions in the currents, with a period of $\frac{2\pi}{f}$, where $f = 2\Omega sin(\phi)$ is the Coriolis parameter, which is a function of the Earths rotation rate $\Omega$ and latitude $\phi$ [160, 109, 50]. These are known as inertial oscillations, and are a common feature seen in the tracks of drifting objects [111].

While wind-driven currents form a foundational aspect of global circulation patterns, this thesis is limited in scope to mesoscale and sub-mesoscale motions, i.e. those on the scale of drifting objects being tracked on time scales of days to weeks and travelling across spatial scales of less than 1000 km. For a review of global scale wind driven motions the reader may refer to texts such as [50, 153, 81].

### Motions due to surface gravity waves

Stokes drift is the most commonly cited wave-induced motion of drifting objects [cf. 122, 16, 154, 144]. It is a Lagrangian phenomenon that results from the fact that the vertical orbits of water parcels in the surface wave field are not closed, but rather describe a small forward motion at every rotation [140]. The resulting drift is not captured by stationary instruments, or the Eulerian reference frame used by most numerical ocean circulation models [14]. This motion decays quickly with depth. Where waves behave as deep-water waves the Stokes drift velocity profile can be calculated from the directional variance density spectrum of sea surface height [64]. The details of the calculation are given in Section 4.2.1, where Stokes drift is described as part of a drift trajectory model with uncertainty propagation.

Since two-dimensional wave spectra are not always available, several parameterizations of Stokes drift have been developed. These are usually based on an assumed Stokes drift profile, or even just wind strength or dominant wave period [cf. 106, 150, 168, 11]. These profiles have been found to yield adequate estimates of Stokes drift, given that the higher frequency contributions of the local wind field are generally well represented compared to remotely generated swell with longer period. The Stokes drift profile is proportional to amplitude squared and frequency cubed, hence higher frequency motions have a greater contribution to Stokes drift [21]. Stokes drift derived from parametric wave spectra is further discussed in Section 4.2.1.

In addition to the time-averaged Stokes drift, inertial forces due to orbital accelerations in individual waves apply on time scales of the order of the dominant wave period. The forces are described as the Froude-Krylov force (pressure gradient force due to water level differential across the width of the object), added mass (added inertia due to water being displaced by the moving object), and drag due to the orbital velocity of individual waves. These forces do not generally exhibit a time-mean, and therefore do not contribute to object drift on the time scales on which wave spectra are measured (generally 20 minutes). The time steps of particle tracking models considered in this thesis are longer than the time scale on which these inertial forces result in significant motion, and therefore they will not be considered in detail in this thesis. However it is noted that these forces will result in potentially significant motions at shorter time steps, which will become important when forcing data from hydrodynamic models with very high resolution necessitates the use of these short time steps.

### 2.1.2 Fjords and Inland Waters

Near the coast, wind-driven flows still form a significant portion of the currents, however in this setting its importance is on the same order as tidal currents and currents due to density gradients induced by freshwater runoff from the surrounding landmasses [cf. 116, 157]. Tidal currents are generally amplified by interaction with the surrounding landmasses, especially in areas with tight constrictions and complex topography such as the British Columbia coast [147]. Freshwater-induced currents vary regionally with the amount of runoff, but are generally non-negligible in coastal environments.

Interaction with nearby landmasses gives rises to several other considerations relevant to surface drift prediction, namely the topographic steering and amplification of local winds and the associated currents [cf. 88], and modulation of the surface gravity wave field by amplified currents [cf. 158]. It is also important to note that where topographic constriction is strong and density stratification is significant, the rotational dynamics of currents, wind-driven or otherwise, mentioned in the previous section may be altered or not applicable depending on the magnitude of the internal Rossby radius of deformation in relation to the scale of the flow constriction. Where the topographic constriction is smaller than the mode-1 internal Rossby radius (i.e. the length scale at which rotational effects become important in a stratified fluid), cross-channel currents induced by Earth's rotation will be diminished though some cross-channel flow due to higher internal modes may persist [152, 115, 156].

Tidal currents result from global scale waves driven by the gravitational fields of the moon and sun acting on the water in the earths oceans, and the interaction of these waves with the seafloor, coastlines, and each other. On the coast of British Columbia tidal currents are primarily related to the lunar semi-diurnal component M2, though diurnal components have significant secondary influence in some parts of the coast (depending on the modulation of the tidal phase by local bathymetry). Tidal currents in tight constrictions in channels and passages along the coast can be some of the strongest in the world [147]. Barotropic (invariant with depth) tidal currents are catalogued in tidal atlases such as WebTide [46] and FES2022 [75].

For the development of hydrodynamic models to generate information on currents in topographically complex regions, it is desirable to achieve a horizontal resolution of at least five grid points across features whose impact shall be resolved in order to minimize adverse impacts of friction parameterizations at the boundary on the

solution. Similarly, in order to provide adequate representation of sheared currents in the upper ocean and their effect on a drifting object a vertical resolution of two grid cells or more across the draft of the object is desirable. However exact values for both horizontal and vertical effective resolution, i.e. the scale of the smallest features represented in the solution rather than the size of the numerical grid on which the solutions are obtained, is also strongly dependent on model setup and sub grid-scale turbulence parameterizations in particular [134].

**Freshwater dynamics and estuarine flow**

Freshwater runoff into the ocean can influence surface currents in a number of ways. Perhaps the most notable is estuarine circulation, which occurs when pressure gradients induced by inflowing freshwater drive a current away from the freshwater source (generally towards the open ocean). This outflowing freshwater transports salt away from the freshwater source, and thereby induces a current at depth flowing towards the freshwater source to replenish this salt and maintain established horizontal density gradients [41]. This phenomenon commonly occurs along the British Columbia coast, with examples detailed in the Strait of Juan de Fuca [149] and Douglas Channel [162, 157].

Where freshwater enters a relatively unconstrained ocean basin directly, coastal or river currents form insofar that the lower density freshwater will attach to the coastline on its right (in the northern hemisphere) to form an outflowing current. The horizontal scale of this current depends on the volume of inflowing freshwater, and the local Rossby radius of deformation [164]. If the density stratification is sufficient, wind-driven currents may also be enhanced here as strong stratification inhibits downward mixing of momentum and results in a condition known as 'slippery water' [147].

Finally, the presence of a stratified layer of fresher, lighter water near the ocean surface can also modulate surface tidal currents. Here interactions between a sloping seafloor and the barotropic tide generate vertical oscillations in the stratified ocean resulting in internal, or baroclinic, tidal motions [26].

The vast number of generating mechanisms for currents along the coastline of British Columbia make for an interesting, and challenging, environment for applying this knowledge to the drift of floating objects.

## 2.2 Methods for Drift Prediction

It is generally accepted that the drift of floating objects is driven by surface currents, winds, and wave-induced circulation, as described in the preceding section. Here common methods for combining these driving forces into a drift trajectory model are reviewed. However to date no universally applicable combination of these driving forces has been found, and uncertainties in trajectory predictions remain high [11].

### 2.2.1 Governing Equations

From Newton's $2^{nd}$ law of motion in a rotating reference frame, the basic force balance on a surface drifter in the ocean can be expressed as follows [30, 10, 122].

$$(m + m')\left(\frac{d^2\vec{x}}{dt^2} - f\vec{k} \times \frac{d\vec{x}}{dt}\right) = \vec{F_o} + \vec{F_a} + \vec{F_w} + \vec{F_g} \tag{2.1}$$

On the left-hand side of Equation 2.1, $\vec{x}$ is the position vector of the drifting object, $m$ is its mass, $m'$ is the mass of water displaced by friction along its surface, $\vec{k}$ is a unit vector in the vertical, and $f = 2\Omega sin(\phi)$ is the Coriolis parameter where $\Omega$ is the Earth's rotation rate, and $\phi$ is the object's latitude. On the right-hand side, $\vec{F_o}$ is the drag force exerted on the submerged portion of the object by the ambient water, and $\vec{F_a}$ is the drag force of ambient air on exposed parts of the object. $F_w$ is the force transferred to the ambient object through excitation by the ocean surface wave field, and $F_g$ is the force of gravity acting on the object along the sea surface slope.

The terms will now be examined in detail, beginning with the two drag force terms. The drag exerted on the drifter by the ambient water may be expressed as a function of the square of the relative velocity between the drifter and the water.

$$\vec{F_o} = \frac{1}{2}A_o\rho_o C_o\Big(\frac{d\vec{x}}{dt} - (\vec{u_E} + \vec{u_S})\Big)\Big|\frac{d\vec{x}}{dt} - (\vec{u_E} + \vec{u_S})\Big| \tag{2.2}$$

In Equation 2.2, $A_o$ is the cross-sectional area of the submerged portion of the drifter oriented perpendicular to the relative velocity, $\rho_o$ is the density of the ambient water, $C_o$ is an empirically determined drag coefficient, $\vec{u_E}$ is the Eulerian (fixed-in-space) current as measured by a stationary instrument or predicted by a numerical model of the governing hydrodynamics, $\vec{u_S}$ is the Stokes drift, or the residual circulation induced by the surface wave field, and $\frac{d\vec{x}}{dt}$ is the velocity of the drifter. By definition the drag force acts in the direction of the relative velocity. Expressing the two-dimensional

vectors as complex numbers and defining $\vec{u_{R_o}} = \frac{d\vec{x}}{dt} - (\vec{u_E} + \vec{u_S}) = |u_{R_o}|e^{i\theta_{R_o}}$, Equation 2.2 can be rewritten as

$$|F_o|e^{i\theta_{F_o}} = \frac{1}{2}A_o\rho_o C_o|u_{R_o}|^2 e^{i2\theta_{R_o}} \tag{2.3}$$

Similarly the drag force exerted by the wind on the exposed portion of the drifter can be expressed as follows.

$$\begin{aligned} \vec{F_a} &= \frac{1}{2}A_a\rho_a C_a\Big(\vec{u_w} - \frac{d\vec{x}}{dt}\Big)\Big|\vec{u_w} - \frac{d\vec{x}}{dt}\Big| \\ |F_a|e^{i\theta_{F_a}} &= \frac{1}{2}A_a\rho_a C_a|u_{R_a}|^2 e^{i2\theta_{R_a}} \end{aligned} \tag{2.4}$$

Definitions of variables are similar to above, except in that the subscript $a$ denotes values with respect to the ambient air rather than water. For brevity, the definition $\vec{u_{R_a}} = \vec{u_w} - \frac{d\vec{x}}{dt} = |u_{R_a}|e^{i\theta_{R_a}}$ is used, where $\vec{u_w}$ is the ambient surface wind velocity.

For a surface drifting buoy, used to measure object drift and generally quite small, the approximate magnitude of these terms is a as follows:

$$\begin{aligned} |F_o| &= \frac{1}{2}A_o\rho_o C_o|\vec{u}_{R_o}|^2 \\ &\approx 10^{-1}m^2 \times 10^3\frac{kg}{m^3} \times (10^{-1} \to 10^0) \times (10^{-2} \to 10^{-1}m/s)^2 \\ &\approx (10^{-3} \to 10^0)N \\ |F_a| &= \frac{1}{2}A_a\rho_o C_a|\vec{u}_{R_a}|^2 \\ &\approx 10^{-2}m^2 \times 10^0\frac{kg}{m^3} \times (10^{-1} \to 10^0) \times (10^0 \to 10^1m/s)^2 \\ &\approx (10^{-3} \to 10^0)N \end{aligned} \tag{2.5}$$

It is evident that both drag due to ambient water and air have similar magnitudes, suggesting a balance between the two terms. What follows is an examination of the magnitude and relative importance of the remaining terms in Equation 2.1.

The wave forcing on an object is determined by the amplitude, $a$ of waves with wavelengths comparable to the object dimension, $D_o$, [82, 59], and given by the following expression for a fixed object (note that the force on a floating object would be less, but is not as readily determined) [30, 10].

$$|F_w| = \frac{1}{4}\rho_o g a^2 D_o \tag{2.6}$$

Here $\rho_o$ is the water density, and $g$ is the acceleration due to gravity. Equation 2.6 can be further simplified by the conservative estimate that $a = \frac{D_o}{7}$, i.e. the amplitude in the spectral band where wavelength corresponds to $D_o$ is limited by wave steepness. From this, and assuming the small dimension of a typical drifting buoy, the approximate magnitude of the term can be determined as follows.

$$\begin{aligned} |F_w| &= \frac{1}{196}\rho_o g D_o^3 \\ &\approx 5 \times 10^{-3} \times 10^3 \frac{kg}{m^3} \times 10^1 \frac{m}{s^2} \times (10^{-1}m)^3 \\ &\approx 5 \times 10^{-2} N \end{aligned} \tag{2.7}$$

Given that two conservative assumptions were made in the estimation of this term (fixed vs. floating object, and $a$ at the limit of wave steepness), it appears unlikely that $\vec{F_w}$ will be of leading order magnitude for the purposes of this study. It will therefore be neglected for the purpose of this thesis, as has traditionally been done for small objects ($D_o < 30$m) [10]. The term may however become important when making predictions about the drift of larger objects.

The potential impact of gravity acting on a drifting object along a sea surface slope, $\Theta$, can be readily determined as follows assuming that the object is oriented normal to the water surface.

$$\vec{F_g} = mgsin\Theta \tag{2.8}$$

For leading-order importance, i.e. $\vec{F_g} \approx 1N$, we therefore require that $sin\Theta > \frac{1}{mg}$. For an object with mass $m$ ≈1 kg this implies a sea surface slope $\Theta > 5°$, i.e. a difference in sea level of ∼87 m over 1 km. This is clearly unrealistic, and hence $\vec{F_g}$ is neglected for the purpose of this thesis.

Measurements or modelled values of $\vec{u_E}$ and $\vec{u_w}$ include their respective contributions from the Coriolis effect. However unless these models are coupled to a surface wave model, the hydrodynamic Coriolis component is only based on Eulerian currents, and the Coriolis-Stokes force, i.e. the apparent Coriolis acceleration of the Stokes drift velocity [122] is neglected. The acceleration of currents due to this force can be written as,

$$\frac{d\vec{u_{CS}}}{dt} = -f\hat{k} \times \vec{u_S} \tag{2.9}$$

Here $f = 2\Omega sin\phi$ is the Coriolis parameter, defined previously. A brief order-of-

magnitude analysis reveals that for the mid-latitudes ($f \approx 10^{-4}\frac{1}{s}$) this acceleration is quite small.

$$\begin{aligned}\frac{d\vec{u_{CS}}}{dt} &= -f\hat{k} \times \vec{u_S} \\ &= 10^{-4}\frac{1}{s} \times (10^{-2} \to 10^{-1})\frac{m}{s} \\ &= 10^{-6} \to 10^{-5}\frac{m}{s^2}\end{aligned} \tag{2.10}$$

While this value is small and the Coriolis-Stokes force is generally neglected in drift trajectory models, its effect on trajectories may be noticeable. The force is not considered further in this thesis, but interested readers are referred to further work on the topic such as [159].

The inertial term in Equation 2.1 (left-hand side) has been neglected in previous work with the justification that research has shown liferafts to accelerate to a quasi-steady velocity from rest within 20 seconds (i.e. $m\frac{d^2\vec{x}}{dt^2} = 0$ after 20 seconds) in strong winds, and therefore it can be inferred that objects of lesser inertia, such as drifting buoys, would respond to changes in velocity even more quickly [10]. This assumption is well supported where time steps in drift trajectory models are significantly longer than the time required for the tracked object to respond to changes in the forcing velocity.

Quantifying the impact of the inertial term through observation is difficult, as drifting buoys generally only transmit their position every few minutes to avoid compounding the effect of positional uncertainty in the GPS fix. Therefore these buoys do not resolve the period between the change in forcing and arrival at the new quasi-steady velocity. However for a small drifting buoy with a mass (including added mass) on the order of 1 kg, approximate values of $\frac{d^2\vec{x}}{dt^2} = 1$ and/or $f\vec{k} \times \frac{d\vec{c}}{dt} = 1$ would be required in order for the inertial term to have similar magnitude to the drag terms in a low wind environment. This corresponds approximately to a change in drift speed of 100 m/s over 2 minutes to satisfy the acceleration term and a steady velocity of order $10^4$ m/s at mid-latitudes ($f \approx 10^{-4}$) to satisfy the Coriolis term. Such drift velocities and accelerations are most improbable in a marine environment. Therefore the drifters' inertia is generally neglected [30, 10, 122], and will be neglected for the purposes of this thesis. It is noted however that this presents a potential limitation when making predictions about the drift of very large and/or heavy objects.

Given these simplifications, Equation 2.1 reduces to a balance between water and

air drag on the drifter, where water drag opposes the motion and air drag forces the motion in a quasi-steady state. In mathematical form,

$$\frac{1}{2}A_o\rho_o C_o|\vec{u}_{R_o}|^2 e^{i2\theta_{R_o}} = \frac{1}{2}A_a\rho_a C_a|\vec{u}_{R_a}|^2 e^{i2\theta_{R_a}} \tag{2.11}$$

From this form, a common further simplification is to assume that $\vec{u}_w >> \frac{d\vec{x}}{dt}$, such that $\vec{u}_{R_a} = \vec{u}_w$. This reduces Equation 2.11 to the commonly known linear vector addition for drift as the sum of the Eulerian current, Stokes drift, and a fraction of the wind velocity [122].

$$\begin{aligned} \frac{d\vec{x}}{dt} &= \vec{u}_E + \vec{u}_s + \alpha\vec{u}_w \\ \alpha &= \sqrt{\frac{\rho_a A_a C_a}{\rho_o A_o C_o}} \end{aligned} \tag{2.12}$$

Here $\alpha$ is known as the windage, and is derived assuming constant values of $C_aA_a$ and $C_oA_o$. These assumptions imply that the drag coefficients are constant in the limit of high Reynolds number [122], and that object orientation is constant (either through isotropic geometry or absence of rotation with respect to the relative motion between the drifter and the water). The validity of these assumptions remains subject to further research. The windage term is discussed further in the following section, and is explored in detail in Chapter 4.

It is however clear that some physical processes unaccounted for in Equation 2.12 may be of significance, though not to leading order. An accurate accounting of uncertainty in the solution of Equation 2.12 will need to address turbulent motions at scales unresolved by the provided $\vec{u}_E$ and $\vec{u}_w$, uncertainty in the provided input data (from numerical model errors and observation limitations), uncertainty in the object geometry, and the impact of missing physics such as the inertial term, direct wave forcing ($F_w$), and Coriolis-Stokes forcing. This thesis aims to establish the necessary framework to perform this uncertainty accounting for the terms of leading-order importance, and the impact of missing physics outlined in the latter point is deferred to future work.

### 2.2.2 Windage

In the form given by Equation 2.12, $\alpha$ only accounts for the drag force imparted by the wind at the ocean surface on parts of the drifting object that protrude above

the water line. However, in its most commonly applied form, $\vec{u}_E$ is taken from the uppermost layer of a numerical ocean circulation model and $\vec{u}_w$ is taken as the wind speed at 10 m from an atmospheric prediction system. In some studies $\vec{u}_s$ is taken from a numerical wave prediction system, or it is neglected as it is considered to be correlated to the wind speed [144]. In this formulation all terms on the right-hand side of Equation 2.12 are subject to considerable uncertainty.

Uncertainty in $\vec{u}_E$ arises primarily from differences between currents predicted by a numerical model and analogous values observed in the real world. Primarily, these differences arise from limitations in models' ability to resolve their domain within reasonable computational effort as well as the fact that they are by necessity an abstraction of reality. For example, [16] and [143] found that increased detail in the representation of the ocean surface in a numerical model yielded significantly improved drift predictions, as gradients in the currents are generally strongest in the upper few meters of the ocean. A secondary source of uncertainty is in the observations of currents that are compared to modelled values, or applied to a drift trajectory model directly as in Chapter 4. All instruments used to make such measurements are subject to some inherent random uncertainty, and placement of the instrument may have an impact on representativeness for a comparison. Additionally, instruments for current measurement are generally based on acoustic principles, and have difficulty measuring currents at the very surface of the ocean due to reflections of acoustic signals. Therefore data must be carefully quality controlled to avoid introducing erroneous measurements into an analysis [cf. 146, 133].

For winds, uncertainty in $\vec{u}_w$ arises from similar sources as for $\vec{u}_E$ however additional uncertainty is introduced by the convention to report wind velocities at 10 m above ground. Due to atmospheric boundary layer effects, these winds can be up to $\sim$30% larger than those at the immediate ocean surface, depending on atmospheric stability and the total wind speed [132]. Periodic sheltering of a drifting object by incident waves in large seas are an additional source of uncertainty that is beyond the scope of the current work to address. Boundary layer effects are explored further in Chapter 4.

In the context of these uncertainties, it is often considered most appropriate for practical applications to treat $\alpha$ as an empirically fitted parameter. The contribution of Stokes drift is in some cases contained within this fitted parameter [cf. 54]. The major drawback of this approach, which will be addressed in Chapter 4, is that $\alpha$ becomes dependent upon the forcing data and is therefore not readily extrapolated if

forcing fields change, say through an update of operational environmental prediction systems, or a drifting object with a previously unconsidered geometry is encountered.

### Direct Windage

A recent example of how windage is derived in an operational context is given in [141]. Here observed tracks of various types of drifting buoys were fitted to virtual analogues created using the output from two different sets of ocean and atmospheric prediction systems, using windage as a fitting parameter. This experiment was conducted both with explicitly added Stokes drift from a wave model, and without. The authors found that explicitly including Stokes drift reduced the windage coefficient by ∼0.01 - 0.015 (1-1.5&). This value, as well as the windage coefficient itself, varied between the sets of prediction systems by ∼0.005 (0.5%) which highlights the dependence on forcing data in this approach. These results are in good agreement with those of [29] who performed similar experiments.

In the context of observations, [122] provides a frequently cited example of the derivation of windage. Here the authors used surface drifting buoys in conjunction with measurements of surface currents from a 2 MHz ADCP deployed specifically for this purpose, which was able to measure currents at only 0.5 m below the ocean surface, wave data from a directional wave buoy, and wind measurements from their research vessel to solve Equation 2.12 for $\alpha$. Similarly to [141], they performed the calculation both with and without inclusion of Stokes drift. They found that values of $\alpha$ of 0.0099 (0.0031) gave the best results with (without) explicit inclusion of Stokes drift. Additionally they found that the importance of Stokes drift is on the same order as the Eulerian surface current, and the explanatory power of Equation 2.12 is diminished without it.

### Leeway Formulation

In search and rescue scenarios a formulation of the drift trajectory model which prescribes down-wind and cross-wind components of windage is generally used, known as the leeway model. The rationale for this is that objects with irregular geometry, for example sailboats, will not respond isotropically to the wind but rather exhibit significant cross-wind motion as well as down-wind. The following operational definition for the term leeway has been put forward by [12]:

> "Leeway is the motion of the object induced by wind (10 m reference height) and waves relative to the ambient current (between 0.3 and 1.0 m depth)."

Through extensive field testing, a database of leeway parameters for 86 common search and rescue targets has been established [2, 12, 28]. These parameters describe the observed object motion relative to the 10 m wind speed ($L \propto W_{10}$) as a set of nine linear regression coefficients describing the down-wind ($d$), cross-wind ($c^+$, to the right), and cross-wind ($c^-$, to the left) components of the motion [10].

$$\begin{aligned} L_d &= a_d W_{10} + b_d + \epsilon_d \\ L_{c^+} &= a_{c^+} W_{10} + b_{c^+} + \epsilon_{c^+} \\ L_{c^-} &= a_{c^-} W_{10} + b_{c^-} + \epsilon_{c^-} \end{aligned} \tag{2.13}$$

Here $a$ and $b$ are the slope and intercept of the linear regression, and $\epsilon$ is an error term that is assumed to be normally distributed about the regression. This error term is usually evaluated through a Monte Carlo simulation, with additional parameterizations to evaluate the probability of jibing, i.e. an object of non-isotropic geometry changing orientation [12, 28].

The relatively simple and standardized nature of the leeway method has led to reasonably good performance in operational search and rescue scenarios [11]. However it is limited in application to objects for which parameters have been derived, or adequate proxies of these objects. Treatment of uncertainty is also limited to the linear addition of normally distributed error, which may not account for errors in the forcing wind and current data. In Chapter 4, an implementation of Equation 2.12 using a fuzzy number-based uncertainty model is compared to the leeway method to evaluate its performance in tracking objects of variable geometry.

### 2.2.3 Dispersion

In addition to the advection by coherent flow features described by Equation 2.12, drift predictions often include an additional stochastic dispersion component to account for turbulent flows that are removed by averaging from observations and/or not resolved by model outputs. Given that the magnitude of turbulent fluctuations is often unknown, this may also be considered an ad-hoc uncertainty treatment.

In this stochastic modelling the displacement, velocity, or acceleration of a particle are taken as the sum of the coherent, deterministic component and a stochastic

component that is simulated by Monte Carlo methods. The particles trajectory is then stepped forward in time based on the determined displacement, velocity, or acceleration. The choice of noised quantity significantly affects the model results and their suitability for a particular application [72].

Models with noised displacement, or zero-order models, reduce to the common advection-diffusion equation provided that the stochastic component of the displacement increments is normally distributed and uncorrelated. While the assumption of homogeneous, isotropic turbulence inherent in the use of an advection-diffusion equation with a constant diffusivity has been brought into question, this modelling approach is still commonly used in operational oceanography as it does offer appreciable savings in computational cost over higher-order Lagrangian stochastic models [13, 72]. Some recent examples of applications are given in [36, 127, 19, 63].

An alternative zero-order model is given by fractional Brownian motion (fBm), which allows for correlation between the noised increments based on a fractal (or Hausdorff) dimension $D$, bounded as $1 < D \leq 2$ for motion in two-dimensional space [79]. Values of $D$ between 1.2 and 1.4 have repeatedly been found to correspond to natural phenomena [78, 99, 125, 128]. Fractional Brownian motion exhibits self-similarity at scales where the motion exhibits fractal characteristics, for example as determined by the 'yardstick' method [99, 148]. Following [125], this self-similarity of positions can be expressed as,

$$[x(t + \lambda\Delta t) - x(t)] \stackrel{d}{=} \lambda^{\frac{1}{D}}[x(t + \Delta t) - x(t)] \tag{2.14}$$

Here $\stackrel{d}{=}$ implies equality in the sense of distributions, i.e. equality of statistical properties [99]. Further, this self-similarity can be exploited to derive a relationship for particle dispersion as $< X^2 >=< [x(t_0 + t) - x(t)]^2 > \propto t^{\frac{2}{D}}$ [79]. This stands in contrast, but not contradiction, to the model proposed by [145], which proposes $< X^2 > \propto t^2$ at short time scales where autocorrelation of the motion increments is near perfect (the ballistic regime, Kolmogorov microscales in the context of fluid dynamics [125]) and $< X^2 > \propto t$ at longer time scales where motion increments are uncorrelated (the diffusive regime). Noting that fBm asymptotes to the ballistic regime for $D$=1 and the diffusive regime for $D$=2, fBm can be viewed as an intermediate regime of single particle dispersion where motion increments are self-correlated, but not perfectly so, and this self-correlation does not decay exponentially as assumed by [145]. In the context of surface drift, this theory has been successfully applied by [99, 125], and

[53] reports promising results in an application to oil spill tracking. In this thesis, fBm is relevant to the studies presented in Chapters 3 and 5.

Higher-order Lagrangian stochastic models randomize components of the particle's velocity (first-order), or acceleration (second-order). They offer increased accuracy over zero-order models though the representation of motion still contains many assumptions, for example the spectral shape of the particle velocities in the first-order case [72]. The increased accuracy of higher-order models comes at appreciable computational cost, with a reported 10 - 100 fold increase from zero-order to first-order. Also, all Lagrangian stochastic models rely heavily on output data from numerical circulation models, which adds appreciable uncertainty to the solution [13].

Regardless of the details of the chosen modelling approach, successful application and analysis of dispersion models relies on appropriate removal of coherent flow signals to ensure that the zero-mean assumption is met. This requires careful consideration for data from both observations and numerical modelling. For numerical models, coherent flow features are dependent on both the resolution of the spatial discretization employed in the model as well as the parameters of the chosen turbulence closure [134]. Any changes to these aspects of a numerical model require re-evaluation of dispersion models used with the outputs. Further, with application to both models and observations, the record length for the removal of tidal signals through harmonic analysis must be carefully considered, as a record of 18.6 years is required to remove all periodic signals from the data, provided that tidal signals are stationary [cf. 103]. In the vicinity of significant freshwater sources stationarity is not guaranteed, as the barotropic pressure gradient due to the inflowing freshwater may modulate the tidal signal [cf. 80]. Due to these challenges, proper de-meaning and interpretation of dispersion models for drift requires careful preparation of the input data.

### 2.2.4 Research Opportunities

Predicting the drift of floating objects is a challenging, yet important, endeavour and from the summary presented in this section several opportunities for research contributions can be identified.

Primary among these is an accounting and aggregation of uncertainty affecting various stages of the study and modelling of surface drift. The latter relies heavily on output data from numerical ocean and atmospheric models, which introduce uncer-

tainty as the results from these models generally differ to some extent from reality, despite continuing advances in the field. But preceding this uncertainty is the fact that numerical models with sufficient detail do not exist for all of the worlds oceans. This is especially true near the coastline, where interactions between the ocean and landmasses require additional detail to be captured in the numerical models and the pressures for accurate predictions are strongest due to more frequent human use of these areas.

Further uncertainty is introduced in the model for drift trajectories itself, especially in the treatment of wind-related components of drift, as only terms of leading order importance are accounted for in Equation 2.12. As detailed in Section 2.2.2, object geometry has a strong influence on drift trajectories. This influence is not readily accounted for in a deterministic sense, as the mathematical formulation to do so requires precise knowledge of currents and winds at the ocean surface. Further, as noted by [122], the drag coefficients used in the calculation of $\alpha$ by Equation 2.12 are dependent on the ambient Reynolds number and drifting objects may be periodically sheltered from wind by large incident waves. Therefore a singular possible value is likely inadequate for a full deterministic description of $\alpha$. Additionally, the derivation of Equation 2.12 assumes that the wind velocity relative to the drift velocity is large. This assumption may not hold in coastal waters, where winds are often weak in the summer time but flow constrictions by the surrounding landmasses result in strong tidal currents year-round. These uncertainties may be addressed through appropriate formulation of the uncertainty associated with input data, however further work to improve drift trajectory models beyond Equation 2.12 may also be fruitful.

In Chapter 3 of this thesis novel work is presented on data from ocean surface drifters in a coastal fjord system where numerical model results were not available for the period required to aid in interpretation of the drifter data. Based on the drifter data, a simple seasonally-varying statistical model for their motion is formulated and found to be consistent with fractional Brownian motion. This model is shown to capture the effect of seasonal changes in the wind patterns, as well as the effect of drifters grounding in the strongly constricted topography. Historical observations of sheening from an oil spill in the region are well reproduced, and the presented results give an adequate model for drift in the study region that is independent of forcing data.

Chapters 4 and 5 present work that addresses the need for accounting and propagating uncertainty through the drift trajectory model given by Equation 2.12. To do

this, the theory of fuzzy numbers is employed, which will be described in the following section. Through the use of fuzzy numbers, an accounting of instrument uncertainty in observations of current and wind at a single mooring can be added to estimates of turbulence at scales unresolved in these measurements and propagated through Equation 2.12, with parameterizations for oceanic and atmospheric boundary layer effects. The results compare well to observations of various types of drifting buoy, with their geometry deterministically accounted for using $\alpha$ as per Equation 2.12. In Chapter 5 this approach is extended to spatially variable forcing fields, in a manner that allows for accurate descriptions of the differences between observations and numerical forcing models to be propagated through the trajectory model.

## 2.3 Fuzzy Numbers

Fuzzy numbers were chosen for this research as they offer well established algorithms for describing and aggregating uncertainty from various sources, and propagating this uncertainty through complex mathematical systems. They do not require assumption of an underlying distribution, in contrast to the common assumption of uncorrelated, normally distributed error statistics in probabilistic uncertainty propagation [148]. Uncertainties in coastal oceanic properties have previously been reported to deviate from Gaussian distributions [cf. 114], and hence fuzzy numbers are thought to offer a promising alternative for uncertainty propagation here. Lastly, data on both forcing and direct measurement of ocean surface drift is generally sparse, and fuzzy numbers offer the capability to produce accurate, though not necessarily precise, results with sparse input data.

Fuzzy sets extend classical set theory to continuous, rather than binary, membership or degree of belonging. Quantitatively put, if $U$ is the set of all values, also known as the universe of discourse, and $A$ is a subset of $U$ then in classical set theory values from $U$ have a membership of either 0 or 1 in $A$, i.e. they either belong to $A$, or they don't. In fuzzy set theory the membership of values from $U$ in $A$ can take any value on the interval $(0, 1]$, indicating varying levels of possibility that the value belongs, or does not belong, to $A$. The function describing the varying level of membership in $A$ is known as the *membership function*, and is denoted $\mu_A$. The concept of fuzzy sets was first proposed by [170], and has seen widespread application throughout information science and various natural sciences since then. Much of the popularity of fuzzy sets arises from the fact that the theory allows for mathematical

treatment of vague statements such as "low", "high", or "might be possible" while relaxing some of the more strict requirements of similar work in probability theory which renders it well-suited for work where data is sparse [**?** ].

In the natural sciences, methods based on fuzzy sets have previously been used to detect and track storm systems [84], describe oceanic water masses [42], study groundwater flow [129, 55], quantify uncertainties in the stage-discharge relationship in rivers [131], and model dissolved oxygen concentrations in rivers [69], amongst others. There has also been recent theoretical work on using fuzzy methods in the context of data assimilation in physical oceanographic modelling, to estimate parameters based on small, biased samples [107].

The present work will make use of the concept of fuzzy numbers. A fuzzy number is defined as a special case of a fuzzy set. Here the support of the number is defined as the interval of all possible values, i.e. all values where $\mu > 0$. The membership function increases monotonically from the lower bound of the support to a single value where $\mu = 1$, known as the height, core, or kernel of the number, and then decreases monotonically back to zero at the upper bound of the support, i.e. the fuzzy set of possible values is convex with respect to membership of the elements of the fuzzy set [170, 65].

Two key concepts required to extend traditional arithmetic to fuzzy numbers are the extension principle [171], and the concept of $\alpha$-cuts, specifically level $\alpha$-cuts [117, 39]. While they are known as $\alpha$-cuts in the literature, they will be referred to here by their synonym 'membership levels' to avoid confusion with the windage coefficient $\alpha$. The extension principle states that if a fuzzy set $\tilde{B}$ results from a function $F$ acting on fuzzy sets $\tilde{A_1}, \tilde{A_2}, ..., \tilde{A_N}$, then the membership of $\tilde{B}$ at value $b = F(a_1, a_2, ..., a_N)$ is,

$$\mu_{\tilde{B}}(b) = sup\{min\{\mu_{\tilde{A_1}}(a_1), \mu_{\tilde{A_2}}(a_2), ..., \mu_{\tilde{A_N}}(a_N)\}\} \tag{2.15}$$

Here $b, a_1, a_2, ..., a_N$ are elements of $\tilde{B}, \tilde{A_1}, \tilde{A_2}, ..., \tilde{A_N}$. Given the extension principle, membership levels can be used to discretize a fuzzy number into crisp intervals describing the support for varying degrees of belief. For example, if element $x \in \tilde{U}$ belongs to fuzzy set $\tilde{A}$, then the membership levels of $\tilde{A}$ are defined as $\tilde{A}_{\alpha 1} = \{x \in \tilde{U}, \mu(x) \geq \alpha_1\}$. In words, $\tilde{A}_{\alpha 1}$ is the crisp set consisting of all elements $x$ with a membership in $\tilde{A}$ that is greater or equal to than the specified threshold $\alpha_1$ [39]. Basic arithmetic can then be extended to these intervals, while retaining the membership level $\alpha 1$. Given two

fuzzy numbers described by their intervals at a membership level, say $\tilde{A_{\alpha 1}} = [a_1, a_2]$, and $\tilde{B_{\alpha 1}} = [b_1, b_2]$, the basic arithmetic operations are defined as follows [65]:

$$\begin{aligned}
\tilde{A_{\alpha 1}}(+)\tilde{B_{\alpha 1}} &= [a_1 + b_1, a_2 + b_2] \\
\tilde{A_{\alpha 1}}(-)\tilde{B_{\alpha 1}} &= [a_1 - b_2, a_2 - b_1] \\
\tilde{A_{\alpha 1}}(.)\tilde{B_{\alpha 1}} &= [a_1 * b_1, a_2 * b_2], \text{ for } \{a_1, a_2, b_1, b_2\} > 0 \\
\tilde{A_{\alpha 1}}(:)\tilde{B_{\alpha 1}} &= [\frac{a_1}{b_2}, \frac{a_2}{b_1}], \text{ if } [b_1, b_2] \in \Re^-, \text{ or } [b_1, b_2] \in \Re^+
\end{aligned} \tag{2.16}$$

The results at various membership levels on the interval $(0, 1]$ can then be used to re-construct a fuzzy number.

The shape of the membership function $\mu$ determines the width of the various membership levels, and therefore proper choice of membership function is clearly of significant importance to proper implementation of fuzzy algorithms. Triangular membership functions have often been used extensively in previous work, primarily due to their simplicity and ease of construction, requiring a minimum of only three data points to formulate. However it has been shown that for modelling applications in the natural sciences results may be improved significantly by choosing more complex membership functions, for example those based on Gaussian or Gumbel distributions [69].

Deriving membership functions requires consideration of possibility theory, introduced by [172], which should not be confused with traditional probability theory (though the two are weakly related). The basic tenets of the theory are as follows:

- If $F$ is a fuzzy subset of the universe $U$, that is characterized by the membership function $\mu_F(u)$ then the distribution function of possibilities that a variable $X$ is equal to $u$, denoted by $\pi_X$ (the associated distribution is $\Pi_X$), is numerically equivalent to $\mu_F(u)$.

- The necessity of an event $A$ is denoted $N(A)$, and defined as 1 minus the possibility of the contrary of the event, $A'$. I.e. $N(A) = 1 - \Pi(A') = inf\{1 - \pi(x), x \notin A\}$.

- Both the possibility of an event and its necessity are required to describe likelihood using possibility theory. This is because $\Pi(A) \geq N(A)$, i.e. an event has to be possible before it can be necessary, which leads to $\Pi(A) + \Pi(A') \geq 1$, and $N(A) + N(A') \leq 1$. Knowing the possibility of an event by itself does not give

complete information about the possibility of the contrary event (which is the case in probability theory, as $P(A) + P(A') = 1$). Therefore possibility theory is less restrictive than probability theory.

- $\Pi(A) \geq P(A)$, i.e. an event has to be possible before it can be probable.

Note that [70] has called the validity of the latter tenet into question for any processes involving the 'or' operator. For two events $A$ and $B$, this may be expressed as follows.

$$\Pi(A \vee B) = max(\Pi(A), \Pi(B)) \leq P(A \vee B) = P(A) + P(B) - P(A \wedge B) \quad (2.17)$$

This thesis does not make use of any logical operations on fuzzy numbers for which this criticism applies. Hence it will not be further considered here, but is mentioned for completeness.

If a probability density function (PDF) corresponding to a variable that is to be fuzzified is known, it can then be used to construct the membership function by the following steps:

1. Discretize the PDF into a series of intervals between 0 and the maximum probability
2. At each level, find the interval over which the PDF indicates a probability greater than or equal to this level
3. compute the probability that the value is either below the lower bound of the interval, or above the upper bound
4. This probability corresponds to the membership level of the interval determined in Step 2

The proof of this transformation is given in [38]. Examples of fuzzy numbers derived in this manner are shown in Figure 4.2.

For systems where a fuzzy-valued term appears multiple times, arithmetic should be performed using the transformation method described by [56]. This method is used in Chapter 4, and the description given there is reproduced here for completeness. A thorough summary of the method can be found in [56, 55]. In the transformation method, the fuzzy-valued terms in the system of equations are discretized into arrays

describing all possible combinations of all possible values of these terms. The system can then be solved using element-wise operations on these arrays.

Consider a system of equations with $n$ independent fuzzy variables $P_i, i = 1, ..., n$. Each $P_i$ consists of $m + 1$ membership levels $X_i^{(j)}$. Each membership level $X_i^{(j)}$ is described by the interval $[a_i^{(j)}, b_i^{(j)}]$, which is then discretized into an array $\hat{X}_i^{(j)}$ according to the following equations.

$$P_i = \{X_i^{(0)}, X_i^{(1)}, ..., X_i^{(m)}\} \tag{2.18a}$$

$$X_i^{(j)} = [a_i^{(j)}, b_i^{(j)}], \quad a_i^{(j)} \leq b_i^{(j)} \tag{2.18b}$$

$$\hat{X}_i^{(j)} = \overbrace{(\gamma_{1,i}^{(j)}, \gamma_{2,i}^{(j)}, ..., \gamma_{(m+1-j),i}^{(j)}, ..., \gamma_{1,i}^{(j)}, \gamma_{2,i}^{(j)}, ..., \gamma_{(m+1-j),i}^{(j)})}^{(m+1-j)^{i-1} \text{ tuples of length } (m+1-j)} \tag{2.18c}$$

$$\gamma_{l,i}^{(j)} = \overbrace{(c_{l,i}^{(j)}, ..., c_{l,i}^{(j)})}^{(m+1-j)^{n-i} \text{ elements}} \tag{2.18d}$$

$$c_{l,i}^{(j)} = \begin{cases} a_i^{(j)} & \text{for } l = 1 \text{ and } j = 0, 1, ..., m \\ \frac{1}{2}\left(c_{l-1,i}^{(j+1)} + c_{l,i}^{(j+1)}\right) & \text{for } l = 2, 3, ..., m - j \text{ and } j = 0, 1, ..., m - 2 \\ b_i^{(j)} & \text{for } l = m - j + 1 \text{ and } j = 0, 1, ..., m \end{cases} \tag{2.18e}$$

A graphical example of this discretization, adapted from [55], is given in Figure 4.3 for a hypothetical fuzzy number described by four membership levels, i.e. $m = 3$. The final result of the transformation of all $n$ fuzzy input variables $P_i$ at one point in time is a set of $m + 1$ arrays $\hat{X}_i^{(j)}$ for each $P_i$. To illustrate explicitly, consider a system of equations with two fuzzy input variables (i.e. $n = 2$), each similar to the example shown in Figure 4.3. The final arrays describing the membership level $j = 1$ are constructed as follows.

$$\hat{X}_1^{(1)} = [c_{0,1}^{(1)}, c_{0,1}^{(1)}, c_{0,1}^{(1)}, c_{1,1}^{(1)}, c_{1,1}^{(1)}, c_{1,1}^{(1)}, c_{2,1}^{(1)}, c_{2,1}^{(1)}, c_{2,1}^{(1)}] \tag{2.19a}$$

$$\hat{X}_2^{(1)} = [c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}, c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}, c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}] \tag{2.19b}$$

Using these arrays the system of equations for the membership level can be solved as it would be in a crisp manner, but using element-wise arithmetic on the arrays $\hat{X}_1^{(1)}$ and $\hat{X}_2^{(1)}$.

The result of this element-wise arithmetic for each membership level ($\hat{X}_{result}^{(j)}$) can

then be re-transformed into a fuzzy number by taking the minimum and maximum values of each $\hat{X}_{result}^{(j)}$ to be the bounds of the membership level $j$ of the fuzzy result, while ensuring that the fuzzy result remains convex.

$$P_{result} = \{Z^{(0)}, Z^{(1)}, ..., Z^{(m)}\} \tag{2.20a}$$

$$Z^{(j)} = [a^{(j)}, b^{(j)}],\ j = 0, 1, ..., m \tag{2.20b}$$

$$a^{(j)} = min(a^{(j+1)}, min(\hat{X}_{result}^{(j)})) \tag{2.20c}$$

$$b^{(j)} = max(b^{(j+1)}, max(\hat{X}_{result}^{(j)})) \tag{2.20d}$$

This method can be used to solve any mathematical system provided that only operations defined for element-wise arithmetic are used.

# Chapter 3

# Surface drift and dispersion in a multiply-connected fjord system

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## 3.1 Introduction

Increasing development of coastal regions requires improved knowledge of complex local oceanographic processes in order to support development while protecting ecosystems in the face of increased shipping and other human activity. Understanding of drift and dispersion at the ocean surface is a key requirement for these endeavours, to predict the transport and dispersion of buoyant contaminants such as oil [cf. 54, 44], larval organisms and other biological material [120], and targets of search and rescue operations [30, 10, 32].

Surface circulation in coastal fjords is driven by tidal currents, wind forcing, and estuarine circulation induced by fresh river inputs and terrestrial runoff [147, 41]. The combination of these forcings and the associated stratification often results in complex flow patterns with significant spatial and temporal variability. When considering the drift of objects at the ocean surface, wave-induced circulation and direct wind forcing acting on exposed parts of the object must also be taken into account [122].

Freely drifting buoys at the ocean surface (drifters) can reveal insights into the dynamics and statistics of ocean surface flow. Certain types of drifters can also be used as a direct proxy for the transport of spilled oil [44], though this is not guaranteed as behaviour may vary significantly between drifters that are drogued at different depths within the upper meter of the water column [121]. Data from surface drifters are also a valuable resource for validation of numerical models, which are often used to estimate surface circulation in operational applications.

Drifter studies in the open ocean and on continental shelves are well documented, both for surface circulation [cf. 34, 35, 24, 108] and deeper circulation using drogues [cf. 47, 112, 95, 124]. These studies infer mean circulation patterns and speeds as well as dispersion characteristics, both for the absolute dispersion of the drifters as well as the relative separation between them. They make use of significant amounts of data and long time series collected by drifters well away from the coastlines. In areas closer to the shoreline long time series are more difficult to collect as drifters often ground on the shoreline [142, 104].

Drifter data from fjords and other topographically restricted coastal regions is usually limited to deployments of 24 hours or less [cf. 37, 151, 89, 136, 138, 135, 137]. These studies provide useful estimates of mean drift and dispersion for time scales on the order of a tidal cycle, but do not provide information on longer time scales. [104] report on statistics from expendable drifters released in the Salish Sea, an inland sea in southern British Columbia, Canada. The lifetimes of their drifters sometimes exceeds 30 days, though they generally ground ashore within three days. However the Salish Sea is an estuarine system with a channel width that exceeds the internal Rossby radius of deformation appreciably, implying that the rotation of the Earth plays a significant role in the evolution of drift trajectories here. Therefore this region differs from the interconnected narrow fjords that characterize much of the coastline in northern regions of the world. Here we aim to address a gap in the literature by analyzing a set of longer drifter trajectories in a system of multiply connected, narrow coastal fjords.

Estimates of absolute dispersion from drifter data are often based on the theory of [145], which considers 1-dimensional dispersion in stationary, isotropic, and homogeneous turbulent flow. The theory has been extended to 2-dimensional anisotropic applications and several drifter data sets have been described using this theory [35, 33, 112, 136]. The theory predicts that the drifters' displacement variance (which is a proxy for occupied area) will grow proportionally to time raised to the power of 2.

In a fjord system, any dispersion model that is applied on length scales comparable to the channel widths should account for the fact that dispersion characteristics may vary significantly across the channel [142], and that boundaries may induce anomalous (non-Taylorian) dispersion [4, 72].

An alternative dispersion model based on fractional Brownian motion (fBm) has produced good results over the length scales on which drifter trajectories exhibit a fractal character (100+ km in the open ocean) [125]. FBm can be used to describe dispersion where persistent autocorrelations in the flow result in non-Taylorian behaviour. It predicts displacement variance growth proportional to time raised to the power of 1.5 for a fractal dimension of 1.3. This fractal dimension has been repeatedly reported in drifter literature [99, 126, 125]. In this context the fractal dimension may be thought of as a measure of the portion of the available space that a drifter occupies. A fractal dimension of 2 implies pure Brownian motion which eventually fills a plane entirely, while a fractal dimension of 1 implies motion along a straight line [125]. Fractional Brownian motion was used to describe drifter dispersion and residence time in a coastal embayment by [37], and has been used to model pollutant dispersion in coastal regions with some success [1, 53].

In this paper we present a study of drifter data from the Kitimat fjord system in northwestern British Columbia, Canada, where 206 surface drifters were deployed between March 2014 and July 2016 [100]. Their tracks cover the study region well, and allow us to derive general results about mean drift patterns and dispersion on time scales up to ten days. This was not previously possible in narrow, interconnected fjord systems due to the short duration of drifter tracks. The observed along-channel dispersion pattern is then contextualized by analyzing the effects of cross-channel drift and interaction with the shoreline. We then investigate the general dynamics of the drifter motion further and provide evidence that the drifter tracks exhibit fractal characteristics. Finally, we confirm that the tracks of the observed drifters are an adequate proxy for spilled oil as they correspond well with observations of diesel sheen made after the passenger ferry *Queen of the North* sank in the study area in 2006. On this basis we propose a statistical model for the possible distance over which oil spills in the Kitimat fjord system may be transported within ten days.

## 3.2 Observations

### 3.2.1 Physical Setting

The Kitimat fjord system (Figure 3.1) is located in northwestern British Columbia, Canada and spans from the coastal waters of Hecate Strait to the town of Kitimat, approximately 140 km inland. Douglas Channel is the main channel in the system and reaches from Kitimat to the community of Hartley Bay, at the northern edge of Wright Sound. Here it splits into Squally Channel to the west and Whale Channel to the east. These channels eventually connect to Hecate Strait. When discussing regional differences within the system Douglas Channel will hereafter be referred to as DC, the smaller channels east of Douglas Channel as EAST, and the channels south of Wright Sound as SOUTH.

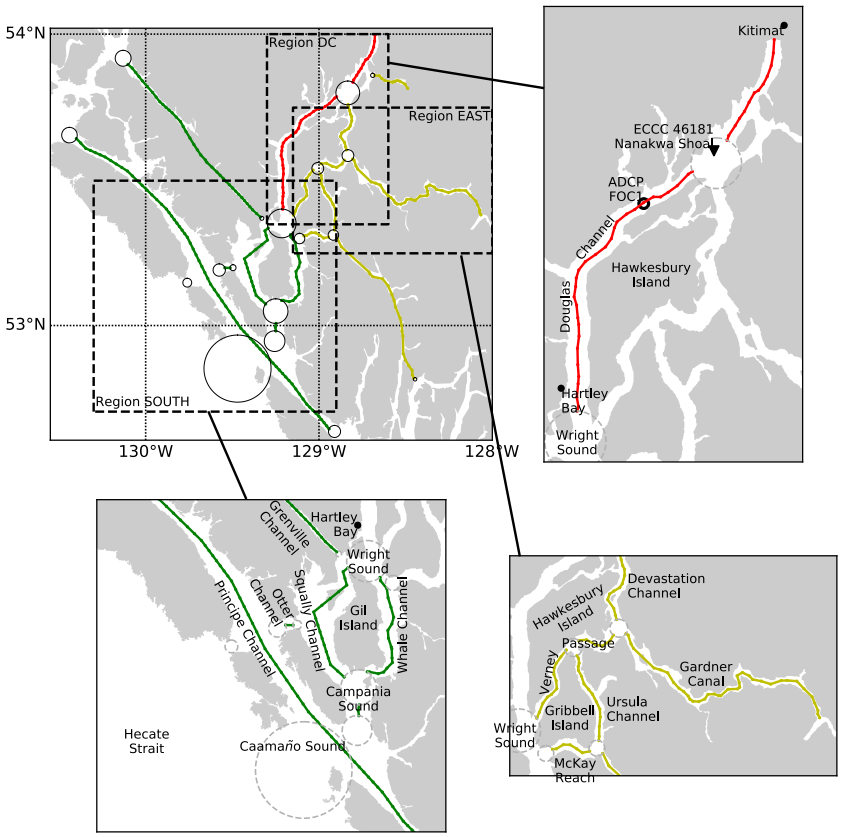


Figure 3.1: Map of the Kitimat fjord system showing sub-regions DC, EAST, and SOUTH, and grid system for analysis of along-channel dispersion. Channels in region DC are shown in red, region EAST in yellow, and region SOUTH in green.

Circulation of surface water in the Kitimat fjord system is driven by winds, tides, and estuarine circulation set up by freshwater input from rain and runoff mixing with oceanic waters beneath [162, 77]. During the period July 2013 - July 2016 near - surface currents were measured by moored acoustic Doppler Current Profilers (ADCP) at a number of locations in the system (Figure 3.2 [167, 165, 166]). Depth of the uppermost resolved velocities ranged from 7 to 12 m. The moorings in Douglas Channel were in place for the entire three years, while the other moorings were moved every 1 - 2 years. Therefore all statistics of velocity recorded by ADCP are based on at least one year of data, but the seasonal values outside of Douglas Channel lack representation of interannual variability.

The mean near-surface currents are directed seaward year-round at most locations, with overall magnitudes ranging from 11 - 17 cm/s in Douglas Channel and 3 - 12 cm/s south of Wright Sound. The only exception here is a mooring near the confluence of Verney Passage, Gardner Canal, and Devastation Channel, where a weak mean flow (2 cm/s) towards Kitimat is recorded. Mean flow speeds are reasonably consistent seasonally, with one noteable exception being the mooring in Otter Channel and two adjacent moorings in Principe Channel where the strength of mean flows in the summer is reduced to $\sim$50% of spring and fall values.

Fluctuations in the measured currents are generally anisotropic, with the major axis directed along the centerline of the channels (dashed ellipses in Figure 3.2). Along-channel RMS fluctuations in the currents range from 19 - 30 cm/s. Cross-channel RMS currents range from 6.5 - 9.5 cm/s except in Campania/Caamaño Sound and Hecate Strait, where fluctuations are less anisotropic and minor-axis values range from 11.5 - 14.5 cm/s. Seasonal variability in the fluctuations is less than 5 cm/s, with fall values exhibiting the largest fluctuations. Data from the ADCP at the confluence for Verney Passage, Gardner Canal, and Devastation Channel is an exception to these patterns. Here the RMS fluctuations are smaller and closer to isotropic, ranging from 9 - 13 cm/s. Combined with the small mean flows recorded here this suggests that currents in this area are less energetic than in Douglas Channel and the southern reaches of the Kitimat fjord system.

Based on summertime driftpole tracking and ship-based current meter records [62] showed that throughout the Douglas Channel system surface currents are generally directed seaward throughout an M2 tidal cycle with the exception of Devastation Channel, where flows are consistently landward. This is in good agreement with the data recorded by the ADCPs discussed above. Flows may also reverse in

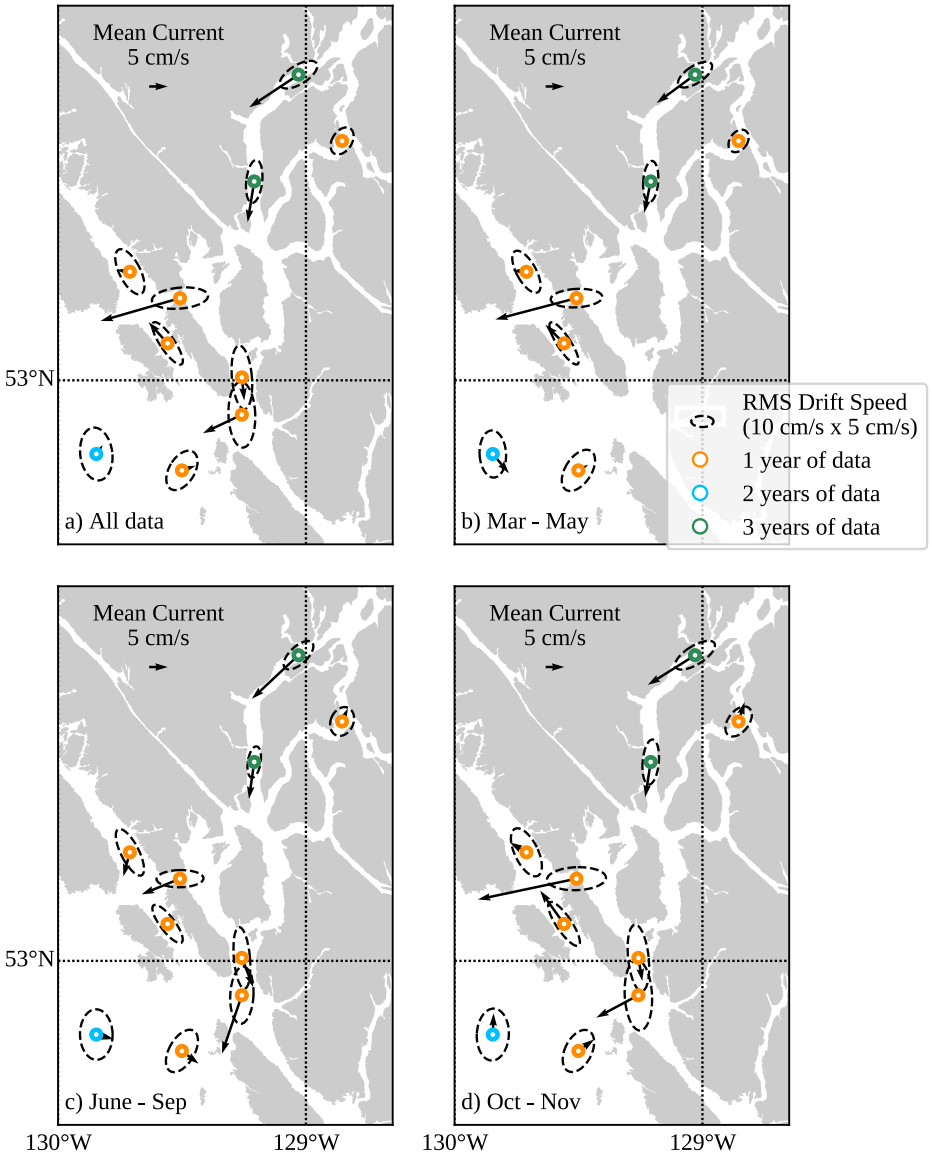


Figure 3.2: Mean (arrows) and RMS (ellipses) velocities recorded by acoustic Doppler current profilers deployed in the Kitimat fjord system between July 2013 and July 2016. Statistics are shown for the entire available record (a), as well as seasonal statistics separated into spring (b), summer (c), and fall (d). The record length at each location is indicated by the marker color. Size of the ellipse axes indicates one standard deviation.

Whale Channel, McKay Reach, and the lower reaches of Verney Passage and Douglas Channel. [62]'s recorded values of hourly averaged currents range from 10 - 100 cm/s.

Incidentally both extremes in this range are reported simultaneously in Wright Sound, at locations separated by ∼10 km.

The tides in the Kitimat fjord system have an average range of 4 meters, though ranges as large as 6.5 meters are observed in some parts of the system. They are dominated by the M2 constituent, which has a near-surface current amplitude of 0.2 m/s. K1 is the primary diurnal constituent, with an energy content of roughly 30% of M2. Tidal currents throughout the system are strongly baroclinic, due to interactions with topography resulting in the formation of internal waves as well as the interaction between tides, wind forcing, and strong stratification near the surface. Near the surface wind-driven currents dominate the energy of the currents (60 - 70% [157]), with semi-diurnal tides being the second most important factor (15 - 20%). The remainder of the energy is distributed between the diurnal tides and high frequency fluctuations. Near the surface tidal ellipses are not strictly rectilinear, suggesting that some tidal currents exist in the cross-channel direction [162, 116]. The tidal excursions, derived by integrating tidal currents calculated using UTide [22], show evidence of some cross-channel movement of water and floating material with the tides but this movement does not exceed 500 m. It is small compared to the width of the Channels.

Freshwater input from rivers and direct runoff manifests as a strongly stratified surface layer with a thickness of ∼10 m in the upper reaches, as seen in seasonal climatologies of temperature and salinity (Figure 3.3). Based on a simplified watershed model, total annual freshwater influx into Douglas Channel and Gardner Canal has been estimated to be 1270 $m^3 s^{-1}$, with an interannual variability of ∼10%. This represents most of the freshwater input into the Kitimat fjord system [157]. Peak riverine freshwater input due to snowmelt occurs between May and August and results in significantly stronger stratification in summer than in spring (as the spring freshet occurs near the end of spring as we define it), which increases further into fall. This stratification gives rise to the appreciable baroclinic current structure mentioned earlier, which will be explored in more detail later in this paper. It may also results in the 'slippery water' phenomenon mentioned by [147], which occurs when strong near-surface density gradients results in increased wind-driven near surface currents. Based on ADCP and wind records the magnitude of the estuarine surface outflow in Douglas Channel has been estimated to range from 14 - 23 cm/s, superimposed on a subtidal wind-driven circulation corresponding to ∼5% of the wind speed[157]. Isopycnals gradually shoal between Kitimat and Wright Sound. Stratification is much

weaker south of Wright Sound, likely due to strong mixing in the sound, where water masses from 6 different channels meet.

Measurement of wind and waves in the Kitimat fjord system are available from Environment and Climate Change Canada's (ECCC) buoy 46181 (Nanakwa Shoal) which is located south of Kitimat at the confluence of Douglas Channel and Devastation Channel. Summaries of the seasonal wind statistics Nanakwa Shoal are given in Figure 3.4. Winter values are not shown as no drifters were released during the winter months.

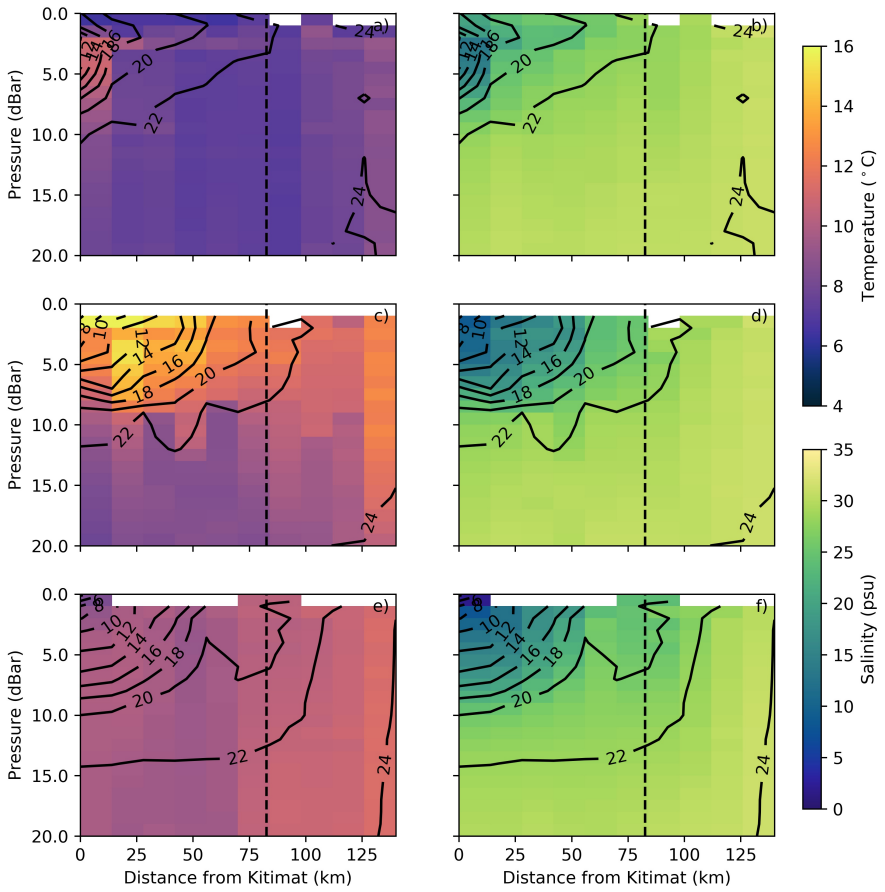


Figure 3.3: Cross-section of mean temperature (left column) and salinity (right column) in the upper 20 m of the water column between Kitimat and Hecate Strait, based on CTD casts taken between July 2013 and July 2016. Values are shown for spring (March - May, a-b), summer (June - August, c-d), and fall (September - November, e-f). Isopycnals (lines of constant potential density, $\rho_0$) are contoured as $\rho_0 - 1000$ kg/m$^3$. The location of Wright Sound is indicated by the dashed vertical line.

The dominant winds are directed nearly rectilinearly along the main axis of nearby Douglas Channel, which suggests that topographic steering of the flows by the fjord walls is significant. Mean winds are weak in the spring and fall, and strongly up-channel in the summer. Strong winds are infrequent during the times that drifters are active, with the strongest recorded wind being 15 m/s (30 knots). Based on the historical buoy record, a 30 knot wind occurs once every 15 years in the period April - October. In wintertime winds of this magnitude occur frequently however; often more than once per month. In conjunction with low temperatures these strong winds can result in temporary failure of the anemometer, and therefore more strong wind events may go unrecorded.

The wave record from Nanakwa Shoal indicates that the local wave climate consists of locally generated short-period wind sea with periods less than 5 seconds and significant wave heights less than 1 m (Figure 3.5). Islands at the entrance to the system isolate it from larger non-locally generated swells in Hecate Strait. Therefore wave-induced Stokes drift is unlikely to contribute significantly to the motion of the drifters within the Kitimat fjord system. This will be confirmed in future work, when a numerical hydrodynamic model is available to inform the prediction and assessment of individual drifter trajectories.

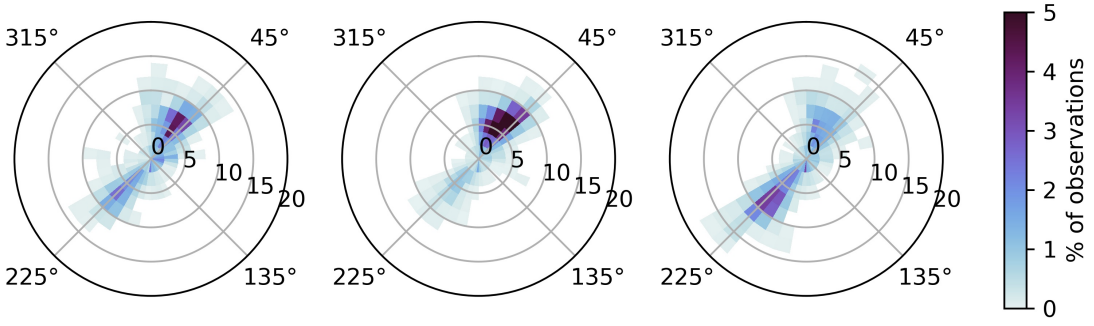


Figure 3.4: Seasonal wind distributions at Nanakwa Shoal (ECCC Buoy 46181)for spring (Mar - May, left), summer (Jun - Aug, middle), and fall (Sep - Nov, right).

A more detailed description of the ADCP and CTD data collected in the Kitimat fjord system between 2013 and 2016 is given in [167, 165, 166] and [157]. Further descriptions of the Kitimat fjord system are given by [162], [77], and [130].

### 3.2.2 Drifter Program

Novel, low-cost, surface circulation tracking drifters were developed as part of this study (Figures 3.6 & 3.7). Their shallow draft (375 mm) serves to mimic the behaviour of objects drifting in the uppermost part of the ocean under the influence of winds,

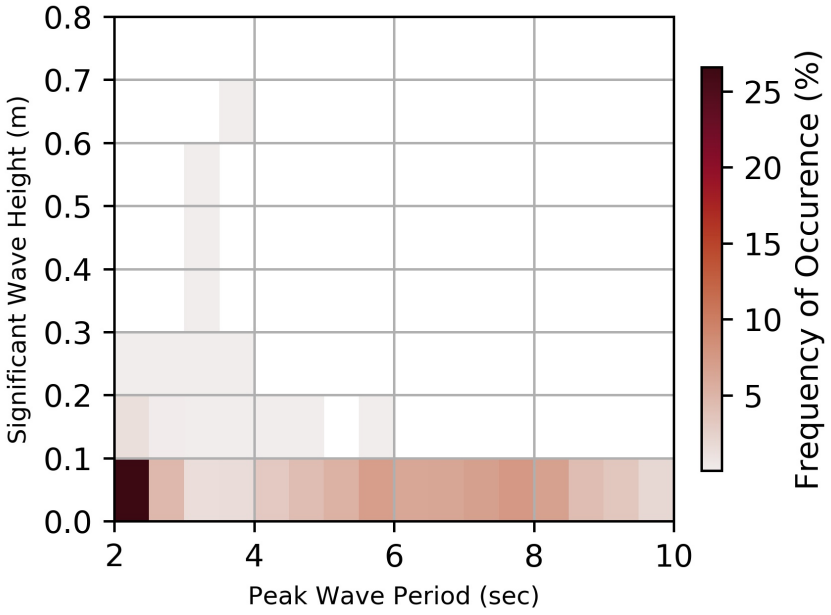


Figure 3.5: Distribution of significant wave height vs. peak wave period observations recorded at Nanakwa Shoal (ECCC Buoy 46181) between Mar 1, 2014 and Sep 1, 2016.

waves, and currents. The drifter's position is tracked using the motion-activated, commercially available SPOT asset tracking unit (SPOT LLC). The unit is attached to the top of the drifter by a stiff spring, to ensure frequent motion of the tracking unit even when the drifter is in calm water. Buoyancy is provided by a reinforced cylindrical cellulose sponge, and an array of cylindrical aluminum vanes serves to increase submerged surface area. In its final design the water surface is approximately 3 - 5 mm below the top of the sponge. This results in a 15:1 ratio of submerged to exposed cross-sectional area weighted by drag coefficient (following [94], further discussion in Section 4.4.1). The drifters are described in more detail in [100].

The drifter deployments were designed to maximize the seasonal and spatial coverage of the tracks. Drifters were deployed in groups of two to five throughout the study area in March, June, and October of 2014, March, July, and November of 2015,

and May and July of 2016. In total 206 drifters were deployed within the Kitimat fjord system, and they collected a total of 2741 drifter-days worth of data (Spring: 991 drifter-days, Summer: 1037 drifter-days, Fall: 713 drifter-days). The longest track in the data set is 94.5 days long, recorded in March 2014. Some drifters travelled outside of the study region, but only the positional fixes recorded within the Kitimat fjord system are considered in the analysis. Analysis is performed for time scales up to ten days, as few drifters remained freely drifting within the fjord system after this time. The drifters transmitted their position on irregular time intervals with an average of $\sim$25 minutes once the drifter design had been optimized (Fall 2014 - Summer 2016).

Figure 3.6: The final drifter design.

Three tests of GPS position accuracy were conducted on the SPOT GPS transmitters used on the drifters. Tests were carried out in various ranges of tree coverage, and a total of 186 positions were collected. The corresponding deviations from the mean position are reported in Figure 3.8. The component-wise standard deviations are 7.25 (11.5) m for the E-W (N-S) direction. The total deviation is $9 \pm 10$ m within

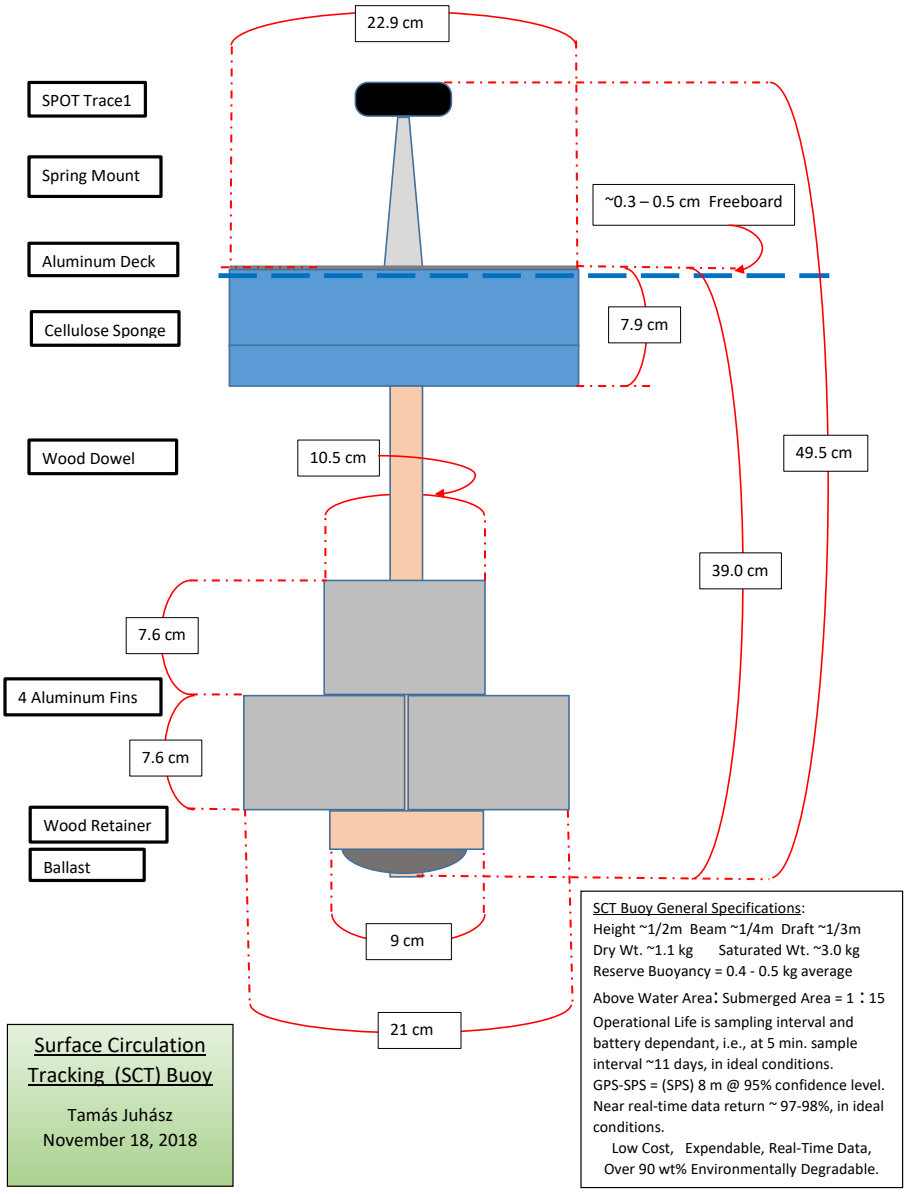


Figure 3.7: Sketch of the final drifter design.

one standard deviation, which suggests that component-wise deviations are rarely zero simultaneously. We take the position uncertainty to be 9 m, as this value agrees well with both the mean total deviation and the mean of the component-wise standard deviations. Since two fixes are required to obtain velocity from drifter positions, the probability that instrument error on both fixes exceeds the mean total deviation is only 25%, assuming that instrument error is approximately normally distributed.

Taking the mean total deviation plus one standard deviation as the GPS uncertainty would eliminate all reported drift speeds below 4.2 cm/s, which was deemed likely to eliminate some good data and unnecessarily conservative since tree cover (which is the likely cause of outliers) is not a factor at the ocean surface. As both positional fixes used in the calculation of a velocity are affected by the GPS uncertainty, a value derived from two fixes recorded 15 minutes apart with 9m GPS uncertainty results in a speed uncertainty of 0.02 $ms^{-1}$ (=2(9m/900s)).

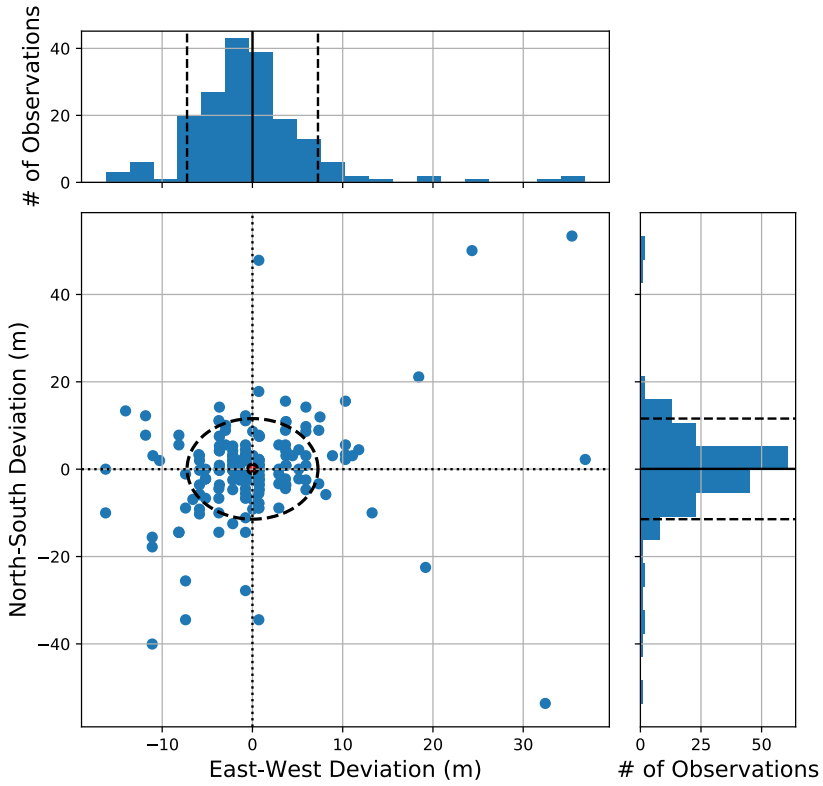


Figure 3.8: Deviation between 186 positions reported by at-rest SPOT GPS transmitters and the mean position. The component-wise mean is shown by the red marker, and component-wise standard deviations are shown by the dashed ellipse.

## 3.3 Methods

### 3.3.1 Channel Coordinates

Drifter motion is decomposed into along-channel and cross-channel coordinates for analysis of long-term dispersion patterns. We only consider seaward (positive, to-

wards Hecate Strait) and landward (negative, towards Kitimat) motion as we cannot statistically resolve the dispersion between individual channels. Drifter positions are mapped to a grid system defining along-channel direction throughout the study region (Figure 3.1). This system divides the region into 16 channels and 18 confluences. Each channel consists of waypoints along the centreline, which are spaced no more than 3 km apart (coloured lines in Figure 3.1). Within channels, displacement associated with pairs of drifter positional fixes is determined by locating the closest waypoint to each fix and projecting the displacement vector between the two fixes onto the down-channel direction (from waypoint to waypoint). If there are intermediate waypoints between those closest to the fixes the along-channel displacement is taken as the distance between the intermediate waypoints and the distance between the closest waypoint and the projection of the positional fix onto the along-channel axis.

At a confluence of two or more channels (black circles in Figure 3.1) the down-channel direction is determined based on the drifters proximity to the adjacent channels. If a drifter is close to a particular channel for at least one hour it is considered to be 'in' that channel, and its motion is calculated using the channels closest down-channel direction. The radius of each confluence is chosen so that the confluence spans the entire width of the channel in every possible direction. If a confluence lies between two successive positional fixes the along-channel displacement through the confluence is considered to be the linear distance between the terminal points of the initial and final channel. If a drifter traverses more than one intermediate channel between two successive positional fixes (i.e. the drifter traversed more than two confluences) the drifter track is split between these two fixes, into two separate tracks.

We quantify the along-channel dispersion of drifters throughout the system by mapping the positional data to the grid system described above. Dispersion can be studied by considering time series of along-channel displacement, $X$, over a range of time lags $\tau$.

$$\Delta X(\tau) = X(t+\tau) - X(t) \tag{3.1}$$

We assume $\Delta X(\tau)$ to be a property of the drifter track that is independent of initial time $t$, i.e. statistically stationary. All non-overlapping segments of a drifter track are therefore used in the analysis. For example, a 200 hour long drifter track will yield 400 values of $\Delta X(\tau$ =0.5 hours), 200 values of $\Delta X(\tau$ =1 hour), etc.

The time-evolving mean, $\mu(\tau)$, and variance, $\sigma^2(\tau)$, of $\Delta X(\tau)$ can then be used to estimate the limits of displacement by using the 95% confidence intervals of a Gaussian distribution. This is a proxy for the area within which we expect to find 95% of the drifters, and will be extended for application to oil spill response in Section 3.6. Mathematically this can be expressed as follows:

$$\Delta X(\tau)_{min/max} = \mu(\tau) \pm 2\sqrt{\sigma^2(\tau)} \tag{3.2}$$

To study cross-channel motion, we define the drifters' proximity to the shoreline, $d_{shore}$ to be the minimum distance between the drifter and the closest polygon in the GSHHS coastline database [163]. We further define the local channel width, $w$, as the minimum distance between the nearest point on a coastline polygon and the next closest polygon (which is the other side of the channel). We then derive the normalized cross-channel position, denoted $\chi$, as

$$\chi = \frac{2d_{shore}}{w} \tag{3.3}$$

such that $\chi = 1$ refers to mid-channel, and $\chi = 0$ refers to the shoreline.

### 3.3.2 Drift Velocity Statistics

We estimate velocities from the drifter positions by interpolating positions to even 15 minute intervals using cubic splines, and then taking centered differences, with a forward (backward) difference at the start (end) of each track [148]. Points where the estimated drift velocity is less than 2 cms$^{-1}$ are discarded in order to reduce the effects of position uncertainty and to eliminate drifters that have grounded ashore.

Drift velocity statistics can be estimated in the Eulerian sense by binning points into sub-regions. Binning is done subjectively, with the aim to ensure adequate sample size in each bin while also minimizing the number of sharp changes in channel geometry within a sub-region. No statistics are reported for sub-regions with less than ten statistically independent samples of drift velocity.

To achieve statistical independence between sampled drift velocities we compute the mean Lagrangian integral timescale, $T_i$, from tracks in each sub-region and only consider estimates separated by time intervals longer than $< T_i >$ [142]. Subscripts $i$ denote orthogonal components of velocity. $T_i$ is derived by integrating the mean normalized autocovariance function, considering times up to 25% of the length of the

drifter track [148]. Tracks available for this analysis are generally short, as drifters frequently move between sub-regions or ground on the shoreline. Since the integration does not converge readily at the longest available lags we adopt the practice of integrating to the time of first zero-crossing, $T_{i0}$ [112]. The arithmetic mean of the integral time scale in the $u$ and $v$ (N-E coordinates) directions is used as $T_i$.

### 3.3.3 Fractional Brownian Motion

We consider fractional Brownian motion as a possible description of the single-particle dispersion statistics of the drifters on short space and time scales, following [125]. A brief overview of the concepts that are relevant to the analysis in this paper are given. For a more thorough treatment of the topic the reader may refer to works such as [79, 86] and [74].

Fractional Brownian motion (fBm) is a stochastic model built on ordinary Brownian motion with the additional feature that the motion increments are infinitely self-correlated, i.e. the process has an infinite memory of its own history. Mathematically, it is equivalent to a moving average of incremental ordinary Brownian motion, in which previous increments are weighted by a time-dependent kernel [79]. The resulting variance of particle displacements, $x$, at a time $t$ after initial time $t_0$ can be written as

$$< [x(t_0 + t) - x(t_0)]^2 >= AV_H t^{\frac{2}{D}} \tag{3.4}$$

Here $H$ is related to the fractal dimension $D$ of the 2-dimensional drifter motion by $D = min[1/H, 2]$, assuming isotropic statistics [125]; $A$ is an amplitude coefficient after [74]; and $V_H$ is the weighting kernel as defined by [79] which, along with the time dependence, controls the self-similar evolution of displacement statistics.

A surface drifter trajectory is restricted to a fractal dimension between a line ($D = 1$) and a plane ($D = 2$) which corresponds to the parameter space $1 > H > 0.5$. When $H = 0.5$ (and $D = 2$) the position variance grows linearly in time (Equation 3.4) and the motions will fill the two-dimensional space available to it if given enough time, whereas the trajectories from fBm will look increasingly like a line as $H$ increases from 0.5 to 1 (Figure 3.9 [125]).

Fractional Brownian motion can only yield a valid description of ocean surface drift trajectories over a finite range of space and time scales. This is because the relevant scales of turbulence are bounded above by the length of the drifter track (larger scales

are filtered out as mean motion) and below by the small scales where one expects the dispersion to scale as $t^2$ [145]. The range of spatial scales over which fBm may be applied is generally thought to correspond with the range over which a drifter trajectory exhibits fractal characteristics [99, 125]. This range can be determined using the 'yardstick' method [cf. 99, 148], where the length of the trajectory $L$ is measured as a function of the length of the 'ruler' (or 'yardstick') with which it is measured, denoted as $\Delta$. If the trajectory exhibits fractal characteristics the length $L$ will scale logarithmically with 'yardstick' length as

$$L(\Delta) \propto \Delta^{1-D} \tag{3.5}$$

where $D$ is the fractal dimension (also referred to as the 'divider dimension' $D_L$ in the literature). Therefore the range of scales over which the trajectory has fractal characteristics can be determined by plotting $L(\Delta)$ vs. $\Delta$ in log-log space and determining whether $L$ evolves linearly over any particular domain of $\Delta$. If such a domain exists, the slope is $1 - D$ and within this domain fractional Brownian motion may provide a suitable model to explain dispersion statistics where dispersion is observed to evolve proportionally to $t^{\frac{2}{D}}$ rather than the proportionality to $t^2 \rightarrow t$ expected in Taylorian dispersion.

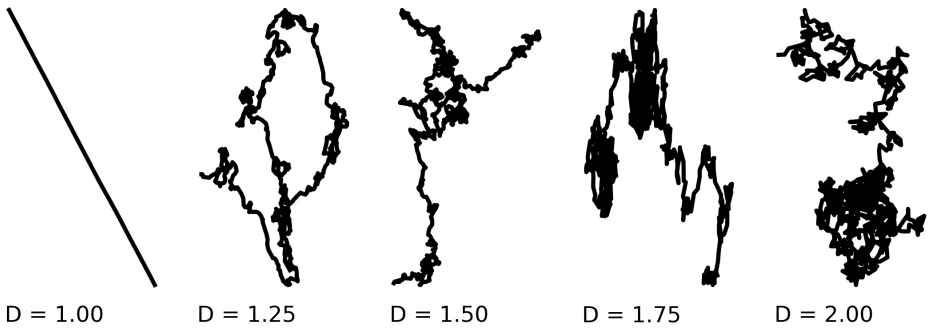


Figure 3.9: Illustration of simulated fractional Brownian motion tracing a dispersing particle's position for various fractal dimensions using the method of [61] over 1000 time increments.

Spectral methods for analyzing fBm are less well-defined. The spectral properties of the position statistics of fBm, which are non-stationary, are not easily defined using traditional Fourier transforms and are not considered in detail in this paper. However,

it is generally accepted that the power spectrum of drifter positions with fractal dimension $D$ is proportional to $f^{-(2/D+1)}$ [99, 125]. This relationship might therefore be used to provide an additional estimate of fractal dimension if the power spectrum of drifter displacements is known. For example, if $D = 1.3$ we expect a position variance spectrum $\propto f^{-2.5}$. However [99] note that spectral slope alone cannot be considered indicative of fractal behaviour. This must be confirmed by the yardstick method or other equivalent techniques. Readers interested in the relationship between fractals and spectral analysis may refer to [74] and the references therein for a more detailed discussion.

## 3.4 Results

We now begin our analysis of the data described in Section 3.2, using the methods outlined in Section 3.3.

### 3.4.1 Mean Drift Patterns and Driving Forces

Ocean surface drift in the Kitimat fjord system exhibits significant seasonal variability, but is generally directed seaward as is expected in an estuarine system (panel a of 3.10). Mean drift speeds generally range from 5 - 10 cms$^{-1}$ though they may reach magnitudes up to 24 cms$^{-1}$. In the spring and fall mean drift is directed seaward everywhere except a few channels north of Wright Sound. In the summertime mean drift reverses direction everywhere north of Wright Sound and slows south of Wright Sound (panels b, c, and d of Figure 3.10). The mean and largest RMS fluctuations of the drift velocities are well aligned with the main axis of the channels, however appreciable fluctuating cross-channel drift is also evident. Along-channel RMS fluctuations of velocity range from 22 - 52 cms$^{-1}$, while cross-channel fluctuations range between 8 - 26 cms$^{-1}$.

The mean and RMS drift patterns shown in Figure 3.10 are quite different from similar maps of near-surface currents (7 - 10 m depth) (Figure 3.2). Mean surface drift velocities regularly exceed mean currents by a factor of 2 or more, and sometimes mean drift is directed opposite to recorded mean near-surface currents. RMS drift velocities are also larger than RMS currents, both in the along-channel and cross-channel directions. This suggests that currents in the upper $\sim$10 m of the water column are strongly sheared, which could be attributed to strong stratification, wave-

induced currents, and some wind-induced slippage of the drifters.

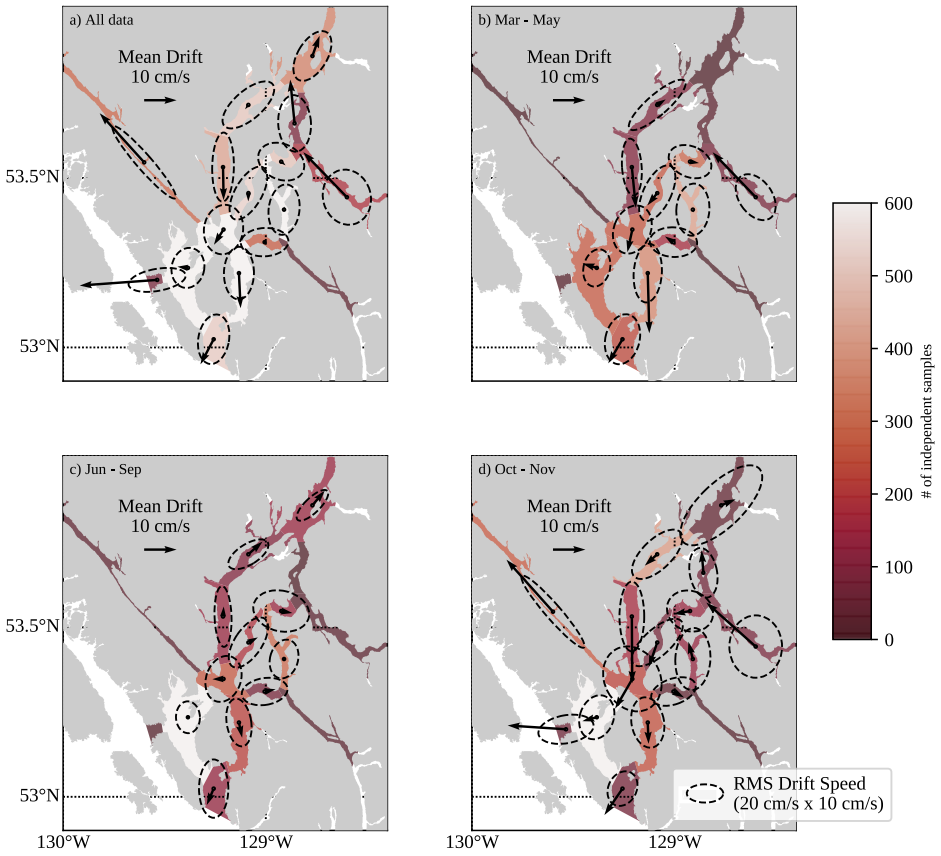


Figure 3.10: Distribution of mean (arrows) and RMS (dashed ellipses) drift velocities in the Kitimat system. Results are shown the average over the data set (a) along with seasonal averages (b - d). Shading indicates the number of independent samples of drift velocity in the regions over which statistics were derived.

Near-surface velocity shear can be quantified by a single-parameter regression analysis comparing along-channel drift velocity $u_{drift_X}$ near mooring FOC1 to currents recorded at 7 - 10 m depth ($u_{current_X}$) and winds recorded 5 m above the water surface at Nanakwa Shoal ($u_{wind_X}$) (see Figure 3.1). Fitting all available drifter data to a linear model of the form $u_{drift_X} = \alpha u_{wind_X} + u_{current_X}$ explains 50% of the variance in the along-channel drift with a regression coefficient $\alpha = 0.042$. The coefficient varies seasonally, from 0.024 in summer to 0.048 (0.05) in spring (fall). These values are similar to those found by [157], who investigated the relationship between winds and currents at 4 - 7 m depth for a subset of the time period considered

here. Similar to their results, we find that our regression model has more explanatory power in spring/fall ($r^2 = 0.58/r^2 = 0.43$) than in summer ($r^2 = 0.34$) (Table 3.1 and Figure 3.11). Regression using information on both wind and currents yields a better description than using only currents ($r^2 = 0.34$) or only winds ($r^2 = 0.32$). Therefore knowledge of both currents and winds is required for drift prediction. Note that the results discussed here are limited to wind speeds below 7 $ms^{-1}$, as there were no drifter observations coinciding with higher wind speeds.

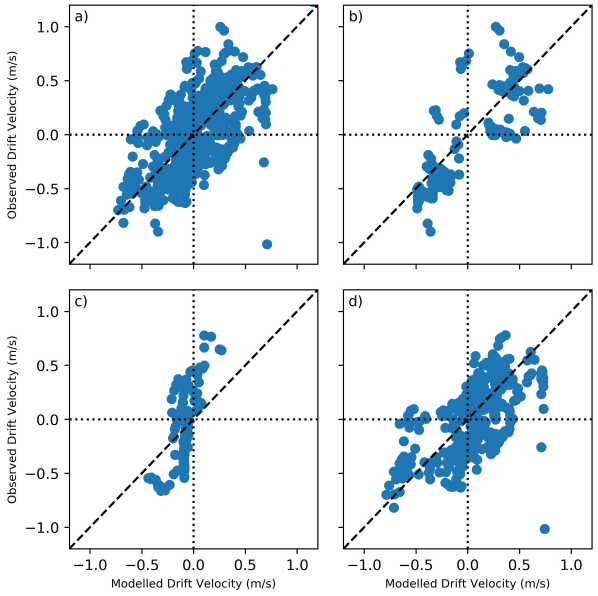


Figure 3.11: Observed along-channel drift velocity (seaward positive) near mid-channel around mooring FOC1 against drift velocity predicted by $u_{drift_X} = \alpha u_{wind_X} + u_{current_X}$ using values of $\alpha$ from Table 3.1 for the complete data set (a), spring (March - May, b), summer (June - August, c), and fall (September - November, d).

Table 3.1: Seasonally resolved regression coefficients, $\alpha$, with number of samples, $N$, and fraction of variance explained by the model $\vec{u}_{drift_X} = \alpha\vec{u}_{wind_X} + \vec{u}_{current_X}$

| **Case** | **Regression Coefficient,** $\alpha$ | $r^2$ | $N$ |
|---|---|---|---|
| **All Data** | 0.042 | 0.5 | 740 |
| **Spring** | 0.048 | 0.58 | 206 |
| **Summer** | 0.024 | 0.34 | 156 |
| **Fall** | 0.05 | 0.43 | 378 |

### 3.4.2 Along-Channel Drift and Dispersion

The along-channel drift and dispersion of all drifters in the data set can be visualized as the distribution of displacements as a function of time scale (Figure 3.12). The displacements are one-dimensionalized as described in Section 3.3.1, and taken as positive towards the ocean and negative towards land. The distributions indicate a seasonally varying mean seaward drift, consistent with Section 3.4.1, and very rapid dispersion within the first 24 hours, as indicated by the increase in the width of the distribution. Dispersion begins to slow noticeably after 24 hours and displacements become increasingly uniformly distributed with time.

The mean seaward drift ranges from 1.4 - 7.5 $\text{cms}^{-1}$ through the seasons, with a mean value of 4.4 $\text{cms}^{-1}$ (Figure 3.12). It is strongest in the fall and spring and insignificant ($< 2$ $\text{cms}^{-1}$) in the summer. This is consistent with the previous observation that up-channel winds can effectively stop, or even reverse, the estuarine outflow at the very surface of the water column in Douglas Channel [157]. In region SOUTH (south of Wright Sound) mean seaward drift is twice as fast as in regions DC (Douglas Channel) and EAST (east of Douglas Channel) (Figure 3.12). The mean drift appears to be constant, as evidenced by the good linear regression fit to the time-evolving mean displacement values, except when considering the regional subsets DC and EAST (Figure 3.13). The exception is likely due to an increasing number of drifters entering the less energetic regions near the shoreline in these smaller channels, while some fast-moving drifters closer to the channel centreline exit the region more quickly. The calculated mean displacement at longer times would therefore be biased towards slow-moving drifters. This is evidenced in the increasingly 'deterministic' character of the displacement distributions for these sub-regions, where trajectories of individual drifters are clearly visible at longer times (panels e and f of Figure 3.12).

Dispersion of the drifters about the mean (indicated by the width of the distributions of $\Delta X$ in Figure 3.12) suggests that drifters may disperse over the length of the Kitimat fjord system ($\sim$140 km) over a period of 7 days. The dispersion is stronger in the fall than in the spring, and weakest in the summer (Figure 3.12), similar to trends in the mean drift. Dispersion is also stronger in the region SOUTH than in the regions DC and EAST. This may also be due to the relatively larger area covered by SOUTH and the resulting bias towards slow-moving drifters at longer times in DC and EAST.

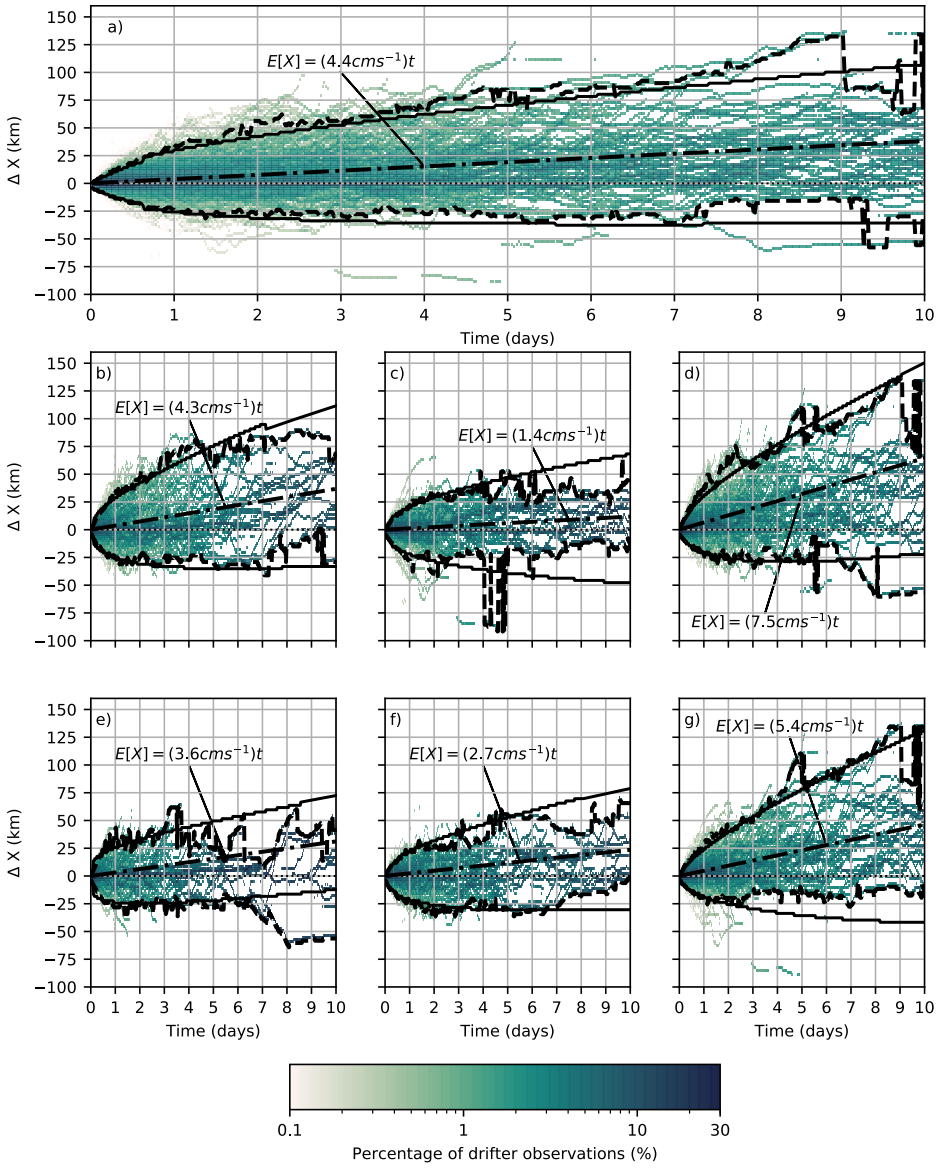


Figure 3.12: Distribution of drifter displacements, $\Delta X$, over ten days with mean (dash-dotted line, with implied mean drift speed), 95% limits (dashed line), and 95% limits of the corresponding normal distribution (solid line). Distributions are derived by binning observed displacements into 2 km by 0.5 hour bins, and for each half hour interval reporting the percentage of observed displacements in each displacement bin. Results are shown for the complete data set (a), as well as seasonal subsets of data from March - May (b), June - September (c), and October - November (d), and regional subsets for DC (e), EAST (f), and SOUTH (g).

The displacement variance (or dispersion) evolves with rapid initial growth (the ballistic regime) which slows at longer times (the diffusive regime). This is consistent with previous studies of drifter dispersion in the open ocean [cf. 112] and the dispersion theory of [145]. The latter theory suggests that particles are dispersed by coherent turbulent structures in the flow field while in the ballistic regime, and enter the diffusive regime once the extent of the dispersion exceeds the largest coherent scales in the flow. We fit power laws of the form $At^B$, where $A$ and $B$ are free parameters, to the observed time-evolving displacement variance and delineate the time of transition between the ballistic and diffusive regimes, $t_T$, as the time which minimizes the overall error in the regression fit (Figure 3.13). In the ballistic regime variance grows according to $\sigma^2 \propto t^{1.5}$, which slows to $\sigma^2 \propto t^{0.85}$ in the diffusive regime. A transition time of 15 hours yields the best regression fits for the complete data set, as determined by varying the transition time to optimize the overall coefficient of determination. There is some variation in these values among seasonal and regional subsets of the data (see Table 3.2). The 95% confidence intervals of a Gaussian distribution calculated using the time-evolving mean and variance from Table 3.2 are a good match to the range within which 95% of the drifters are observed (Figure 3.12).

The skewness and kurtosis of the displacement distributions indicate that up-channel displacements are more frequent while down-channel displacements are larger in the sense of extremes (consistent positive skewness), and that for times $< 3$ days extreme displacements are more likely than a normal distribution would predict, i.e. the distributions are leptokurtic (kurtosis $> 3$) (Figure 3.14).

### 3.4.3 Effects of Cross-Channel Dispersion

Over the course of the experiments it was noted that drifters tended to ground ashore quite frequently, despite being released near mid-channel. To quantify this tendency, we examine the distribution of normalized proximity to the shoresline, $\chi$. This indicates that on average a drifter will move from mid-channel ($\chi > 0.8$) to near the shoreline ($\chi < 0.1$) within 13.5 hours (panel a of Figure 3.15). The frequency distribution has the character of an extreme value distribution, with short times being more frequently observed (Median = 9.9 hours, Mode = 5.3 hours), though some drifters remain near mid-channel for more than 48 hours. The statistics of $\chi$ are relatively consistent seasonally (Figure 3.16).

Nearshore drift velocity statistics differ appreciably from velocity statistics near

mid-channel. For example, in upper Douglas Channel mean and RMS along-channel velocities near the shore are approximately half of those in mid-channel. Kurtosis increases near the shore even though extreme values are comparable, indicating that slow drift velocities occur more frequently in the nearshore. Cross-channel velocities remain much more consistent than along-channel velocities (panels b and c of Figure 3.15). Averaging results from all drifters supports the notion that mean drift speed in mid-channel is approximately twice as large as speeds near the shoreline (Figure 3.17). Therefore the significant decrease in along-channel dispersion after ∼15 hours observed in Section 3.4.2 can be explained by majority of drifters' tendency to move into nearshore regions, where along-channel transport occurs more slowly, within that time.

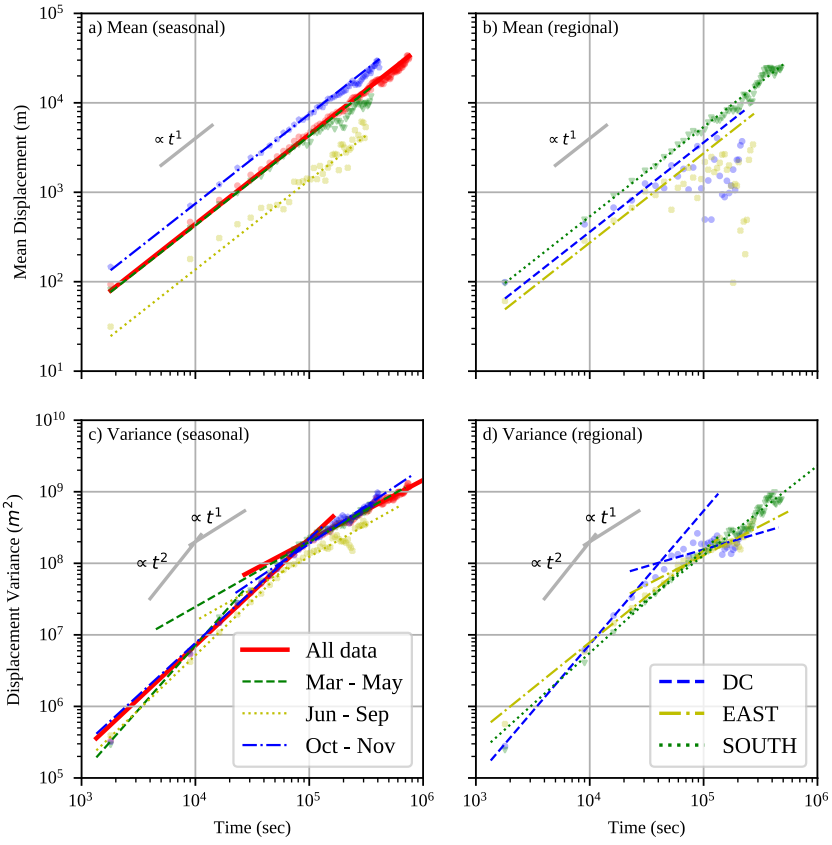


Figure 3.13: Mean and variance of drifter displacement, $\Delta X$, as functions of time for the seasonally - resolved (a, c) and regionally - resolved (b, d) data. Optimal linear (power-law) regression fits are superimposed on the plotted mean (variance). Fit parameters are given in Table 3.2.

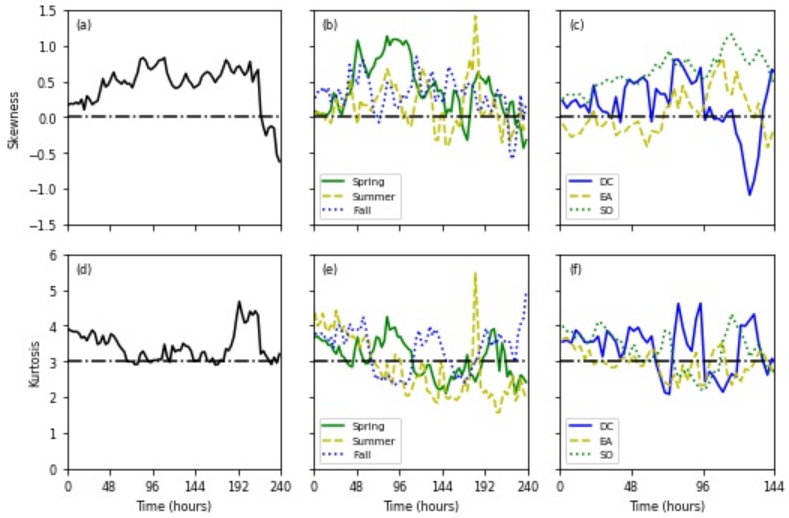


Figure 3.14: $3^{rd}$ and $4^{th}$ moments of the observed displacement distributions for the complete data set (left) as well as the seasonal (middle) and regional (right) subsets. For clarity values have been smoothed using a moving average with a 3 hour window and outliers more than 3 standard deviations from the mean are excluded.

### 3.4.4 Spectra and Fractal Dimension

To further characterize the dynamics of the drifter motion, we estimate the energy spectrum $P_e = 0.5(P_{uu}+P_{vv})$ of drift velocity. Here $P_{uu}$ and $P_{vv}$ are the power spectra of drift velocity in the east-west and north-south directions. Global coordinates are used here rather than along- and cross-channel coordinates since the energy spectrum is invariant under coordinate rotations, and along- and cross-channel directions are not clearly defined at channel confluences. Spectra are estimated using Welch's average periodogram method [148]. In this method, time series of velocity are separated into segments consisting of 512 data points at 15 minute intervals (128 hours), with 50% overlap, and a Hanning window is applied to reduce leakage as recommended by [148]. Only time series longer than 24 hours are considered, and they are zero-padded where required. A discrete Fourier transform is applied to the segments to derive the associated periodogram, and all periodograms are then averaged to arrive at the spectral estimate.

The peak at the semi-diurnal (M2) frequency in $P_e$ indicates appreciable influence of the semi-diurnal tide and decreasing energy towards higher frequencies, with the mean spectrum scaling as $\propto f^{-1.9\pm1.69}$ at frequencies higher than 0.1 cycles per hour

(cph) (top panel of Figure 3.18). The uncertainty estimate on the spectral slope indicates one standard deviation. At the M2 frequency a widened peak is evident in the spectrum, but no peak is evident at the diurnal (K1) frequency. This may be due to the fact that the majority of the drifters come in close proximity of the shoreline within less than 24 hours, which exposes them to a velocity field that differs from mid-channel. Summertime drifter motion is less energetic than motion in the spring and fall, most likely due to lessened storm activity and seasonal up-channel winds slowing the estuarine currents. As a result, the semi-diurnal tidal peak is more pronounced, and the overall lower energy level at low frequencies suggests lessened dispersion on longer timescales [72], which is consistent with Figure 3.12.

Table 3.2: Regression fit parameters describing the evolution of the first two moments of observed displacement probabilities. Linear fits describe the mean, and power laws of the form $\sigma^2 = A\tau^B$ describe the two regimes of variance growth (ballistic, subscript b, and diffusive, subscript d). $\tau_T$ is the time of transition between the regimes, determined as the time which minimizes the total error in the regression fits to the observed variance.

| **Season** | **Region** | **Mean** ($\mathbf{cms^{-1}}$) | $\mathbf{A}_b$ | $\mathbf{B}_b$ | $\mathbf{A}_d$ ($\times 10^3$) | $\mathbf{B}_d$ | $\tau_T$ **(hrs)** |
|---|---|---|---|---|---|---|---|
| **All** | **System** | 4.42 ± 0.02 | 7.7±0.91 | 1.49 ± 0.01 | 12.9 ± 1.86 | 0.84 ± 0.01 | 15.0 |
| **Spring** | **System** | 4.26 ± 0.02 | 0.44 ± 0.15 | 1.80 ± 0.04 | 5.88 ± 0.91 | 0.90 ± 0.04 | 2.5 |
| **Summer** | **System** | 1.37 ± 0.04 | 3.98 ± 1.73 | 1.53 ± 0.04 | 4.11 ± 2.12 | 0.90 ± 0.04 | 6.0 |
| **Fall** | **System** | 7.46 ± 0.03 | 11.3±2.7 | 1.46 ± 0.02 | 0.931 ± 0.261 | 1.06 ± 0.02 | 12.5 |
| **All** | **Douglas** | 3.60 ± 0.07 | 0.25 ± 0.04 | 1.87 ± 0.02 | 710±396 | 0.47 ± 0.05 | 2.0 |
| **All** | **Eastern** | 2.73 ± 0.11 | 57.2 ± 22.1 | 1.28 ± 0.04 | 9.94 ± 4.72 | 0.82 ± 0.04 | 12.0 |
| **All** | **Southern** | 5.37 ± 0.05 | 9.98 ± 2.79 | 1.44 ± 0.03 | 0.072 ± 0.036 | 1.25 ± 0.04 | 15.0 |

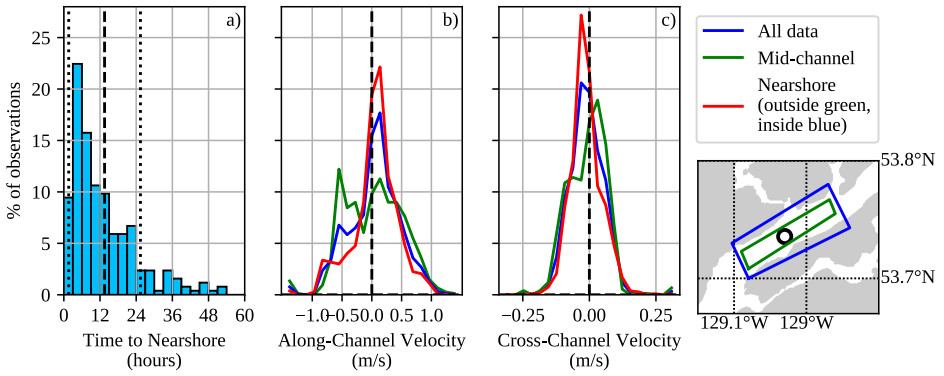


Figure 3.15: a) Distribution of times taken for a drifter to move from mid-channel ($\chi > 0.8$) to near the shoreline ($\chi < 0.1$). Mean time is indicated by the dashed black line and dotted lines indicate one standard deviation around the mean. b,c) distributions of drifter along- and cross-channel velocities near mid-channel (shown in green; inside green polygon), near the shore (shown in red; inside blue polygon, outside green polygon), and for all data (shown in blue; inside blue polygon). Along-channel velocity is positive down-channel (approximately southwest) and cross-channel velocity is positive 90° clockwise from down-channel (approximately northwest).

The ratio of cross-channel: along-channel velocity amplitude spectra suggests that within the channels drifter motion is isotropic on timescales shorter than 3.5 - 5 hours to within $\sqrt{E[P_{cross}]}/\sqrt{E[P_{along}]} = 1 \pm 0.3$, following [125]'s argument for isotropy (middle panel of Figure 3.18). Repeating this calculation in global u,v coordinates suggests isotropic motion on timescales up to 10 hours (Figure 3.19). We therefore conclude that drifter motion is sufficiently isotropic to potentially be described by fractional Brownian motion on timescales up to at least 3.5 hours, and up to 10 hours in areas where the distinction between along-channel and cross-channel direction is not clear (for example in Wright Sound or Squally Channel). Potential driving forces of cross-channel motion at these short timescales are discussed in Section 3.5. On longer timescales along-channel motion becomes significantly more energetic as cross-channel motion is limited by drifters reaching nearshore areas.

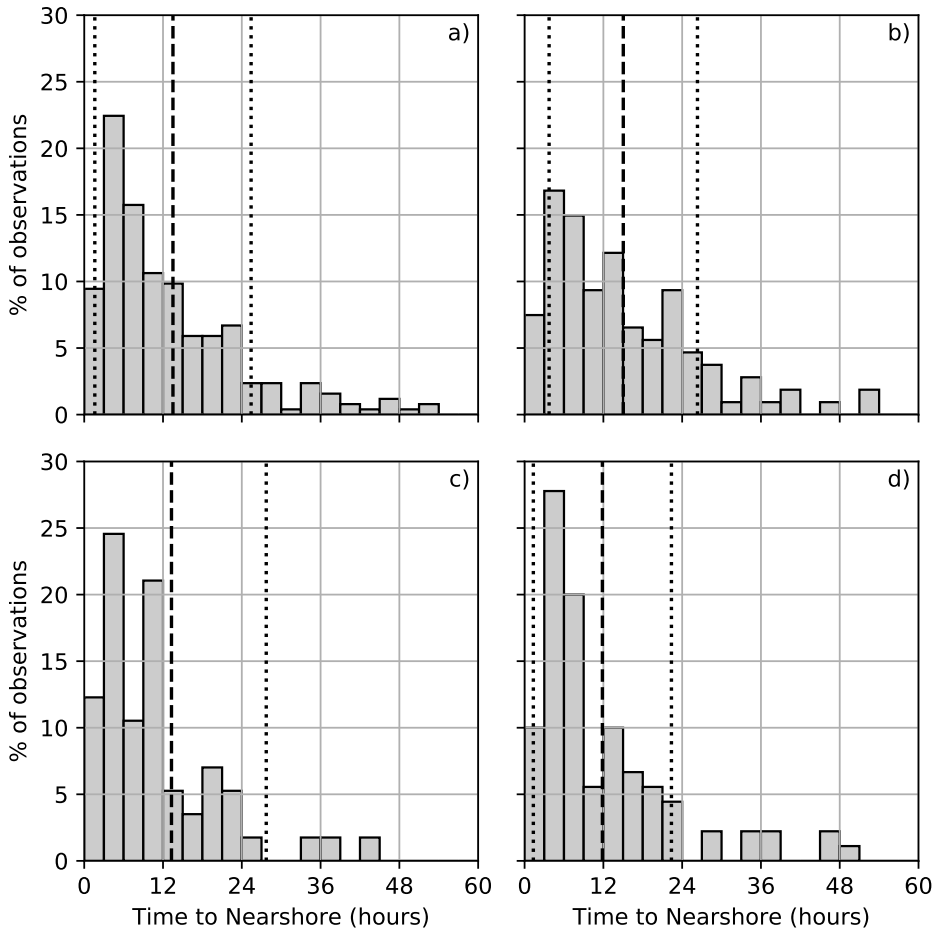


Figure 3.16: Distribution for times required for cross-channel transport from $\chi > 0.8$ to $\chi < 0.1$. Distributions are shown for the complete data set (a), spring (b), summer (c), and fall (d). Mean transit time is denoted by the dashed black line and dotted lines denote one standard deviation.

For later comparison with the theory of fractional Brownian motion, we note that corresponding energy spectra of displacement (bottom panel of Figure 3.18) scale between $f^{-2.23\pm0.46}$ (spring) and $f^{-2.47\pm0.66}$ (summer) with an average of $f^{-2.31\pm0.53}$.

The drifters display fractal behaviour with a dimension, D, of 1.34 over scales ranging from 2 - 13 km, which corresponds well to the width of the channels in the Kitimat fjord system (Figure 3.20). The fractal dimension of the shoreline over these scales (D = 1.26), estimated using the yardstick method and coastline data from the GSHHS database [163], is similar to the value reported for the west coast of Britain (one of the 'most irregular in the world', with D = 1.25) [78]. The fractal dimension

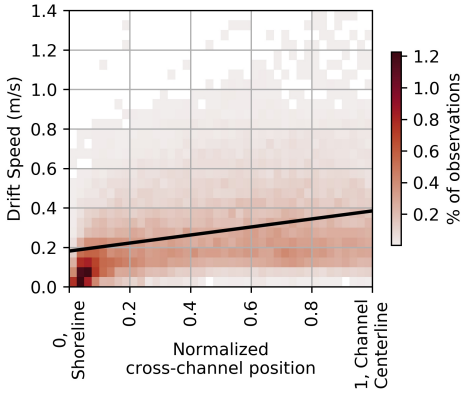


Figure 3.17: Observations of drift speed, $u_{drift}$ as function of cross-channel position, $\chi$. The linear regression fit line (black), corresponds to the equation $u_{drift}(\chi) = 0.18 + 0.2\chi$.

of the shoreline is less than that of the tracks, which range from D = 1.31 (1.30) in the spring (fall) to D = 1.46 in the summer (Table 3.3). This indicates that drifters are occupying space both parallel to and orthogonal to the coastline, and that more cross-channel, space-filling motion occurs in summer than in spring and fall.

## 3.5 Discussion

The above results suggest that in general drifters in the Kitimat fjord system disperse isotropically for up to 10 hours after deployment while being moved seaward by the mean flow. Drifters that come very close to the shoreline slow down and thereby slow the along-channel dispersion. Along-channel dispersion continues in a slowed manner and together with constant mean seaward transport results in drifters being dispersed over scales up to the size of the Kitimat fjord system within 7 days. These dynamics are driven by a combination of tidal currents, density-driven estuarine flow (resulting in mean seaward motion), and wind-induced circulation. None of these processes are negligible for drift prediction, and information on both currents and winds is required to optimally characterize surface drift.

The assumption that the statistics of drifter displacement are stationary is considered tenable, and necessary to the analysis, while acknowledging that the differences between summer and spring/fall, and regions SOUTH and DC/EAST, imply some nonstationarity in the total data set. The mean and variance of $\Delta X$ remain on the same order of magnitude for all subsets of the data. Also, the differences between the

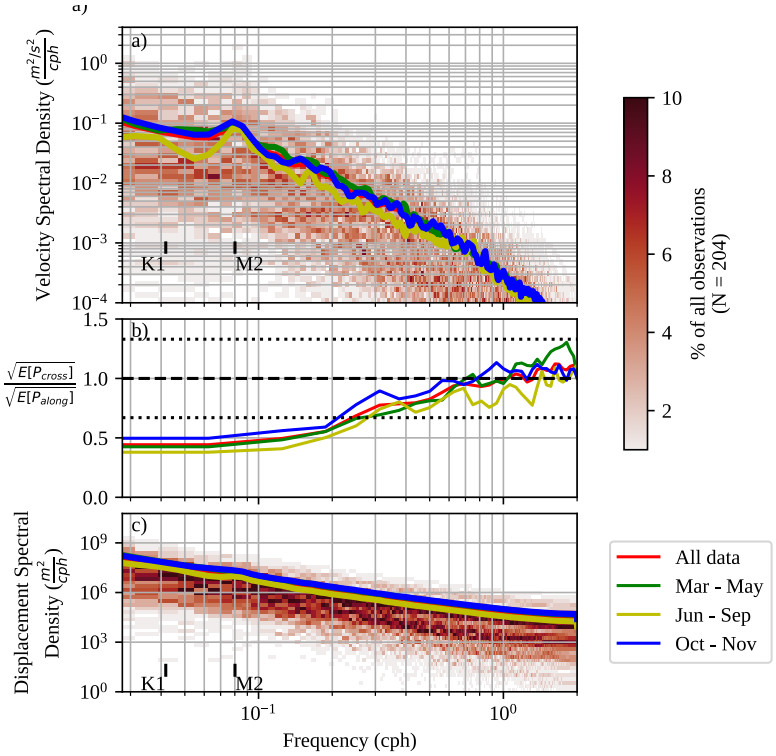


Figure 3.18: a) Mean energy spectra of drift velocity $0.5(P_{uu}+P_{vv})$ for the complete data set (red), as well as spring (green), summer (yellow), and fall (blue). Binned density of observations from individual drifter tracks are shown by the colorbar. b) Ratio of mean amplitude spectra of velocity in the cross-channel: along-channel directions as a function of frequency for all data (red), spring (green), summer (yellow), and fall (blue). Bounds for approximately isotropic motion are indicated by the dashed lines. c) Energy spectrum of drifter displacement in the same style as a).

higher moments for subsets of the data is generally less than the variability of these moments, i.e. the standard deviation of an individual time series of skewness/kurtosis, over ten days. We do not consider the seasonal and regional variability of the statistics to be significant enough to void the assumption of stationary statistics.

The power laws describing the evolution of the drifters' displacement variance exhibit some similarities to previous descriptions of single-particle dispersion. The two-regime structure of variance growth agrees with the theory of [145], though the power law exponents are less than those proposed by this theory. [145] proposes $\sigma^2 \propto t^2$ in the ballistic regime and $\sigma^2 \propto t^1$ in the diffusive regime, whereas we find $\sigma^2 \propto t^{1.5}$ (ballistic) and $\sigma^2 \propto t^{0.84}$ (diffusive). This is hardly surprising, as the theory explicitly assumes homogeneous, isotropic turbulence which does not appear to be

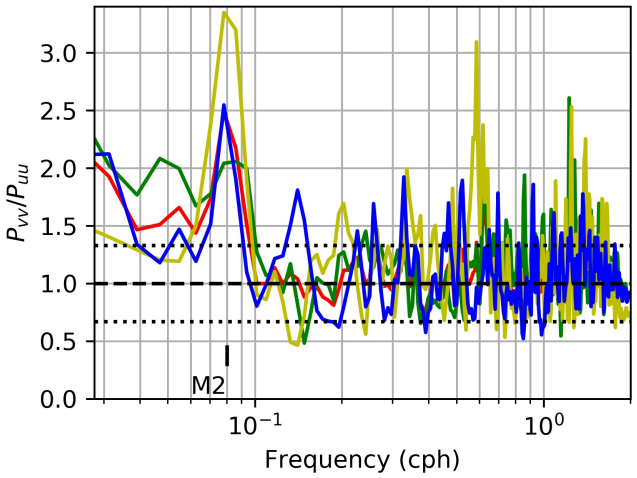


Figure 3.19: Ratio of mean amplitude spectra of velocity in the meridional:zonal directions as a function of frequency for all data (red), spring (green), summer (yellow), and fall (blue). A value of 1 (dashed horizontal line) indicates isotropic motion. The dotted lines represent 30% deviations from isotropy which may be acceptable for fractal analysis, following [125].

applicable across the width of the channels, though we do not preclude the possibility that $\propto t^2$ dispersion holds at very small time and space scales. However, anomalous (non-Taylorian) dispersion can occur in systems where long-term autocorrelations exist in the flow, or the flow is strongly affected by boundaries [4].

Fractional Brownian motion (Section 3.3.3) may provide a suitable model for isotropic single-particle dispersion in the ballistic regime since the drifter trajectories exhibit fractal characteristics (i.e. trajectory length scaling exponentially with the length of 'ruler' used to measure trajectory length) over spatial scales ranging from 2 to 13 km. Turbulent motion over timescales less than ~3.5 - 10 hours is also reasonably isotropic. Therefore the fractal dimension $D$ can be used to estimate the power law for variance growth as $\propto t^{1.49}$, according to Equation 3.4 [125]. This matches the value observed over time lags of up to 15 hours. For seasonal subsets of the data the observed exponents are within 10% of those predicted by fractional Brownian motion when power laws are fitted to variance in the first 15 hours (Table 3.3). Note that these exponents may not be consistent with those presented in Table 3.2, which were derived by optimizing the regression fit through varying the time of transition between the ballistic and diffusive regimes. Given the good agreement of observed and modelled rates, we suggest that dispersion may be reasonably well

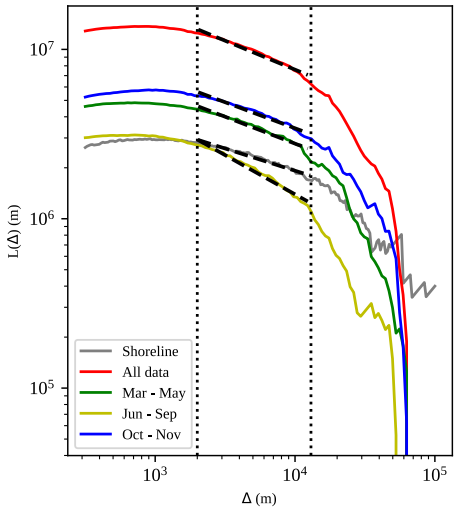


Figure 3.20: Yardstick trajectory length estimates for shoreline, and drifter tracks with seasonal subsets. Dashed lines show log-transformed regression fits to $L(\Delta)$, with slopes $1 - D_L$. Regression results are reported in Table 3.3. Dotted vertical lines indicate the range of spatial scales over which fractal behaviour is evident.

modelled as isotropic fractional Brownian motion for up 3.5 hours within confined channels (middle panel of Figure 3.18) and up to 10 hours after release within less constricted regions (Figure 3.19). In fact, solving for the time required for dispersion across a 5 km wide channel by assuming that the ballistic power law for along-channel dispersion from Table 3.2 applies and solving $\sigma^2 = \pi(2500m)^2 = 7.7t^{1.49}$ results in $t = 5.5$ hours which corresponds well with the most frequently observed time required for a drifter to reach nearshore areas (the mode of distribution shown in panel a of Figure 3.15). Leptokurtic displacement distributions, as were described in Section 3.4.2, are also consistent with fractional Brownian motion. Similar to [125], we observe excess kurtosis that is even greater than the theoretically predicted value.

The mean slopes of the displacement energy spectra of drifter motion, $s$ (Figure 3.18) suggest fractal dimensions ranging from 1.36 (summer) to 1.63 (spring) with a mean value of 1.53. These values differ from those calculated using the 'yardstick' method described in Section 3.3.3, and the difference is 2 times larger than the variability between fractal dimensions estimated for the same data by various methods reported by [99]. Some variability between methods is expected, and the increased variability may in part be due to the fact that [99] considered mesoscale turbulence while we are concerned with sub-mesoscale, confined motions above the tidal frequencies which are more susceptible to deterministic perturbations like the influence of

local topography. We also note that the relatively large standard deviation of measured spectral slopes implies that the entire parameter space $1 \leq D \leq 2$ is contained within the standard deviation of the spectral slope estimate for summertime data. In spring and fall the parameter space is only slightly more constrained, covering $1.16 \leq D \leq 2$. We therefore conclude that while the measured displacement spectra are not in disagreement with fBm they also do not offer information that could be used to usefully constrain the parameters of fBm. Hence it is possible that fBm does not provide a valid theory for modelling drifter dispersion in some areas of the Kitimat fjord system at some times. However given the evidence presented earlier we consider fBm with a fractal dimension of $D \approx 1.2 - 1.6$ to provide a credible estimate of isotropic drifter dispersion on timescales up to 3.5 - 10 hours and spatial scales from 2 to 13 km.

Table 3.3: Fractal dimensions derived from log-transformed regression fits in Figure 3.20 ($D$=$1 - slope$), power law exponents ($\sigma^2 \propto t^B$) predicted by fractional Brownian motion ($B_{predicted}$), and those observed during the first 15 hours of dispersion based regression fit to observations of displacement variance growth ($B_{observed}$).

| **Case** | $D$ | $B_{predicted}$ | $B_{observed}$ |
|---|---|---|---|
| **Shoreline** | 1.26 ± 0.01 | - | - |
| **All Data** | 1.34 ± 0.01 | 1.49 ± 0.01 | 1.49 ± 0.01 |
| **Spring** | 1.31 ± 0.01 | 1.53 ± 0.01 | 1.54 ± 0.02 |
| **Summer** | 1.46 ± 0.01 | 1.37 ± 0.01 | 1.48 ± 0.03 |
| **Fall** | 1.30 ± 0.01 | 1.54 ± 0.01 | 1.44 ± 0.02 |

Proper characterization of cross-channel dispersion is especially important, as cross-channel motion has a significant impact on the overall transport behaviour of the drifters. We claim that the transition of along-channel dispersion from the ballistic to the diffusive regime is likely driven by cross-channel dispersion of drifters into less energetic nearshore areas, citing the strong correspondence between the transition timing and the mean transport time from mid-channel to the nearshore as evidence. This cross-channel dispersion is likely driven by a number of processes, such as: 1) non-rectilinear tidal ellipses and corresponding cross-channel tidal excursions, though these are generally small (∼500 m); 2) topographically driven eddies, which are likely to occur frequently in the Kitimat fjord system considering the appreciable roughness of the coastline; 3) shear dispersion generated by increased wind-driven currents near the centreline of the channel; 4) cross-channel transport of drifters with winds that are not perfectly aligned with the channel axis.

Internal wave activity (as previously reported on by [161]) may also contribute to cross-channel shear dispersion. Analysis of CTD profiles shows that on average internal modes 2 (and higher) exhibit Rossby radii, $R_i$, less than or equal to the channel width, $W$, (4 - 6 km for Mode 2, 2.5 - 3.5 km for Mode 3) with strong surface intensification (Figure 3.21). This suggests that rotational dynamics may be important but not dominant, as the channels fall into an intermediate regime where $W/2 < R_i < W$ [152, 115]. More work is needed to fully understand the dynamics governing cross-channel dispersion of drifters in the Kitimat fjord system, but this is beyond the scope of this paper and will be addressed in future work.

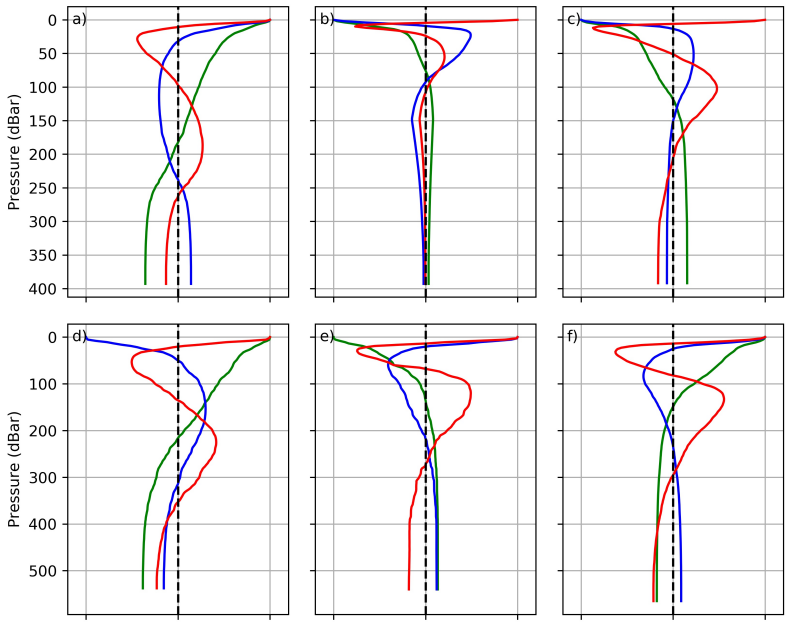


Figure 3.21: Horizontal velocity modes 1 - 3 (Mode 1 = green, $R_i$ = 6.5-9.5 km, Mode 2 = blue, $R_i$ = 4-6 km, Mode 3 = red, $R_i$ = 2.5-3.5 km) as a function of depth in Upper Douglas Channel (top, a - c) and Whale Channel (bottom, d - f). Values are shown for spring (left, a & d), summer (center, b & e), and fall (right, c & f). Modes were calculated using the standard vertical structure equation following [3]. All modes are normalized by their surface value.

## 3.6 Application to Oil Spill Risk Analysis

The results presented in Section 3.4.2 form a useful basis for predictions about the absolute dispersion of drifters in the along-channel direction. This could have important implications for assessing potential oil spill trajectories in the Kitimat fjord

system.

The dynamical response of surface drifters varies considerably with draft, and therefore the behaviour of a drifter must be experimentally compared to that of an oil spill before drifter observations can be used as a proxy for spilled oil. In 2006, the passenger ferry *Queen of the North* sank near the mouth of Douglas Channel. This incident resulted in a spill of ∼220,000 L of diesel fuel and tragically claimed the lives of two passengers. To assess the similarities between spilled oil and the drifters considered here, we compare drifter tracks passing the incident site to observations of oil sheening within 48 hours of the incident [57]. The cumulative density of positional fixes collected by drifters within 48 hours after passing within 2 km and 5 km of the incident site is shown in Figure 3.22. Tracks passing within 1 km and 10 km were also compared, but were not significantly different from the 2 km and 5 km cases (not shown).

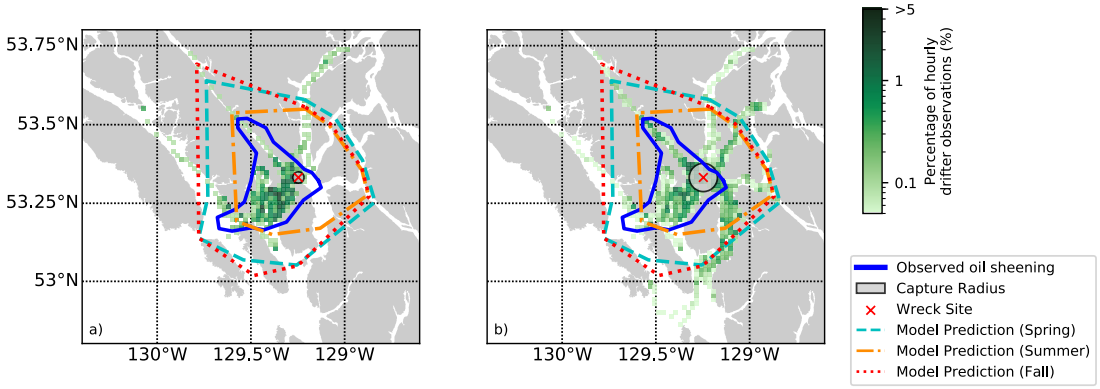


Figure 3.22: Density of hourly drifter positional fixes up to 48 hours after passing within 2km (a) and 5km (b) of the site of the wreck of the *Queen of the North*. Observations of oil sheening resulting from the wreck after 48 hours are shown in blue. Potential seasonal maximum spill extents predicted by applying a normal distribution with mean and variance specified by the regression fits from Figure 3.13 are shown in cyan (spring), orange (summer) and red (fall).

This comparison suggests that the observed drift may be an adequate proxy for the transport of spilled oil. For the 2 km case (left panel of Figure 3.22), the agreement between the drifter tracks and the observations of the diesel sheen is very good. Within 48 hours of passing the wreck site, 88% of the hourly drifter positions are recorded within the area where oil sheening was observed, and the drifters cover 80% of this area during that time. The majority of areas where drifters exited the region of observed oil sheening are areas where appreciable uncertainty in the sheening

observations is noted ([57]). Also, given seasonal and inter-annual variability in the circulation it is not surprising that some drifters travelled outside of the area where the diesel slick was observed. We note however that no drifters passing within 2 km of the incident site travelled into the channels connecting to Wright Sound from the east. This differs for the 5 km case (right panel of Figure 3.22), as the majority of drifters passing within 2 - 5 km of the wreck site occupy the channels connecting to Wright Sound from the east. The number of available drifter tracks approximately doubles for this case, however the percentage of fixes in the area with reported sheening drops to 57%. This suggests that trajectories of surface drifters and oil spills from Wright Sound are highly sensitive to initial location. Local residents' observations of diesel sheen in Whale and Ursula Channel in the year following the incident support the notion that transport of oil from Wright Sound to these regions is possible [57].

As the drifters appear to be a reasonable proxy for spilled oil, we can use the results from Section 3.4.2 to model the possible extents over which oil may be transported, noting a few cautions and assumptions. By taking only the along-channel dispersion into consideration, we assume that oil spreads across the channel quickly. This is supported by the rapid cross-channel motion noted in Section 3.4.3, and we note that gravitational spreading of a large oil spill would speed cross-channel dispersion up further. We also caution readers that taking the drifters described here to be a suitable proxy for spilled oil is likely not a universal result, since the behaviour of different oil slicks may vary appreciably depending on the physical and chemical properties of the slick. Also, the modelling approach proposed here should only be applied for times between the months of March and November. No drifter data is available to inform the choice of model parameters outside of these months, and the effect of changes in wind patterns and stratification on surface drift is not readily estimated without additional data collection or verifiable numerical modelling.

Acknowledging these caveats, Equation 3.2 can be used to model the area within which we would expect to find 95% of the oil spilled from the wreck of the *Queen of the North* within the first 48 hours. We calculate $\mu(\tau)$ and $\sigma^2(\tau)$ using values from Table 3.2. $\mu(\tau)$ is the product of the mean drift (third column) and $\tau = 48$ hours, and $\sigma^2(\tau) = A_d \tau^{B_d}$. We use the diffusive parameters to calculate $\sigma^2(\tau)$ as 48 hours is outside of the ballistic regime, and predict that the maximum drifter (or oil) displacement will be 27 - 32 km upstream of the initial location and 28 - 50 km downstream, with exact values depending on season (Figure 3.22). The area described by these displacements from the wreck site fully encompasses the area within which

oil sheening was observed after the incident, and they are a good match with observed drifter positions within 48 hours of passing within 5 km of the wreck site of the *Queen of the North*.

The area where oil is predicted to be found is larger than the reported oil sheen observation, which can be expected for two reasons. First, by applying Equation 3.2 in the manner presented in Figure 3.22 we assign each channel equal likelihood of drifters (or oil) entering it, as we cannot statistically resolve the dispersion between the channels. This is a major assumption which is supported by the fact that our predictions are in much better agreement with the drifters passing within 5 km of the wreck site than with those passing within 2 km. Since trajectories are sensitive to initial position, and the location of an oil spill is often not known precisely in operational scenarios, we suggest that assuming equal likelihood or oil presence in each channel is appropriate. The second reason we expect the predicted area to be larger than the area of observed oil sheening is that the sheening is only one realization of a stochastic process (dispersion from the wreck site), while the prediction refers to the 95% confidence intervals of this process. Each drifter (or group of drifters) passing the wreck site is an independent realization of the process however, and consequently the footprint of the drifter data is larger than the observed sheened area. In the spring and fall, 91% of the observed drifter data was recorded within the area predicted by the model, though that number falls to 80% in the summer. These values are only slightly smaller than the expected 95%, and the misfit can be attributed to the excess kurtosis of the displacement distributions.

As an alternative to using observationally derived parameters in Equation 3.2, dispersion may be modelled using generalized results from previous work. This approach is attractive as it does not require additional data collection. To the best of our knowledge [98] is the best available source of general dispersion estimates for this application, however some differences between his work and that presented here must be considered. [98] measures dispersion as dye patch size, whereas we consider single-particle statistics. Therefore the apparent diffusivity, $K_a$, at large scales appears to be the reasonable metric for comparison since we expect that $K_a$ from multi-particle statistics converges to $K_a$ derived from single-particle statistics as the patch size and dispersion time approach the decorrelation scales of the velocity field. Our results suggest $K_a = \frac{1}{2}\frac{d\sigma^2(\tau)}{d\tau} = \frac{1}{2}A_d B_d \tau^{B_d - 1} = 7.9$ x $10^6$ cm$^2$s$^{-1}$ at $\tau = 48$ hours (with $A_d$ and $B_d$ from Table 1, in units of meters and seconds). This is near the upper limit of the range of apparent diffusivities at spatial scales of $\sim$100 km presented in Figure 4

of [98], and larger than the diffusivity implied by regression fit to their data.

There are several reasons we expect our data to imply slightly higher diffusivity than the results of [98]. The first is that 48 hours may not be enough time for estimates of $K_a$ from multi-particle and single-particle statistics to converge. Also, the along-channel axis is clearly the major axis of dispersion (see Figures 3.10 and 3.15) and therefore applying dispersion along this axis isotropically will result in a larger estimate of diffusivity than [98] who distinguishes between major and minor axis dispersion. Finally, we note that [98]'s data points for diffusivity at scales larger than 10 km are predominantly from experiments in the North Sea, and currents and turbulence levels during these experiments were likely less than in the Kitimat fjord system where currents and turbulence are amplified by topographic restrictions on the flow.

We propose that single-particle statistics are appropriate for our application, as the purpose of our model based on Equation 3.2 is to predict the farthest extents that oil may have drifted to, rather than the change in size of an oil slick due to dispersion of the oil. Since the relationships proposed by [98] cannot be used to successfully recover the associated diffusivity we conclude that dispersion parameters for the Kitimat fjord system and dynamically similar regions should be derived from observations, as was done here, until sufficient data is available to expand the work of [98] to include multiply connected fjord systems.

The effect of mean drift should also be considered. We note above that our predictions are better in the spring/fall than in the summer, and since the mean drift is nearly negligible in the summer data this suggests that including a mean drift improves predictions. Equation 3.2 allows for a mean drift, and it does not appear that this equation can easily be simplified further. Applying a constant mean and constant dispersion rate would simplify the mathematics, however the observed rate of dispersion decreases with time due to drifters approaching the shoreline, and therefore a constant dispersion rate would result in overestimation. For these reasons we propose that modelling the possible extents of along-channel transport using Equation 3.2 strikes a good balance between simplicity and accuracy.

## 3.7 Conclusions

We have investigated the mean drift and dispersion of 206 novel ocean surface drifters deployed in the Kitimat fjord system between 2014 and 2016. To the best of our

knowledge this is the most extensive set of drifter data collected in a fjord system to date.

The drifters exhibit a mean seaward motion through the study region, which is consistent with previously described estuarine circulation patterns [157]. Mean outflow is strongest in the spring and fall, which has been linked to strong up-channel winds occurring in the summertime. Along-channel dispersion of drifters is also strongest in the spring and fall, with some drifters dispersing over the length of the fjord system ($\sim$140 km) in 7 days. No drifter data is available during the winter months (beginning of December to mid-March).

On timescales longer than $\sim$12 hours interaction with shorelines slows drifter dispersion to rates below those expected from the classical theory of [145]. The close match between the time required for drifters to reach less energetic nearshore regions and the timing of the slowing of along-channel dispersion suggests that rapid cross-channel dispersion is the root cause of slowed along-channel dispersion. On shorter timescales the observed dispersion is consistent with fractional Brownian motion [125], and the tracks are found to display fractal characteristics over scales of 2 to 13 km with fractal dimensions that are comparable to values from the literature.

The observed drift speeds are found to be best described by a combination of uppermost available currents in the water column and wind-driven term parameterizing near-surface current shear, wave-induced circulation and some drifter slippage. Therefore measurements of wind are needed to complement measurements of currents made with moored subsurface ADCPs to fully characterize the drift of objects at the ocean surface in the Kitimat fjord system. A seasonally resolved linear regression model for along-channel drift speed has been proposed.

Finally, the drifters are shown to be a reasonable proxy for oil sheening observed after an oil spill in the study region. A simple dispersion model based on the 95% confidence intervals of a Gaussian distribution fitted to the observed along-channel drifter displacements is found to well reproduce the area within which drifters or oil from a specified location can be found. This model may yield useful information for estimating the potential extent of oil transport after a spill in the Kitimat fjord system.

# Chapter 4

# A fuzzy - based framework for assessing uncertainty in drift prediction using observed currents and winds

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## 4.1 Introduction

The ability to respond to emergencies such as marine contaminant spills and search and rescue efforts is directly linked with the ability to predict the trajectories of objects drifting at the ocean surface [30, 10, 32, 15]. Better predictions of trajectories also enhance our understanding of marine ecosystem functioning and connectivity [18, 49, 120, 139]. Drift trajectories of floating objects are governed by currents and winds at the very surface of the ocean. These remain persistently difficult to measure [133], and the submesoscale variability of these fields can only be measured directly during targeted campaigns of limited duration [108, 31]. Therefore further treatment is generally required to properly characterize uncertainty arising from imprecise

measurements of surface currents and winds. In practice, currents and winds used to force drift trajectory simulations are usually obtained from numerical circulation models, which introduces further uncertainty into the problem as these models are by necessity a simplification of reality. For the purposes of this paper, we consider only observational data and the relevant processes giving rise to uncertainty in this data. Model data and associated uncertainties will be considered in future work, since the uncertainties in observations used to verify the model data must first be established in order to accurately estimate the uncertainties in modelled data.

Uncertainty in drift trajectory models is usually treated by applying a series of random kicks to modelled particles [173, 36, 19], ensemble modelling based on the uncertainty in empirically derived forcing parameterizations [10, 12, 28], or by defining an empirically determined dispersion coefficient [98]. This approach has a long history of relative success. However the specification of the magnitude of random kicks or dispersion coefficients is generally not obvious, and the assumption that the underlying forcing data is free of systematic uncertainties and biases remains. Usually a large number of perturbed trajectories are computed. Other workers have also computed ensemble predictions from a variety of numerical models [119, 155] or systematic variations of forcing data to capture variability on seasonal time scales [43].

The present study takes a different approach, by expressing the forcing data for a drift trajectory model as fuzzy numbers. Fuzzy numbers are used to describe imprecise data. They are described by their membership function, which gives the range of possible values of the true data at varying degrees of belief, or membership levels [170, 172]. Robust frameworks for propagating the uncertainty described by fuzzy numbers through mathematical models have been developed [65, 56]. The formulation of fuzzy numbers does not require assumptions about the underlying character of the uncertainty, they simplify aggregation of uncertainty from various sources, and they can be constructed from sparse data sets. In environmental sciences and engineering, fuzzy numbers have been applied to a variety of problems involving imprecise knowledge of parameters, such as: description of water masses [42]; tracking of storm systems [84]; remote sensing of ocean surface properties [87]; modelling of water quality in rivers [69]; and assessing the frequency of landslide occurrence following rain events [102]. Interval analysis, which can be considered a simplification of fuzzy number arithmetic, has been applied to drift trajectory prediction by [93], who considered a leeway model for drift [10] in the absence of waves, and tested this

model using idealized uniform current and wind forcing.

The leeway model is commonly used to model drift using current and wind data as forcing. Here the motion of a drifting object is described as the superposition of the current and a fraction of the wind speed (often $\sim 3\%$) directed at an angle to the wind (to the right in the northern hemisphere). Encapsulated in this parameterization are the effects of current shear and rotational dynamics in the upper ocean above the level at which currents are measured, direct wind drag on exposed parts of the drifting object, wave-induced circulation, and wind shear below the level at which winds are measured. The origins of this method date back to [90] and [40], however it remains an active field of research [28, 141].

Object geometry has a significant effect on wind response. The work reviewed in [2] showed that empirically fitted leeway parameters (rate and angle) of a variety of drifting search and rescue targets is strongly dependent on the target geometry. Wind response may change quite significantly even for relatively small changes in geometry [121, 141]. Deterministically describing the wind response for a given object geometry remains an active research question, and empirically determined coefficients are more frequently used [2, 12]. A framework for deterministic description of the wind response has been developed based on the force balance on the drifting object [30, 122], and in this paper the wind drag coefficients of five different types of drifting buoys are derived using this framework. These coefficients are used in our trajectory model in conjunction with parameterizations for wave-induced circulation and near-surface wind and current shear, rather than using empirical leeway parameters.

A similar effort is described by [122], who used ship-based wind measurements, near-surface (0.5 m) current measurements from a 1 MHz ADCP, and directional wave spectra in a Norwegian fjord to describe the observations from surface drifters. They found that wave-induced Stokes drift was of first order importance to the trajectory prediction along with winds and currents. Wind effects were parameterized by fitting a drag coefficient to optimize the trajectory description. In a similar vein, [143] considered surface current shear extrapolated from an ocean model of the Gulf of St. Lawrence, coarsely resolved in the vertical direction, to test the effect of adding this shear into a trajectory model. They report noticeably improved predictions.

Here we expand on these efforts by developing a drift trajectory model with uncertainty propagation based on fuzzy numbers, forced with currents and winds measured at Ocean Station Papa (hereafter OSP) in the northeast Pacific. The resulting predictions are verified against observations from five different types of ocean surface

drifters launched nearby. Uncertainty in the model is considered as both uncertainty in the forcing data, which is propagated using fuzzy numbers, and uncertainty about the relevant combinations of forcing parameters. To address the latter, we compute an ensemble of drift trajectory predictions using all possible combinations of forcing terms and identify the combination resulting in the optimal predictions for each drifter type. Results from the proposed method are compared to analogous results derived using the leeway method with empirical coefficients corresponding most closely to the object under consideration.

To begin, we describe the trajectory model and the derivation of relevant forcing terms in Section 4.2. Fuzzy numbers are introduced next, and we describe how they are used to propagate uncertainty through this trajectory model in Section 4.3. Next, the observational forcing data used to test and validate the model is described in Section 4.4, followed by the presentation of results on model performance and the optimal combination of forcing terms in Section 4.5, followed by discussion and conclusions.

## 4.2 Drift Trajectory Model

The trajectory of an object drifting at the ocean surface is determined by the balance of forces acting on it. The derivation of the following linear vector equation for the drift velocity according to [30, 10, 122] from the balance of forces is given in Section 2.2.1.

$$\frac{d\vec{x}}{dt} = \vec{u}_c|_{z=z_{drifterC}} + \vec{u}_{stokes} + \sqrt{\frac{\rho_a A_a C_a}{\rho_w A_w C_w}}\vec{u}_w|_{z=z_{drifterW}} \tag{4.1}$$

Here, as previously, $\frac{d\vec{x}}{dt}$ is the drift velocity, $\vec{u}_c|_{z=z_{drifterC}}$ is the current at the effective depth of the drifter below the water surface, $z_{drifterC}$, $\vec{u}_{stokes}$ is the Stokes drift velocity, and $\vec{u}_w|_{z=z_{drifterW}}$ is the wind at the effective height of the drifter above the water surface, $z_{drifterW}$. For notational simplicity we will write the scaling factor on $\vec{u}_w|_{z=z_{drifterW}}$ as

$$\alpha = \sqrt{\frac{\rho_a A_a C_a}{\rho_w A_w C_w}} \tag{4.2}$$

The density of air and water are taken as $\rho_a = 1.225\ kg/m^3$ and $\rho_w = 1025\ kg/m^3$,

respectively. The airside and waterside cross-sectional areas of the drifter are $A_a$ and $A_w$, and the corresponding drag coefficients are $C_a$ and $C_w$, respectively. Their product can be determined from the vertical profiles of effective width, $w(z)$, and approximate drag coefficient, $C(z)$, of the drifting object.

$$A_a C_a = \frac{\int_0^{Z_{max}} w(z)C(z)dz}{\int_0^{Z_{max}} w(z)dz} \tag{4.3a}$$

$$A_w C_w = \frac{\int_{Z_{min}}^0 w(z)C(z)dz}{\int_{Z_{min}}^0 w(z)dz} \tag{4.3b}$$

The maximum draft of the object below the water surface and its maximum height above the surface are given by $Z_{min}$ and $Z_{max}$ , respectively.

The assumptions implicit in (4.1) are that $\vec{u}_c|_{z=z_{drifterC}}$ is known at the effective depth of the drifter, and $\vec{u}_w|_{z=z_{drifterW}}$ is known at the water surface. This however is generally not the case, as both are measured some distance from the water surface on conventional observation platforms. Therefore some shear in the winds and currents due to boundary layer effects between the measurement level and the effective height (depth) of the drifter is not represented in measurements. Accounting for near-surface current shear has been shown to significantly improve drift predictions based on numerical ocean models [143]. We test whether this holds for drift predictions made using observations of currents at depth $z_c$ and winds at height $z_w$ from OSP, by adding estimates of near-surface current shear, $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$, and wind shear $\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$ to (4.1). Therefore the current at the effective depth of the drifter is $\vec{u}_c|_{z=z_{drifterC}} = \vec{u}_c + \Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$, and the wind at the water surface is $\vec{u}_w|_{z=z_{drifterW}} = \vec{u}_w + \Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$. $\vec{u}_c$ and $\vec{u}_w$ are the measured currents and winds, respectively. This yields the following model for drift trajectories,

$$\frac{d\vec{x}}{dt} = \vec{u}_c + \Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}} + \vec{u}_{stokes} + \alpha(\vec{u}_w + \Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}) \tag{4.4}$$

From $\vec{u}_w$ and observed water and air temperatures, we estimate $\vec{u}_{stokes}$, $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$, and $\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$. The processes used to estimate these quantities are discussed in the following sections.

Both Stokes drift, $\vec{u}_{stokes}$, and current shear, $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$, exhibit vertical gradients with strong surface intensification. Stokes drift decays rapidly within a few meters of the surface [21]. Since the shapes of drifters below the water surface are

often irregular, we investigate whether the method by which these velocity profiles are integrated has a noticeable effect on trajectory predictions. We compare results derived with Stokes drift and current shear taken either as the value at the geometric centroid of the drifter, or as an area-weighted average of the vertical velocity or drag force profile. For a velocity profile $\vec{u}(z)$, these can be expressed as,

$$\vec{u}_{centroid} = \vec{u}(z)|_{z=z_{centroid}=\frac{\int_{z_{min}}^{0} zw(z)dz}{\int_{z_{min}}^{0} w(z)dz}} \tag{4.5a}$$

$$\vec{u}_{velocity} = \frac{\int_{z_{min}}^{0} w(z)\vec{u}(z)dz}{\int_{z_{min}}^{0} w(z)dz} \tag{4.5b}$$

$$\vec{u}_{force} = \sqrt{\frac{\int_{z_{min}}^{0} w(z)C(z)\vec{u}(z)|\vec{u}(z)|dz}{\int_{z_{min}}^{0} w(z)C(z)dz}} \tag{4.5c}$$

Here all symbols are as previously defined.

In order to address the systemic uncertainty in drift trajectory prediction that arises from the uncertainty around the importance of forcing terms in (4.4) and methods for applying sheared forcing, we test the sensitivity of the predicted drift trajectory to the inclusion of the forcing terms in (4.4) and the choice of the three possible methods of applying sheared forcing from (4.5). This is done by first calculating trajectories for each drifter type using all possible combinations of forcing terms and sheared forcing methods, starting with the simplest case of predicting a trajectory using only a single forcing term, either $\vec{u}_c$, $\vec{u}_w$, or $\vec{u}_{stokes}$. Additional terms are added in subsequent predictions. All predictions are then scored against observations from the type of drifter being modelled at each time step of the prediction, and the combination of terms resulting in the highest skill is reported. A term that is in the solution with the highest skill at the majority of the time steps is considered to be part of the optimal combination of parameters for this drifter.

### 4.2.1 Stokes Drift

We estimate Stokes drift [140] from one-dimensional frequency spectra of sea surface height, which we in turn estimate using the a parametric spectrum for fully developed seas by [106]. Stokes drift is assumed to be in the wind direction. Using the classical expression for Stokes drift in non-monochromatic seas given by [66] yields the follow-

ing expression for the vertical profile of Stokes drift, $U_s(z)$, in the deep-water limit of the dispersion relation (i.e. $\omega^2 = gk$ where $\omega$ is angular frequency, $k$ is wavenumber, and $g$ is acceleration due to gravity),

$$U_s(z) = 2\alpha_P g \int_0^\infty \omega^{-2} e^{\left[\frac{2\omega^2 z}{g} - \beta_1 \left(\frac{\omega_p}{\omega}\right)^4\right]} d\omega \tag{4.6}$$

Here $\alpha_P$ is the Phillips constant, set to 0.0081 following [60], and $\beta_1 = 0.74$ is a constant. The peak circular frequency can be approximated as $\omega_p = g/U_{19.5}$, where $U_{19.5}$ is the wind speed at a height of 19.5 m (the same height as the measurements used by [106]). The vertical Stokes drift profile for a given $U_{19.5}$ can be derived by integrating (4.6) at discrete depths, $z$. This profile is the converted to a singular value, $\vec{u}_{stokes}$ for use in (4.4) using the methods in (4.5).

We limit the discussion in this paper to parameterized one-dimensional wave spectra of fully developed wind seas. This is done for generality, as validated estimates of two-dimensional wave spectra are unavailable for many parts of the world's oceans. Recent work by [21], including results from OSP, suggests that Stokes drift is well approximated by parameterized one-dimensional spectra and that the contribution of remotely generated swell is small compared to higher frequency wind seas. This is because Stokes drift for a monochromatic wave is proportional to the cube of the waves frequency, implying exponential increases in Stokes drift with increasing wave frequency (note that (4.6) gives the sum of all frequency bands in the spectrum). Rather than increase the magnitude of Stokes drift, the inclusion of swell bands in the calculation has been noted to change the shape of the vertical profile of Stokes drift [154]. Further, the approach adopted here is applicable wherever wind information is available and the deep-water limit of the dispersion relation applies. Fetch-limited locations may be considered using the parametric spectrum proposed by [58]. Future work will extend the approach presented here to include Stokes drift calculated from validated estimates of two-dimensional wave spectra.

It has been well established that the contribution of the Stokes drift to the Coriolis force, i.e. the Coriolis-Stokes force (c.f. [110]), influences near surface currents. However Coriolis-Stokes forcing is not considered here, since this is an active field of research with evolving understanding of processes and uncertainties, and a detailed contribution to this subject is beyond the scope of the current work. Interested readers may find more information in recent papers and reviews, such as [159], [154] and [21].

### 4.2.2 Current Shear

Wind forcing results in significantly sheared upper ocean currents directed to the right of the wind (in the northern hemisphere), as was first shown by [40]. Near-surface shear is further enhanced in the shallow wave-affected layer in the few meters immediately below the ocean surface [23]. The resulting difference between the current at the effective depth of the drifter and the depth at which currents are measured is given as $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$ in (4.4).

The wind-driven current shear in the upper ocean can be modelled by numerically solving the unsteady momentum balance between rotation and vertical diffusion of momentum with surface wind forcing using the level 2.5 turbulence closure scheme of [83], as described in [23].

$$\frac{\delta u}{\delta t} - \frac{\delta}{\delta z}\left(A\frac{\delta u}{\delta z}\right) = fv \tag{4.7a}$$

$$\frac{\delta v}{\delta t} - \frac{\delta}{\delta z}\left(A\frac{\delta v}{\delta z}\right) = -fu \tag{4.7b}$$

$$\frac{\delta b}{\delta t} - \frac{\delta}{\delta z}\left(lqS_q\frac{\delta b}{\delta z}\right) = lqS_M\left(\left(\frac{\delta u}{\delta z}\right)^2 + \left(\frac{\delta v}{\delta z}\right)^2\right) - \frac{2qb}{Bl} \tag{4.7c}$$

The first two equations describe the balance of momentum in the $u$ and $v$ directions, and the latter describes the balance of turbulent kinetic energy (TKE). The two relate via the eddy viscosity $A = lqS_M$, which requires the prescription of the turbulent length scale $l$. $q = \sqrt{b}$ is the turbulent velocity scale, or square root of TKE. $S_q$ is a dimensionless constant. $S_M$ is a constant in unstratified water, which use here and assume that the well-mixed condition holds in the upper ocean mixed layer. This is supported by profiles of temperature and salinity observed around the time of the drifter deployments, which show well-mixed water down to 80 - 90 m depth (Figure 4.1).

Apart from two minor adjustments, which are described in the next paragraphs, we retain the formulation of [23] and do not reproduce details here. Readers interested in a detailed description are referred to the original paper. We force the system by converting measured winds to friction velocities, $u_*$, defined as $\vec{u}_* = \sqrt{C_d\vec{U}_{10}^2\left(\frac{\rho_a}{\rho_w}\right)}$. $C_d$ is the drag coefficient, interpolated from tables given by [132] and all other variables are as defined previously. The densities of air/water ($\rho_a/\rho_w$) are taken as 1.225 $kg/m^3$ and 1025 $kg/m^3$, respectively. For simplicity we take $\vec{u}$ to be in the downwind

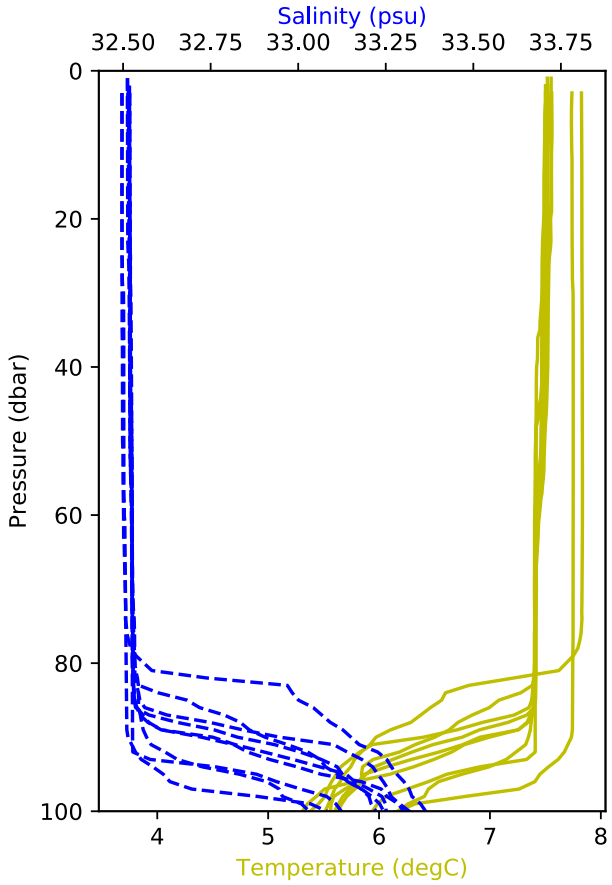


Figure 4.1: Profiles of temperature (yellow) and salinity (dashed blue) taken between Feb 18 - 19, 2015.

direction and $\vec{v}$ in the crosswind direction, and rotate the derived velocities back to geographical coordinates once time-stepping is complete.

Boundary conditions are applied using this friction velocity.

$$\begin{aligned} A\frac{\delta u}{\delta z} &= u_*^2, \text{ at z=0} \\ A\frac{\delta v}{\delta z} &= 0, \text{ at z=0} \\ u = v &= 0, \text{ at z=H} \end{aligned} \tag{4.8}$$

We solve the equations over a depth of $H$=200 m, discretized into 0.5 m bins. This implies the assumption that direct wind-driven currents and TKE are negligible below 100 m as in this formulation the return flow due to mass conservation and bottom boundary layer occur in the bottom half of the domain. These simulated phenomena

are not considered in this study.

Once the simulated wind-driven upper ocean current $\vec{u}(z) = [u(z), v(z)]$ has been modelled, we compute the wind-driven current at the effective depth of the drifter, $\vec{u}(z = z_{drifterC})$ by the methods in (4.5) and then derive $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$ for use in (4.4).

$$\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}} = \vec{u}(z = z_{drifterC}) - \vec{u}(z = z_c) \tag{4.9}$$

### 4.2.3 Wind Shear

Vertical shear in the wind field between the measurement height (4 m) and the level of the exposed parts of the drifter, taken here as 1 m, can be estimated from atmospheric boundary layer relationships. [132] gives a review of the necessary theory and tabulates conversion factors for wind speed from 1, 2, 5, and 20 m to the standard 10 m reference height, for varying degrees of atmospheric stability. Atmospheric stability is represented as the difference between virtual sea surface temperature and virtual potential air temperature at the measurement height, which we obtain from measurements of both air and water temperature. We then use the conversion factors, $C_{UX_Y}$, interpolated from tables in [132] to convert winds at the measurement height, first to the 10 m reference height, and then to 1 m. Single subscripts denote the vertical level in meters, and double subscripts $X_Y$ denote conversion from level $X$ to level $Y$. The conversion to derive $\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$ for use in (4.4) proceeds as follows,

$$\vec{U}_{10} = (\frac{3}{4}C_{U5_10} + \frac{1}{4}C_{U1_10})\vec{U}_4 \tag{4.10a}$$

$$\vec{U}_1 = \frac{\vec{U}_{10}}{C_{U1_10}} \tag{4.10b}$$

$$\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w} = \vec{U}_1 - \vec{u}_w \tag{4.10c}$$

In the following section we present a method for propagating uncertainty in currents and winds measured at OSP through the trajectory model described here, including the derivation of the parameterized forcing terms $\vec{u}_{stokes}$, $\Delta\vec{u}_c|_{z=z_c}^{z=z_{drifterC}}$, and $\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$. The method is based on fuzzy numbers, and we begin by giving a brief overview of the relevant theory.

## 4.3 Uncertainty Propagation

### 4.3.1 Overview of Relevant Fuzzy Number Theory

Fuzzy numbers are used to describe imprecise quantities. At their core, they describe the possible range of values a quantity might take as a function of ones degree of belief in that range of values. This is known as the *membership function*. Membership functions are usually derived through quantitative analysis of the concept being considered, coupled with expert knowledge of this concept. To illustrate the concept of a membership function, an informal hypothetical example of such a derivation is included in the appendix.

In general, fuzzy numbers are a special case of fuzzy sets [170, 172]. The membership function, which describes the fuzzy number, is defined by a unique value that is certainly possible, with a membership of one (the *crisp value*). From that value the function decreases monotonically towards the lowest and highest values thought to be possible, with a decreasing degree of belief as values diverge from the crisp value. This monotonic decrease implies the function is *convex*. The largest range of values thought to be possible is the *support* of the fuzzy number, i.e. the largest interval with non-zero membership.

It is important to note that membership functions, and the fuzzy numbers they represent, are not equivalent to probability distributions. They instead describe the range of values one believes a fuzzy number might take, without making claims about the probability of occurrence of values in this range. Fuzzy numbers are however weakly linked to probability theory through the restriction that the possibility of any value must always be greater than or equal to its probability. This is known as the *consistency principle* [170].

Membership functions are generally given as a set of crisp intervals defining a possible range of values at a given degree of possibility, rather than defining a continuous function. We refer to these intervals as membership levels, however they are also known as $\alpha$-cuts in the literature [117]. Hence a fuzzy number $P$ can be described as a set of membership levels $A^{(j)}$, i.e.

$$P = \{A^{(0)}, A^{(1)}, ..., A^{(m)}\} \tag{4.11a}$$

$$A^{(j)} = [a^{(j)}, b^{(j)}] \text{ for } j = 0, 1, ..., m,\ a^{(j)} \leq b^{(j)} \tag{4.11b}$$

Here $a^{(j)}$ and $b^{(j)}$ are the lower and upper bounds of the membership level $j$. The membership, or degree of belief, at each level is given by $\frac{j}{m}$.

The simplest, and most common, type of membership function is the triangular function, which is defined by a crisp value and two points defining the support. However other types of membership functions exist, and some that are more representative of the uncertainty being modelled have been shown to improve results for geophysical applications [69, 67]. We do not consider triangular membership functions for the purposes of this paper. Instead we use a probability-possibility transform based on the consistency principle to quantitatively derive membership functions for the current and wind measured at OSP using Gaussian and Laplace distributions. For this we must specify a crisp value, or mean $\mu_K$, and possible variability, or standard deviation $\sigma$, as well as the number of membership levels, $j = 0, 1, ..., m$. The functions describing a Gaussian (Laplace) distribution are given by $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu_K)^2}{2\sigma^2}}$ $\left(f(x) = \frac{1}{\sqrt{2}\sigma} e^{-\frac{\sqrt{2}|x-\mu_K|}{\sigma}}\right)$. With this information the symmetric range of possible values around $\mu_K$, denoted as $[\mu_K - a, \mu_K + a]$, can be derived for each membership level by iteratively solving the following integral for $a$ [38, 174].

$$\int_{-\infty}^{\mu_K - a} f(x)dx + \int_{\mu_K + a}^{\infty} f(x)dx = \frac{j}{m} \text{ for } j = 0, 1, ..., m \tag{4.12}$$

The resulting membership functions are symmetric about $\mu_K$ since the Gaussian and Laplace distributions are also symmetric; this is not a general property of fuzzy numbers. The membership levels are also not required to be evenly distributed, though they often are for convenience. Specifying or modifying the shape of a fuzzy number based on expert knowledge is possible, where appropriate, as long as the membership function remains convex. Examples of fuzzy numbers in various forms (triangular, transformed from probability- distributions, and modified by expert knowledge) are shown in Figure 4.2.

Fuzzy numbers can be used to propagate uncertainty through a mathematical model by applying well established rules for performing arithmetic operations on them. Traditionally this has been done by applying interval arithmetic to individual membership levels [65]. However this method has the drawback that uncertainty is overestimated in systems where a fuzzy-valued term appears more than once in the system's equations. This is because all possible values of the fuzzy term are considered for each instance of the fuzzy term, which results in 'double-counting' [55].

For systems where a fuzzy-valued term appears multiple times, arithmetic should

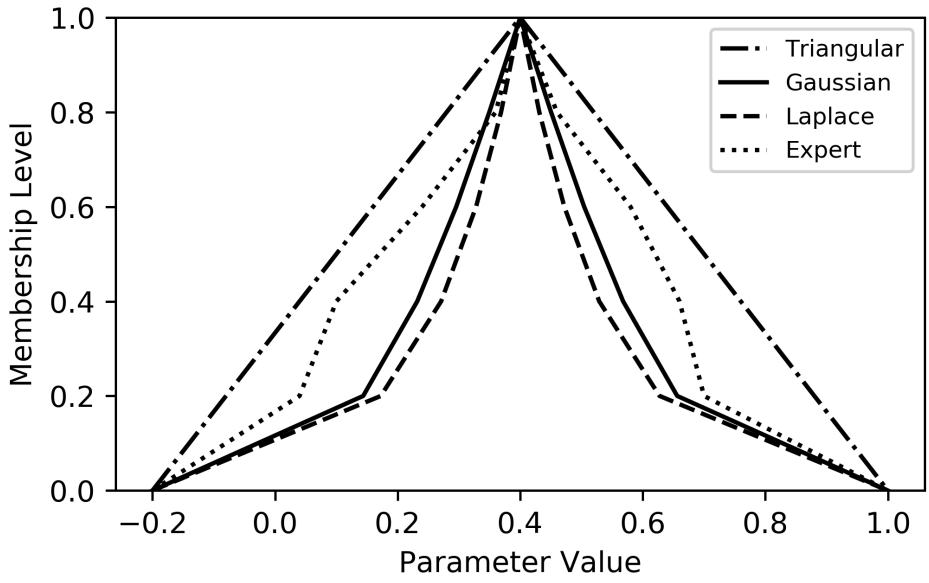


Figure 4.2: Examples of possible shapes of the membership function for a fuzzy number describing possible values of an arbitrary parameter. Functions for a triangular (dash-dotted), transformed Gaussian (solid), transformed Laplace (dashed), and expert knowledge-informed (dotted) fuzzy number are shown. The transformed membership functions are derived using a crisp value $\mu_K = 0.4$, and standard deviation $\sigma = 0.2$. All functions have a support width of $3\sigma$.

be performed using the transformation method described by [56]. This method is used here, since parameterizations for wave-induced circulation, near surface current shear, and wind shear all depend on the fuzzy-valued wind velocity. In the transformation method, the fuzzy-valued terms in the system of equations are discretized into arrays describing all possible combinations of all possible values of these terms. The system can then be solved using element-wise operations on these arrays. We give a brief description of the general transformation method here. Interested readers may refer to [56] and [55] for a thorough summary.

Consider a system of equations with $n$ independent fuzzy variables $P_i, i = 1, ..., n$. Each $P_i$ consists of $m + 1$ membership levels $X_i^{(j)}$. Each membership level $X_i^{(j)}$ is described by the interval $[a_i^{(j)}, b_i^{(j)}]$, which is then discretized into an array $\hat{X}_i^{(j)}$ according to the following equations.

$$P_i = \{X_i^{(0)}, X_i^{(1)}, ..., X_i^{(m)}\} \quad (4.13a)$$

$$X_i^{(j)} = [a_i^{(j)}, b_i^{(j)}], \quad a_i^{(j)} \leq b_i^{(j)} \quad (4.13b)$$

$$\hat{X}_i^{(j)} = \overbrace{(\gamma_{1,i}^{(j)}, \gamma_{2,i}^{(j)}, ..., \gamma_{(m+1-j),i}^{(j)}, ..., \gamma_{1,i}^{(j)}, \gamma_{2,i}^{(j)}, ..., \gamma_{(m+1-j),i}^{(j)})}^{(m+1-j)^{i-1} \text{ tuples of length } (m+1-j)} \quad (4.13c)$$

$$\gamma_{l,i}^{(j)} = \overbrace{(c_{l,i}^{(j)}, ..., c_{l,i}^{(j)})}^{(m+1-j)^{n-i} \text{ elements}} \quad (4.13d)$$

$$c_{l,i}^{(j)} = \begin{cases} a_i^{(j)} & \text{for } l = 1 \text{ and } j = 0, 1, ..., m \\ \frac{1}{2}\left(c_{l-1,i}^{(j+1)} + c_{l,i}^{(j+1)}\right) & \text{for } l = 2, 3, ..., m-j \text{ and } j = 0, 1, ..., m-2 \\ b_i^{(j)} & \text{for } l = m-j+1 \text{ and } j = 0, 1, ..., m \end{cases} \quad (4.13e)$$

A graphical example of this discretization, adapted from [55], is given in Figure 4.3 for a hypothetical fuzzy number described by four membership levels, i.e. $m = 3$. The final result of the transformation of all $n$ fuzzy input variables $P_i$ at one point in time is a set of $m + 1$ arrays $\hat{X}_i^{(j)}$ for each $P_i$. To illustrate explicitly, consider a system of equations with two fuzzy input variables (i.e. $n = 2$), each similar to the example shown in Figure 4.3. The final arrays describing the membership level $j = 1$ are constructed as follows.

$$\hat{X}_1^{(1)} = [c_{0,1}^{(1)}, c_{0,1}^{(1)}, c_{0,1}^{(1)}, c_{1,1}^{(1)}, c_{1,1}^{(1)}, c_{1,1}^{(1)}, c_{2,1}^{(1)}, c_{2,1}^{(1)}, c_{2,1}^{(1)}] \quad (4.14a)$$

$$\hat{X}_2^{(1)} = [c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}, c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}, c_{0,2}^{(1)}, c_{1,2}^{(1)}, c_{2,2}^{(1)}] \quad (4.14b)$$

Using these arrays the system of equations for the membership level can be solved as it would be in a crisp manner, but using element-wise arithmetic on the arrays $\hat{X}_1^{(1)}$ and $\hat{X}_2^{(1)}$.

The result of this element-wise arithmetic for each membership level ($\hat{X}_{result}^{(j)}$) can then be re-transformed into a fuzzy number by taking the minimum and maximum values of each $\hat{X}_{result}^{(j)}$ to be the bounds of the membership level $j$ of the fuzzy result, while ensuring that the fuzzy result remains convex.

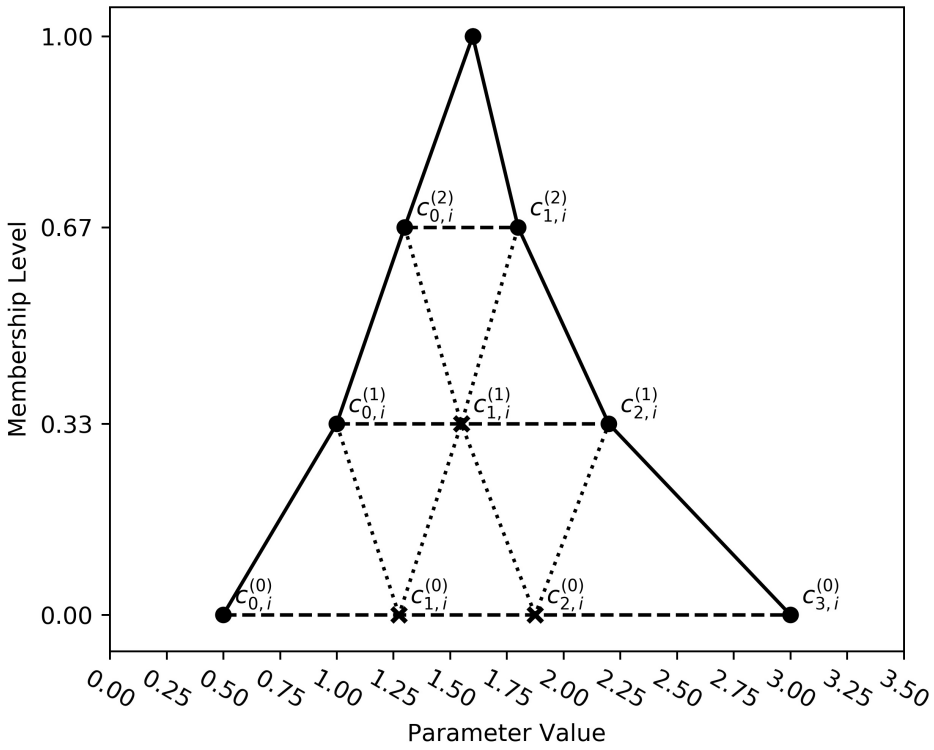


Figure 4.3: Illustration of the discretization of a fuzzy number into array form using the transformation method. Adapted from [55].

$$P_{result} = \{Z^{(0)}, Z^{(1)}, ..., Z^{(m)}\} \tag{4.15a}$$

$$Z^{(j)} = [a^{(j)}, b^{(j)}],\ j = 0, 1, ..., m \tag{4.15b}$$

$$a^{(j)} = min(a^{(j+1)}, min(\hat{X}_{result}^{(j)})) \tag{4.15c}$$

$$b^{(j)} = max(b^{(j+1)}, max(\hat{X}_{result}^{(j)})) \tag{4.15d}$$

To apply this method to the prediction of drift trajectories, we begin by applying a probability-possibility transform to convert time series of measured ocean currents and winds at OSP into fuzzy numbers, using (4.12) as detailed in the following section. Each value in the resulting fuzzy time series is then transformed into arrays describing each membership level of the currents and winds, $\hat{X}_{\vec{u}_c}^{(j)}(t)$ and $\hat{X}_{\vec{u}_w}^{(j)}(t)$, according to (4.13). The fuzzy-valued time series of drift speed, $\hat{X}_{\frac{d\vec{x}}{dt}}^{(j)}(t)$ can then be derived by substituting $\hat{X}_{\vec{u}_c}^{(j)}(t)$ and $\hat{X}_{\vec{u}_w}^{(j)}(t)$ into (4.4) and re-transforming each value in $\hat{X}_{\frac{d\vec{x}}{dt}}^{(j)}(t)$ into a fuzzy number according to (4.15).

$$\hat{X}^{(j)}_{\frac{d\vec{x}}{dt}}(t) = \hat{X}^{(j)}_{\vec{u}_c}(t) + \Delta\vec{u}_c|^{z=z_{drifterC}}_{z=z_c}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big) + \\ \vec{u}_{stokes}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big) + \alpha\big(\hat{X}^{(j)}_{\vec{u}_w}(t) + \Delta\vec{u}_w|^{z=z_w}_{z=z_{drifterW}}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big)\big) \tag{4.16}$$

As described above, (4.16) is solved using element-wise array arithmetic. The term $\Delta\vec{u}_c|^{z=z_{drifterC}}_{z=z_c}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big)$ is derived by applying the procedure described in Section 4.2.2 to all elements of $\hat{X}^{(j)}_{\vec{u}_w}(t)$. Similarly, $\vec{u}_{stokes}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big)$ and $\Delta\vec{u}_w|^{z=z_w}_{z=z_{drifterW}}\big(\hat{X}^{(j)}_{\vec{u}_w}(t)\big)$ are derived by element-wise solution of the procedures in Sections 4.2.1 and 4.2.3, respectively. The procedure by which the resulting fuzzy-valued drift velocities are integrated to give the set of all possible displacements is described in Section 4.3.3.

### 4.3.2 Fuzzification of Input Data

Fuzzy forms of observed currents and winds can be derived by considering the energy at scales unresolved by the observations, as well as the associated instrument uncertainties. To estimate the energy at time scales that are not resolved in the current and wind measurements, we fit a power law $Af^{-B}$ to the high-frequency tail of the energy spectra of the currents and winds, defined as $P(f) = \frac{1}{2}(P_u(f) + P_v(f))$ where $P_{u/v}(f)$ are the component spectra in the $u/v$ directions. The unresolved energy between the Nyquist frequency, $f_N = \frac{1}{2dt_{obs}}$, and the frequency corresponding to the model timestep, $f_C = \frac{1}{2dt_{model}}$, can be estimated by log-linearly extrapolating the fitted power law to $f_C$, integrating from $f_N$ to $f_C$, and finding the associated velocity scale by converting the resulting variance to amplitude, $A_v$.

$$A_v = \sqrt{2\sigma^2} = \sqrt{2}\bigg(\int_{f_N}^{f_C} Af^{-B}df\bigg)^{\frac{1}{2}} \tag{4.17}$$

In order to later derive the possible time series of drift velocity, the amplitude of the maximum unresolved accelerations, $A_a$, can be similarly derived by differentiating the extrapolated velocity energy spectrum. Here,

$$A_a = \sqrt{2}\bigg(\int_{f_N}^{f_C} Af^{-(B-2)}df\bigg)^{\frac{1}{2}} \tag{4.18}$$

Uncertainty due to spatial variability in the currents fields can only be estimated from numerical model results, or similar products, as observations of currents are only available in a single location. This uncertainty is denoted by $\sigma_s$, and the derivation

is discussed in Section 4.5.1 as it is case-specific.

To fuzzify the observed currents and winds we then assume that the uncertainty is isotropic, and that the uncertainties in the $u$ and $v$ directions are uncorrelated. The probability-possibility transform (4.12) can then be applied to each component separately, using the measured value as the crisp value. The standard deviation of the probability density function to be transformed is taken as the sum of $A_v$, $\sigma_s$, and the instrument uncertainty, $\sigma_i$. The approach taken here only requires the assumption of one distribution to describe the total uncertainty (discussed in Section 4.5.1). A support width of $b^{(0)} - a^{(0)} = 6(A_v + \sigma_s + \sigma_i)$, centered on the crisp value was selected, and the bounds of the intermediate membership levels were derived from (4.12). Once the input variables have been expressed in fuzzy form they are used to solve for the fuzzy drift velocity by (4.16), which is then integrated to derive the possible displacement of the drifting object.

### 4.3.3 Integration of Fuzzy Drift Velocities

The fuzzy set of all possible displacements can be derived by integrating all possible time series of drift velocity within a membership level that adhere to the maximum unresolved acceleration $A_a$. For each membership level below the crisp solution these time series can be found by first discretizing the membership level at the initial time into a number of starting points, $n_S$. From each starting point $n_p$ possible values at the next time step are determined based on the maximum unresolved acceleration and the bounds of the membership level. For each subsequent time step $n_p$ further possible values are determined from each possible value at the current time step, until the last time step is reached.

Letting the fuzzy drift velocity $P_{\frac{d\vec{x}}{dt}} = Z$ (from Equation 4.16) for notational simplicity, a possible value $Z_{k,t}^{(j)}$ (i.e. within membership level $j$, originating from starting point $k$, at time $t$) results in the following possible values at the next time step.

$$Z^{(j)}_{k,t+1}(n) = Z^{(j)}_{k,t} + \Delta Z^{(m)} + \Delta Z^{(j)}(n), \text{ for } n = 1, 2, ..., n_p \tag{4.19a}$$

$$\Delta Z^{(m)} = Z^{(m)}_{t+1} - Z^{(m)}_{t+0} \tag{4.19b}$$

$$\Delta Z^{(j)}(n) = 6A_a F^{(j)}(n)\Delta t \tag{4.19c}$$

$$F^{(j)}(n) = \frac{\frac{1}{2}(n_p + 1) - n}{\frac{1}{2}(n_p - 1)} \frac{b^{(j)}_{t+1} - a^{(j)}_{t+1}}{b^{(0)}_{t+1} - a^{(0)}_{t+1}} \tag{4.19d}$$

$$a^{(j)}_{t+1} \leq Z^{(j)}_{k,t+1} \leq b^{(j)}_{t+1} \tag{4.19e}$$

Here $\Delta t = dt_{model}$ is the model time step, $a^{(j)}_t$ and $b^{(j)}_t$ are the lower and upper bounds of membership level $j$ at time $t$, and $Z^{(m)}$ is the crisp time series of predicted drift velocities. At each time step, $Z^{(j)}_{k,t+1}$ is bounded by $a^{(j)}_{t+1}$ and $b^{(j)}_{t+1}$ (Equation 4.19e). All other symbols are as previously defined. A schematic of the process detailed in (4.19) is shown in Figure 4.4.

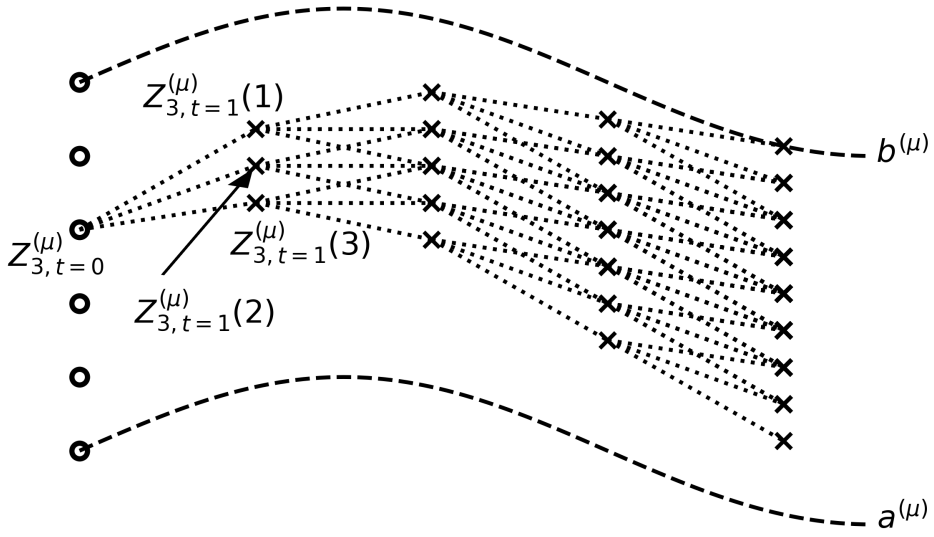


Figure 4.4: Illustration of the scheme to derive all possible crisp time series corresponding to a fuzzy time series of drift velocity. An example of the first four time steps starting from point $m = 3$ with $n_p = 3$ is shown here.

Integrating all unique crisp time series in the resulting set, using the forward Euler method (Equation 4.20), gives the set of all possible drifter displacements within this membership level $P^{(j)}_{\vec{x}}$ at time $T = N\Delta t$. This is written as follows

$$P_{\vec{x}}^{(j)} = \sum_{t=1}^{N-1} \frac{1}{2}\big(Z_{m,t+1}^{(j)}(n) + Z_{m,t}^{(j)}\big), \text{ for } n = 1, 2, ..., n_p, \text{ and } m = 1, 2, ...n_S \tag{4.20}$$

In this case, use of the forward Euler method does not impact the results negatively, due to the assumption that the forcing fields are free of spatial variability. This will change when extending this analysis to include data from numerical ocean and atmospheric models and the associated spatial interpolation.

### 4.3.4 Model Evaluation

#### Skill Score

Drift trajectory models are generally evaluated for their ability to predict a drifting objects position at a snapshot in time [85], as well as their ability to predict the complete path the object took to arrive at this position [76]. The latter is a more difficult test, as it requires that the timing, magnitude, and direction of forcing events are captured correctly, whereas the former only requires that the cumulative effect of forcing events up to the time the snapshot is taken is correct. For the purposes of this paper we only evaluate the model performance at snapshots in time.

Our evaluation of model skill is based on the approach outlined in [85] and also used in [96]. We adapt the approach for application to fuzzy results as follows. Model skill is calculated as a weighted sum across the membership levels, $j = 0, 1, ..., m$. Predictions with no skill receive a score of zero, while a perfect prediction receives a score of one.

$$Skill = \frac{\big(1+\beta\big)\big(1 - min\big(1, \frac{s}{d_{obs}}\big)\big) + \sum_{j=0}^{m-1}\big(f_j \frac{j}{m} + \beta\big) min\big(1, \frac{d_j}{X_{dj}}\big)}{\sum_{j=0}^{m}(\frac{j}{m} + \beta)} \tag{4.21}$$

The distances relevant to (4.21) are illustrated in Figure 4.5. The first term in the numerator is the crisp instantaneous skill score, as in [96]. The distance between the observed and crisp modelled drifter positions, $s$, is normalized by the distance the observed drifter has travelled, $d_{obs}$, since a prediction within a fixed $s$ should be considered more skillful as $d_{obs}$ increases [76]. This instantaneous skill score becomes zero if $s > d_{obs}$. $\beta$ is a positive constant (arbitrarily set to 0.1 here) to ensure some

skill is assigned to the predictions in the lowest membership level ($j = 0$).

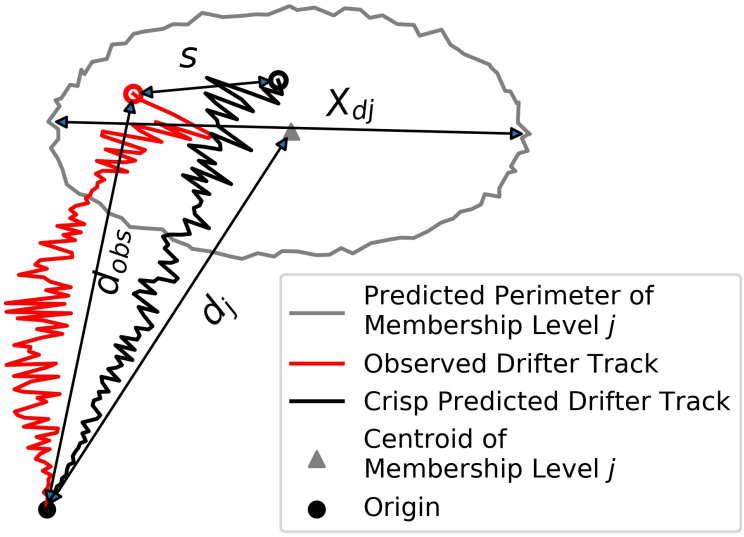


Figure 4.5: Illustration of instantaneous fuzzy skill score for trajectory evaluation. For clarity only a single membership level is shown.

The second term in the numerator is added to assign skill for each membership level $P_{\vec{x}}^{(j)}$ that contains the observed position. $f_j$ is set equal to $-\beta\frac{m}{j}$ if the observed drifter is not within membership level $j$, and equal to 1 otherwise. To ensure that overestimation of uncertainty negatively impacts skill scores, scores are reduced proportionally to the ratio of the distance across the membership level,$X_{dj}$, and the mean displacement of the predicted drifter positions within the membership level $d_j$. A time series of skill can be derived by solving (4.21) for each model time step.

### Leeway Model Comparison

To contextualize the performance of the proposed drift prediction scheme we compare our results to analogous predictions made using the commonly used leeway method [10, 12, 11]. In this method, the drift of an object is described as the sum of it's downwind, crosswind, and current-driven motion. The wind-driven component of the motion is object-specific and described using a set of nine coefficients, empirically determined for each object [2, 12], which describe linear fits of the motion to the downwind, left-crosswind, and right-crosswind components of the motion along with error terms around these fitted values. Uncertainty is considered by perturbing the

error terms using a Monte Carlo scheme [10]. For more details on the leeway method, interested readers are referred to [10] and other works referenced herein.

The trajectories of each drifter type were simulated using the formulation of the leeway model in the OpenDrift trajectory modelling suite [28], by choosing the leeway object class which most closely corresponds to each drifter type (given in Section 4.4). Simulations were performed using an ensemble of 5000 particles released in a 100 m radius around the deployment site of the drifters.

For quantitative comparison of the proposed fuzzy number - based method with the leeway method, the results of the leeway method were described as a fuzzy number to enable scoring using the algorithm presented in the previous subsection. To convert the distributions of the 5000 simulated particles into a fuzzy number, the relative particle density was calculated on a 1 km horizontal grid and this gridded density was then converted to a 2-dimensional fuzzy number by a slight modification of the algorithm presented by [67], which enables the construction of fuzzy numbers from histograms of observed scalar data. Here the binned data comprising the histogram is sorted from largest to smallest bin count, and a membership of 1.0 is assigned to the coordinates of the bin with the largest count. The membership at coordinates corresponding to bins with lower counts is determined as the sum of the relative observation density in all bins lower than that under consideration to satisfy the consistency principle. For a detailed description of this method of constructing fuzzy numbers please refer to [67]. To adapt this method to two dimensional data, the coordinates of the bin centers were expressed as a complex number $\vec{x}_{bin} = x_{bin} + iy_{bin}$ and fixed membership levels were determined by delineating the area encompassing all bins with membership greater than or equal to the chosen level. Scoring of the leeway simulations was conducted according to section 4.3.4 once the simulated particle distribution was fuzzified in this manner.

## 4.4 Forcing and Verification Data

The observational data used to test the methods presented in Sections 4.2 and 4.3 was collected near Ocean Station Papa (OSP) in the northeast Pacific during the period Feb 20 - Mar 3, 2015. OSP (50.1°N, 144.9°W) has been the site of meteorological and oceanic observations since the early 1940's, and currently hosts a moored observation platform operated by the Ocean Climate Stations Office of NOAA/PMEL [48]. Detailed descriptions of the circulation and water properties at OSP are given

in publications such as [105] and [27]. We make use of measurements of currents and winds recorded at the moored observation platform during a release of 44 ocean surface drifters nearby. The drifter tracks, as well as time series of the winds and currents measured at OSP, are shown in Figure 4.6.

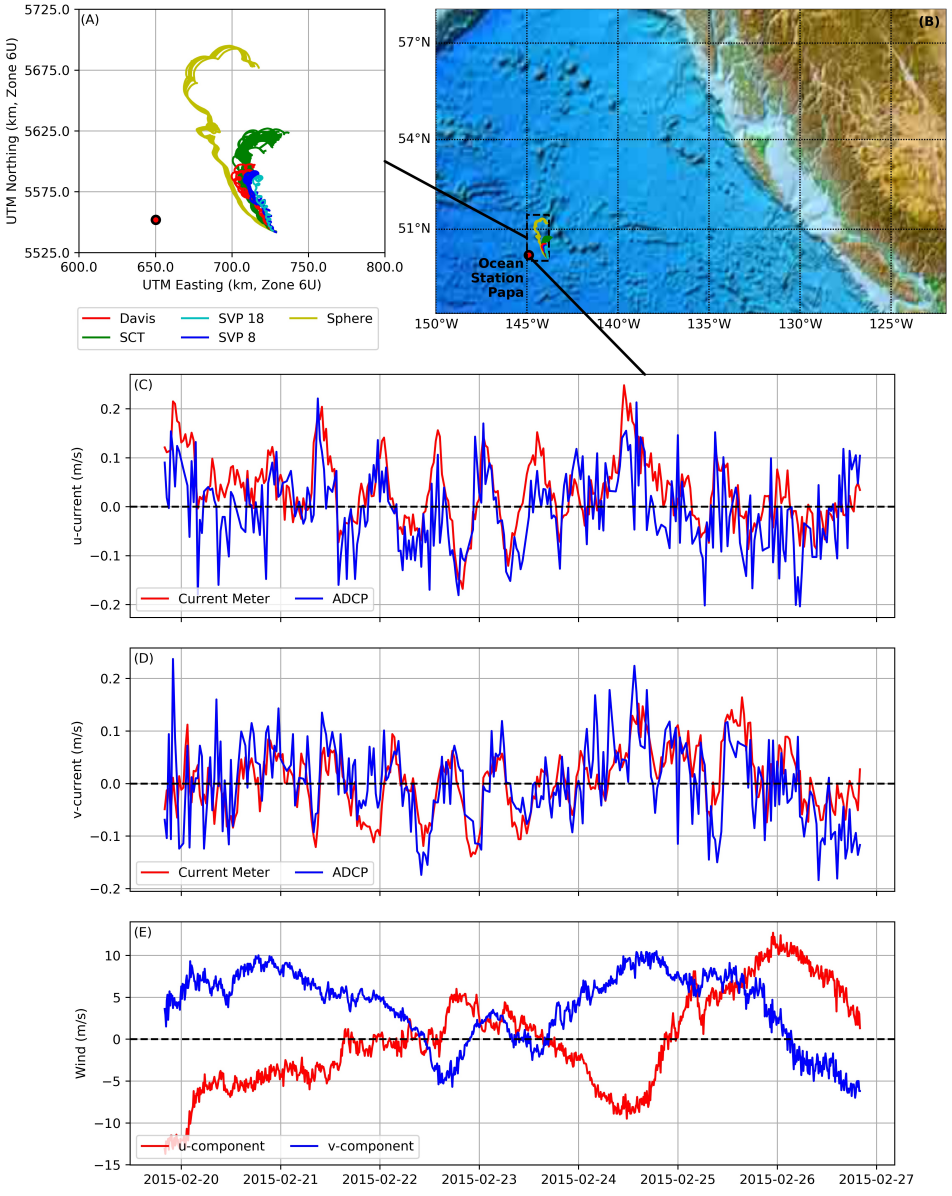


Figure 4.6: Tracks of drifters for seven days after deployment near Ocean Station Papa (panels A & B), and time series of u,v components of currents (panels C & D) and wind (panel E) during the deployment.

Near-surface currents at OSP are recorded by a single-point 2 MHz Nortek Aquadopp

current meter moored in a taut-line configuration at 15 m depth as well as a downward looking 300 kHz RDI Sentinel ADCP at 2 m below the water surface. Both instruments sample data over a period of 2 minutes. The current meter reports data at 10 min intervals, while ADCP profiles are reported at 30 min intervals [91]. The reported current meter accuracy is $\pm$(1% of velocity + 0.5 cms$^{-1}$), while the accuracy of the ADCP is reported as $\pm$ 0.5 cms$^{-1}$. Both sensors have a resolution of 0.1 cms$^{-1}$. Winds are recorded 4 m above the water surface by a Gill Windsonic anemometer and averaged to 10 minute intervals, with a reported accuracy of $\pm$ 2% (3%) of the speed (direction) reading, and a resolution of 0.01 ms$^{-1}$ (1°) [92].

The current meter record at 15 m depth (panels C & D of Figure 4.6) initially shows northeastward flow in response to northwesterly winds (panel E of Figure 4.6), consistent with descriptions of wind-driven flow in the upper ocean dating back to [40]. As winds slacken between midday on Feb 21 and the evening of Feb 23, signals consistent with inertial oscillations appear in the current data.

Visual comparison of the current meter record with data from the closest ADCP measurement (centered at 16 m depth) during the first seven days of the experiment (panels C & D of Figure 4.6) reveals that the current meter indicates higher amplitude of low frequency fluctuations in the flow, and shows less spiking than the ADCP. A more detailed comparison (included below, in Section 4.4.2) reveals further differences between the two instruments. The current meter records a northeastward mean flow of $\sim$ 3-4 cms$^{-1}$, while the mean flow recorded by the ADCP is much smaller and in the opposite direction. This appears to be a systematic discrepancy between the two instruments. Differences between the two instruments could be due to interference from the point current meter on the beams of the ADCP (Nathan Anderson, PMEL, personal communication, Sep 2021) or due to the differences in operating frequency and sampled volume of water (Richard Thomson, DFO, personal communication, Sep 2021). Drifter trajectories are computed and assessed using the data from both instruments, in order to test which instrument provides more relevant information.

### 4.4.1 Drifter Deployment

On Feb 19, 2015 at approximately 20:30H UTC, 44 ocean surface drifters of five different geometries were deployed approximately 75 km east of OSP. These included the following: 8 Sphere drifters which float mostly above the water surface; 28 Surface

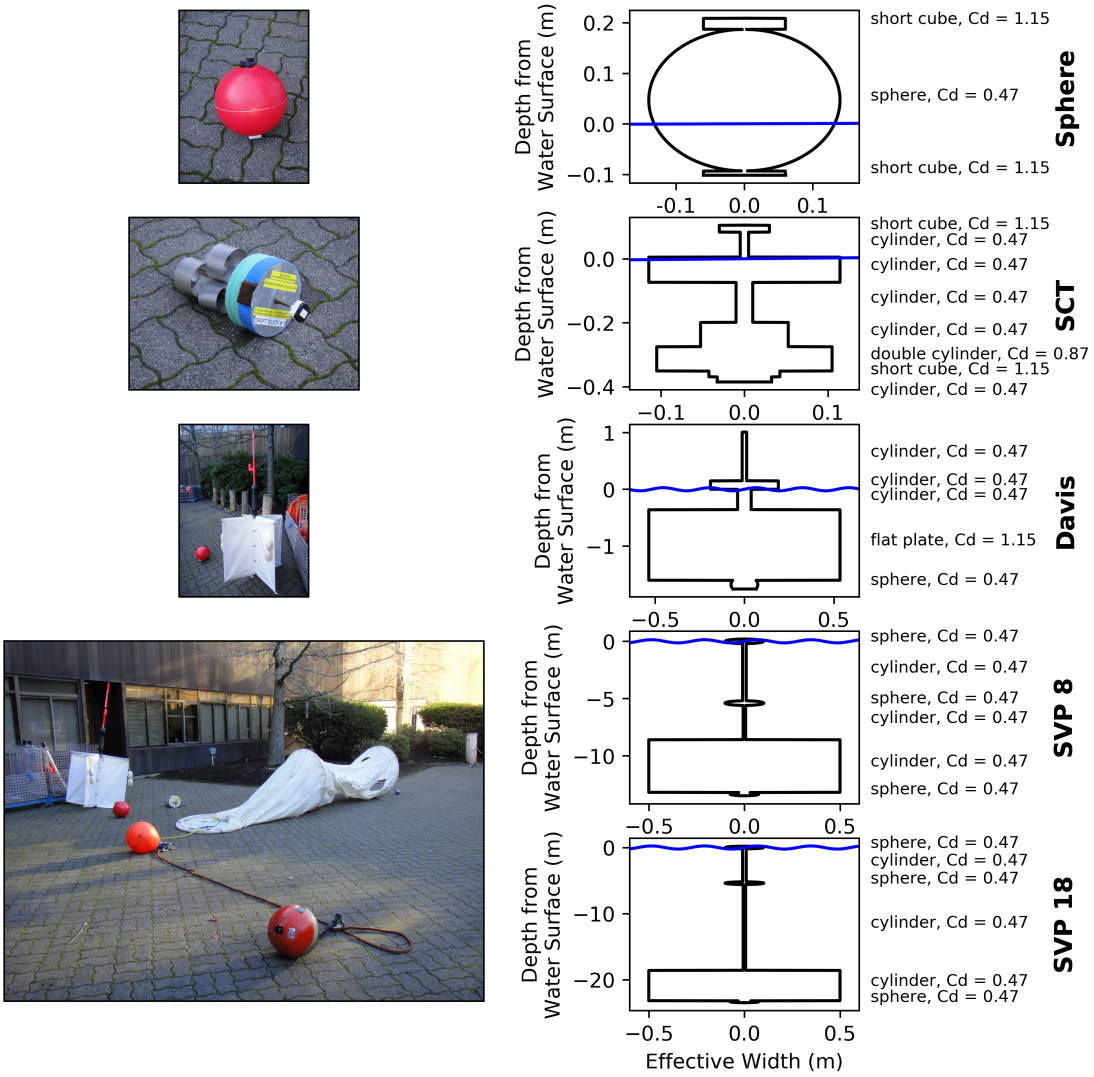


Figure 4.7: The drifters deployed for this study (left), and their idealized geometry, including approximate drag coefficients (right). The SVP 8 (bottom-left photograph) and SVP 18 drifters (no photograph shown) are identical except for the length of the tether.

Circulation Tracker (SCT) [97] drifters which float in the upper 37.5 cm of the water column and have been shown to reasonably reproduce oil sheening [7]; 3 Davis drifters [34] which float in the upper 1.5 m of the water column and are similar to the SLDMB drifters commonly used for search and rescue applications; and 5 drifters based on the Surface Velocity Program (SVP) design [112, 94], 4 with the 4.6 m holey-sock drogue starting at ∼8 m depth (hereafter SVP 8) and one starting at ∼18 m (hereafter SVP

18). SVP drifters are generally used to map near-surface ocean circulation.

One of the SCT drifters failed shortly after deployment. Pictures of the drifters and their idealized geometry and drag coefficients are shown in Figure 4.7. Most drifters transmitted data until Mar 3, 2015. As short-term drift forecasts are of primary interest in emergency response scenarios, we limit our analysis to the first seven days of the deployment in order to maintain reasonable brevity of the presented results. During this time drifters were within 55 - 150 km of OSP (Figure 4.8). To simulate the trajectories of these drifter types using the leeway method, the following object types were chosen as proxies: Sphere $->$ bait box (lightly loaded); SCT $->$ WW2 mine; Davis $->$ oil drum; SVP (8m and 18m) $->$ person-in-water floating vertically. The corresponding leeway coefficients are reproduced in Table 4.1.

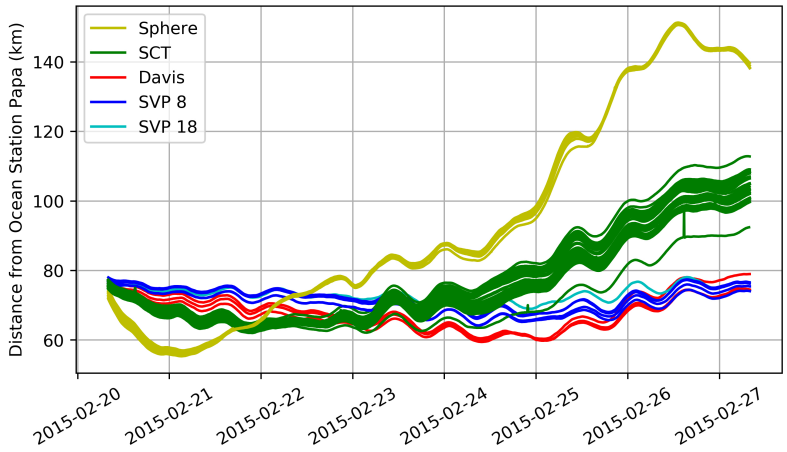


Figure 4.8: Distance of drifters from Ocean Station Papa during the first seven days of the deployment.

All drifters followed a mean northeastward trajectory during the first 5 days of their deployment (panel A of Figure 4.6), consistent with the mean wind direction during this time, with a change to westward drift coincident with a change in the wind direction. The lengths of the trajectories are inversely related to drogue depth of the drifter. The Sphere drifter trajectories are approximately twice as long as those of the SCT drifters, which are in turn longer than the Davis, SVP 8, and SVP 18 drifters. The latter three drifter types, which are drogued deeper in the water column, exhibit clockwise looping which is likely due to inertial oscillations like those noted in the current meter record.

Table 4.1: Leeway coefficients for proxy object types for drifters used in this study (see Equation 2.13). Reproduced from the OpenDrift source code [28].

| | **Bait Box, lightly loaded** | **WW2 Mine** | **Oil Drum** | **Person-In-Water, vertical** |
|---|---|---|---|---|
| $a_D$ (%) | 2.53 | 1.07 | 0.75 | 0.48 |
| $b_D$ (%) | 9.01 | 4.47 | 2.66 | 0.00 |
| $\epsilon_D$ (%) | 3.05 | 6.55 | 2.83 | 8.30 |
| $a_{C^+}$ (%) | 1.09 | 0.41 | 0.48 | 0.15 |
| $b_{C^+}$ (%) | -2.76 | 1.15 | 2.88 | 0.00 |
| $\epsilon_{C^+}$ (%) | 4.14 | 4.13 | 3.92 | 6.70 |
| $a_{C^-}$ (%) | -1.09 | -0.41 | -0.45 | -0.15 |
| $b_{C^-}$ (%) | 2.76 | -1.15 | -1.46 | 0.00 |
| $\epsilon_{C^-}$ (%) | 4.14 | 4.13 | 4.59 | 6.70 |

### 4.4.2 Comparison of Current Meter and ADCP at Ocean Station Papa

To investigate potential differences in the measurements recorded by the single point current meter and the ADCP at Ocean Station Papa, and to check for systematic discrepancies in reported current direction between the two instruments, we plot the vector correlation between the current meter record and the ADCP record at all available depths, following [71]. The vector correlation is expressed as,

$$R = \frac{< w^{`*}_{CM} w^{`}_{ADCP} >}{\sqrt{< w^{`*}_{CM} w^{`}_{CM} >< w^{`*}_{ADCP} w^{`}_{ADCP} >}} \tag{4.22}$$

where $w^{`}_{CM/ADCP}$ are the detrended recorded velocities expressed as a complex number $w = u + iv$. The superscript $*$ denotes the complex conjugate, and $<>$ denotes an ensemble average. From $R$ we can obtain the magnitude of the correlation, $|R|$, as well as the mean angle between the vectors, $Phase(R)$. These are plotted in Figure 4.9 for the raw current meter record as well as the current meter smoothed to 30 minute intervals using a boxcar filter.

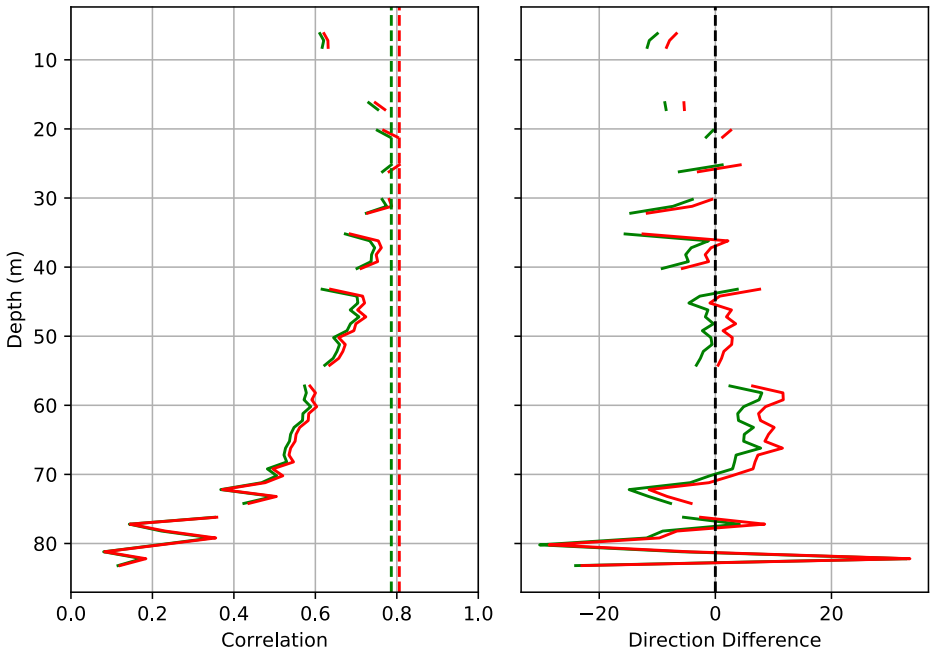


Figure 4.9: Magnitude (left) and phase (in degrees, right) of the vector correlation between the ADCP record at all available depths and the current meter record at 15 m. Red lines indicate correlation with the raw current meter record, while green lines indicate correlations with the smoothed current meter record. Dashed lines in the left-hand panel indicate the maximum correlation for each case.

Clearly the measurements are well correlated throughout the measurement depth of the ADCP (up to ∼70 m) and peaking near the depth of the current meter, however relative flow direction changes throughout the water column by ± ∼10° as would be expected in wind-driven Ekman-type flow. Smoothing the current meter data does not change these results appreciably. Comparing the timeseries of the current meter record and the 6 m ADCP bin during the first seven days of the drifter deployment visually confirms this strong correlation (Figure 4.10).

However the ADCP data does not indicate significant persistent vertical structure in the magnitude of the currents, as indicated by the time-averaged profile of the ADCP data over the course of the simulation. The proposed model for vertical shear (Equation 4.7) shows strong surface intensification. The shear below the surface is relatively consistent with the ADCP, though slightly higher in the northward direction by ∼2 cm/s (Figure 4.11).

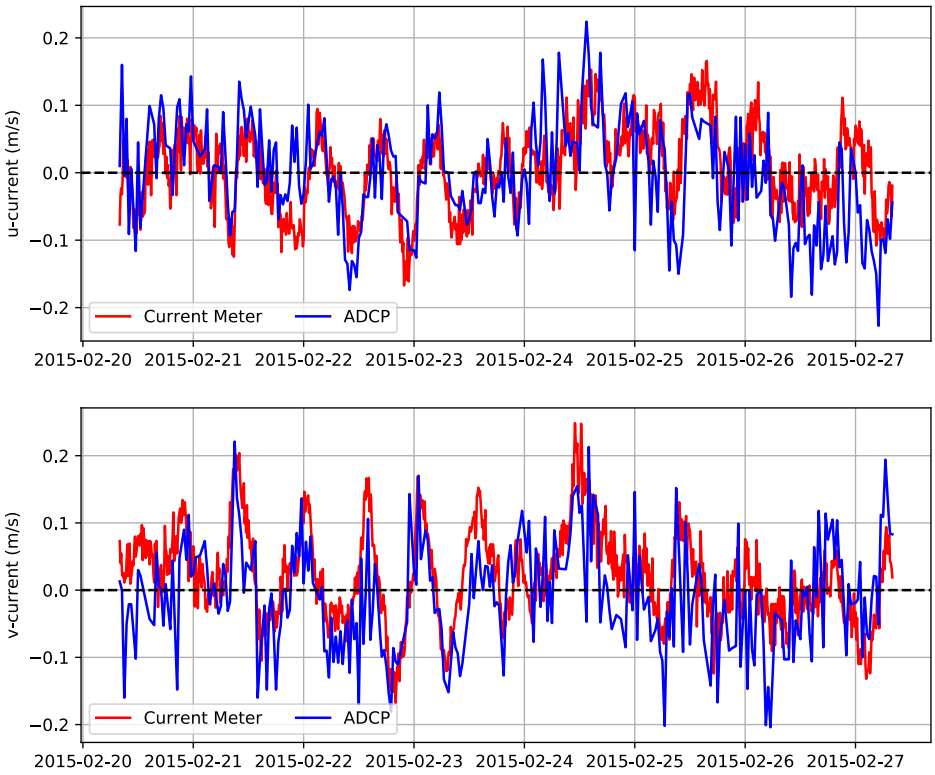


Figure 4.10: Timeseries of currents recorded by the current meter at 15 m depth (red) and the ADCP at 6 m depth (blue). $u$ components (E-W) are shown on top and $v$ components (N-S) below.

The correlation between the current meter and the ADCP data also tracks the depth profile of good data returned relatively well (Figure 4.12). Data return increases from ∼85% in the bins closest to the surface to ∼100% by about 35 m depth. The largest number of bins with no good data are also clustered near the surface. Therefore it is not completely clear whether the ADCP captures an Ekman-like flow near the ocean surface or whether this is an artifact of comparatively sparse data return and significant spiking in the near-surface ADCP data.

In general, the ADCP record shows significantly more high-frequency spiking than the current meter record. The low-frequency signals match well, however the ADCP appears to report a lower amplitude of low-frequency signals at some times. This is similar to the comparison between the current meter and the 16 m bin of the ADCP, shown in Figure 1 of the main paper. To further investigate these differences we plot the distribution of the difference in measurements that are co-located in time in $u, v$ space (Figures 4.13 & 4.14).

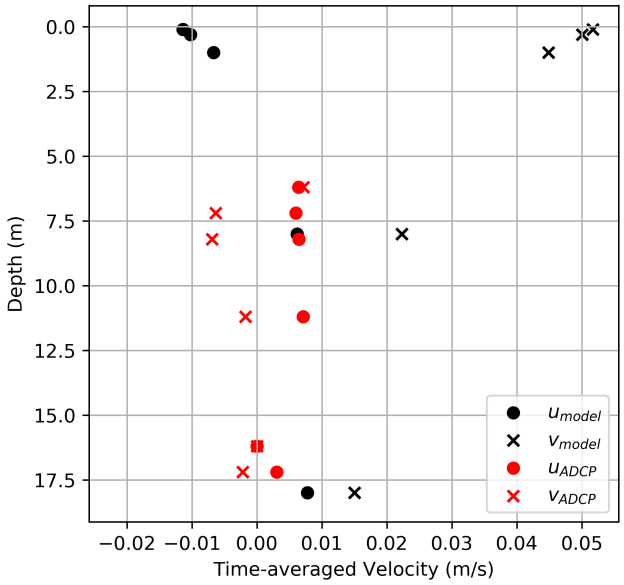


Figure 4.11: Time-averaged profile of velocity components from the ADCP, and the model for current shear (Equation 4.7) over the course of the simulation.

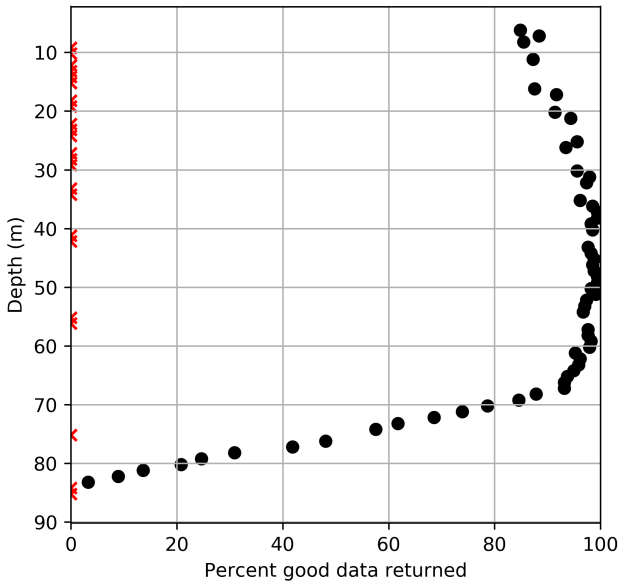


Figure 4.12: Depth profile of data return for the ADCP after quality control flags are applied. Bins with no good data are marked by red x's on the y-axis.

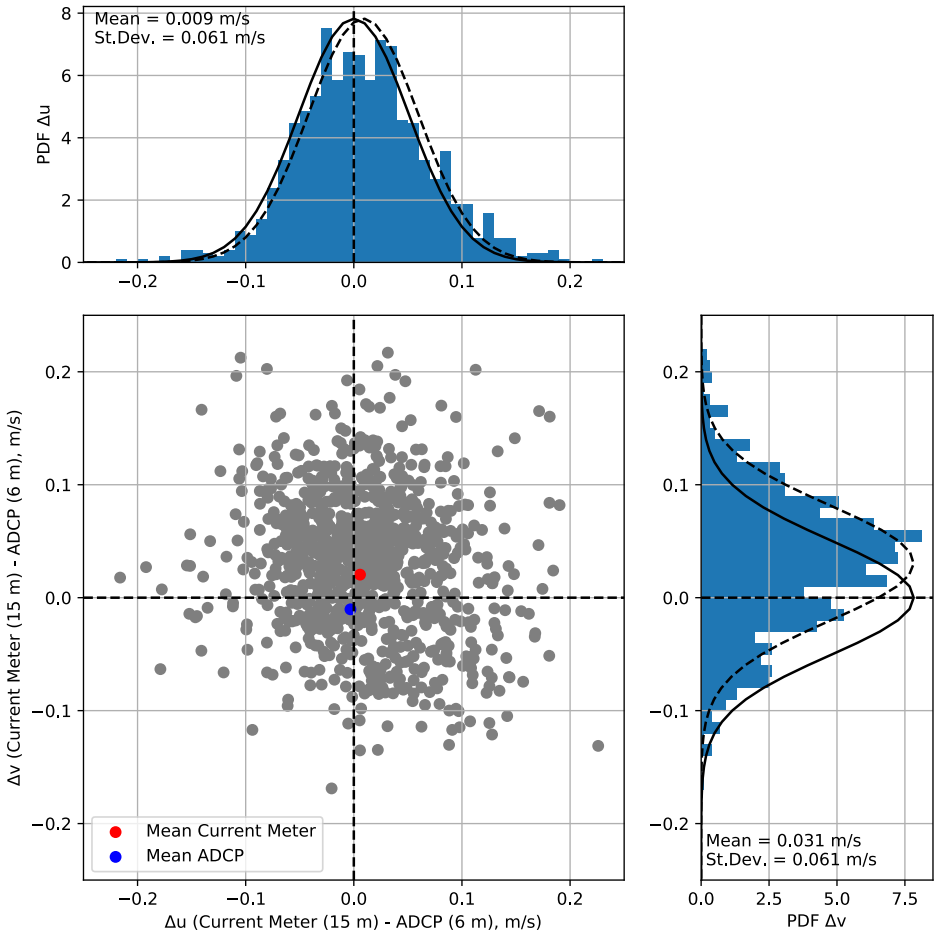


Figure 4.13: Difference between current meter at 15 m and ADCP measurements at 6 m, in u,v space. Distributions of component differences are shown in the side panels, with a Gaussian distribution superimposed. The solid line indicates a zero-mean distribution while the dotted line indicates a distribution that is shifted to the bias between the two data sources. The time-averaged currents recorded by the current meter and the ADCP are shown by the colored markers.

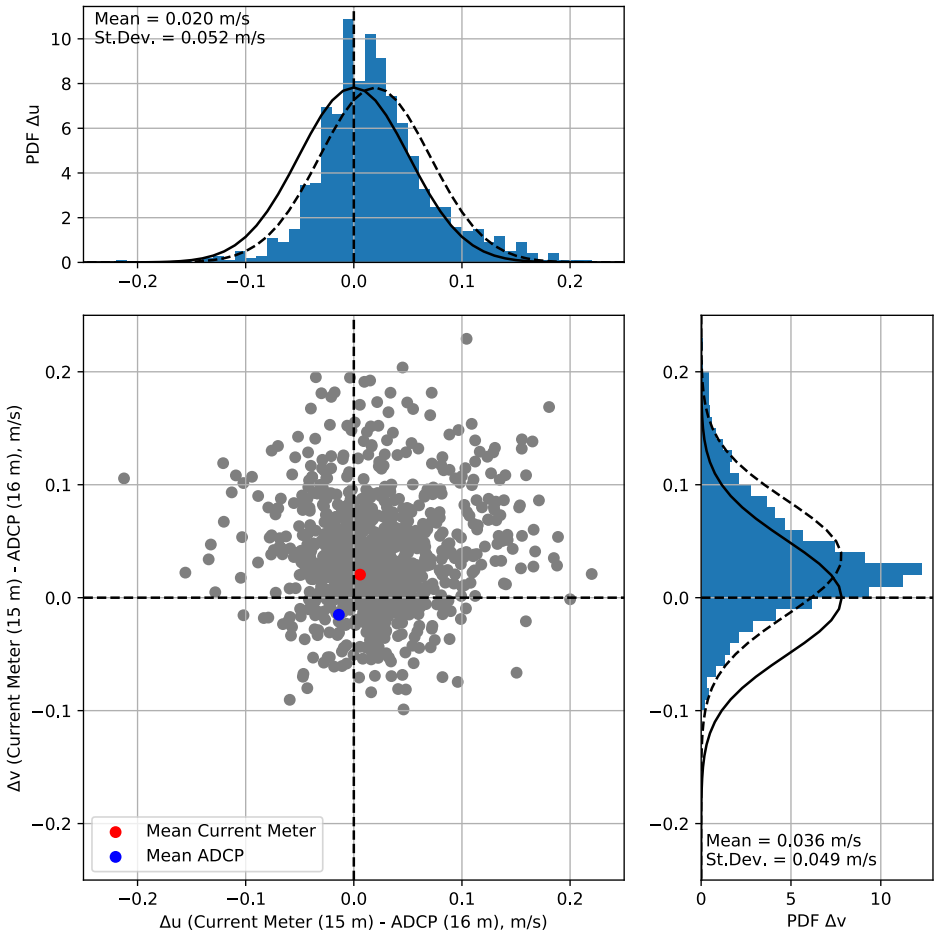


Figure 4.14: Same as Figure 4.13, but for ADCP bin at 16m.

These distributions indicate that there is a significant ($\sim$3.2 cms$^{-1}$) bias between the two instruments, with mean current meter results directed more northeasterly than ADCP results ($E[u_{CM} - u_{ADCP}] > 0$, and $E[v_{CM} - v_{ADCP}] > 0$). This suggests that the mean flow recorded by the current meter is approximately opposite to the much smaller mean flow recorded by the ADCP at a shallower depth. The differences approximately follow a Gaussian distribution with this mean, with the standard deviation being indicative of energy at time scales that are not resolved by the ADCP as well as the spiking noted earlier. The difference noted here is consistent with the difference between the shear recorded by the ADCP and the values estimated by Equation 9, as noted above.

Plotting a similar diagram comparing the current meter record with the closest ADCP bin (16 m) shows that the northwesterly bias of the current meter with respect

to the ADCP persists at this depth and increases in magnitude to $\sim 4.1$ cms$^{-1}$, which suggests that this is a systematic discrepancy between the two instruments rather than a feature of the near-surface current dynamics. The distribution of discrepancies at this depth is significantly more peaked than a Gaussian distribution, which is again consistent with periodic spiking.

## 4.5 Results

### 4.5.1 Fuzzy Forcing

Converting the current and wind record into fuzzy number equivalents using the procedures described in Section 4.3.2 shows that the uncertainty in these measurements is appreciable once aggregated. To begin, we fit power laws to the tail of the wind and current meter spectra using log-transformed linear regression, between frequencies $\frac{1}{2(7200sec)}$ and the Nyquist frequency, $f_N$. The low-frequency threshold is chosen by visual inspection of the energy spectra (Figure 4.15). The parameters to the regression $Af^B$ as well as the amplitude of the unresolved velocities and accelerations according to (4.17) and (4.18) are given in Table 4.2 for a model timestep $dt_{model} =$ 300 seconds. The amplitude of the unresolved energy $A_v$ is approximately twice the instrument uncertainty for the anemometer and the current meter but approximately six times larger for the ADCP. This is at least in part due to the longer output interval of the ADCP but also due to the shallower spectral slope that suggests additional noise at high frequencies, as was noted from visual inspection of the time series in Section 4.4.

Table 4.2: Amplitude of energy in unresolved timescales and associated power law coefficients.

| **Data Source** | A | B | $A_v$ (m/s) | $A_a dt_{model}$ (m/s) |
|---|---|---|---|---|
| **Anemometer** | 0.19 | -0.97 | 0.46 | 0.17 |
| **Current Meter** | 1.77x$10^{-3}$ | -0.73 | 0.020 | 0.0074 |
| **ADCP** | 0.084 | -0.49 | 0.088 | 0.026 |

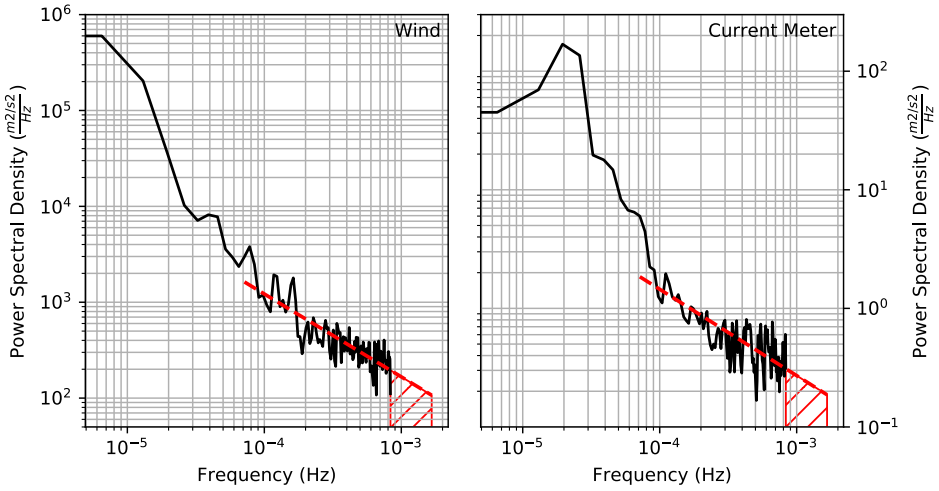


Figure 4.15: Power spectra of measured wind (left) and current (right). Power law fit to the high-frequency tail is shown in dashed red and the area corresponding to the energy in unresolved timescales is hatched in red.

Uncertainty due to spatial variability is estimated by examining the current fields from the GLORYS12 ocean re-analysis. GLORYS12 is a state-of-the art hindcast numerical simulation of ocean conditions from 1993 - present, with data assimilation of satellite altimetry and temperature measurements as well as *in situ* profiles of temperature and salinity [73]. The modelled daily-mean currents at 15 m depth suggest significant spatial structure with maximum fluctuations in current amplitude of ∼0.2 m/s over the duration of the experiment (see Figure 4.16). Comparing the modelled currents at OSP to daily averages of the measurements from the current meter and ADCP suggests that at OSP the modelled currents are biased southeasterly by ∼0.045 m/s with respect to the current meter, and easterly by ∼0.05 m/s with respect to the ADCP. RMS errors of the modelled velocity components are ∼0.04 - 0.06 m/s. A comparison of the modelled currents at the drifter locations with the modelled current at OSP also suggests that currents at OSP are directed more southeasterly than at the drifter locations, with a bias of ∼0.075 m/s and RMS errors of 0.01 - 0.02 m/s for all drifters except the Sphere drifters which travel further and therefore sample more variability (see Figures 4.8 & 4.17). However there is considerable uncertainty about the accuracy of the mean difference in the modelled currents between the drifter location and OSP given that this difference corresponds well with the identified bias in the currents at OSP. A plausible explanation is that the features shown in Figure 4.16 may be slightly misaligned in the model. For the

purpose of this paper, we approximate uncertainty due to spatial variability by taking $\sigma_s = 0.015$ m/s as the standard deviation of the differences between the modelled currents at OSP and the drifter locations. As the modelled currents leave some doubt about the relevance of the currents at OSP to the drifter locations, trajectories are computed with and without forcing from the measured currents as is later discussed in Section 4.5.2. Further assessment of the modelled surface dynamics in GLORYS12 can be found in [73]. As noted previously, the effect of uncertainty in modelled ocean currents on trajectory prediction will be further investigated in future work.

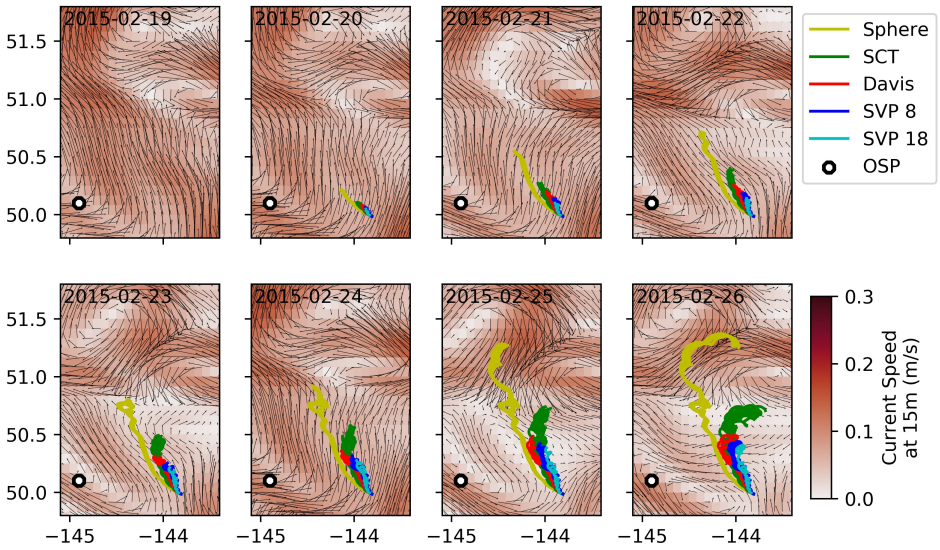


Figure 4.16: Daily-mean currents in the study area at 15 m depth from the GLORYS12 numerical re-analysis for Feb 19 - 26, 2015 [73].

Once uncertainties have been quantified, the current and wind data are converted to a fuzzy number using the probability-possibility conversion in (4.12). We expect that the uncertainty due to unresolved time scales will follow a distribution that is similar to that obtained by simulating a red noise process with an $f^B$ spectral slope and variance corresponding to the sum of $A_v$ and $\sigma_i$ (shown for the current meter record in Figure 4.18). This distribution is better represented by a Laplace distribution than by a Gaussian distribution, though both overestimate the tails. The random instrument uncertainty is more likely to follow a Gaussian distribution. Since the unresolved energy in the current data, especially for the ADCP, is significantly larger than the instrument uncertainty we convert it to fuzzy numbers using a Laplace distribution, while a Gaussian distribution is chosen for the wind record. The effect of

the choice of distribution should be further investigated, however this is beyond the scope of the current study. Applying (4.12) with the above distributions, $A_v$ from Table 4.2, $\sigma_s$ from above, $\sigma_i$ from Section 4.4, and four equally spaced membership levels on the interval $[0, 1]$, i.e. $m = 3$, yields the fuzzy time series shown in panels A - D of Figure 4.19. The number of membership levels is limited by the computational expense of the current shear calculation. This may be mitigated by employing alternative parameterizations for current shear, however a detailed comparison of possible parameterizations is beyond the scope of the current work.

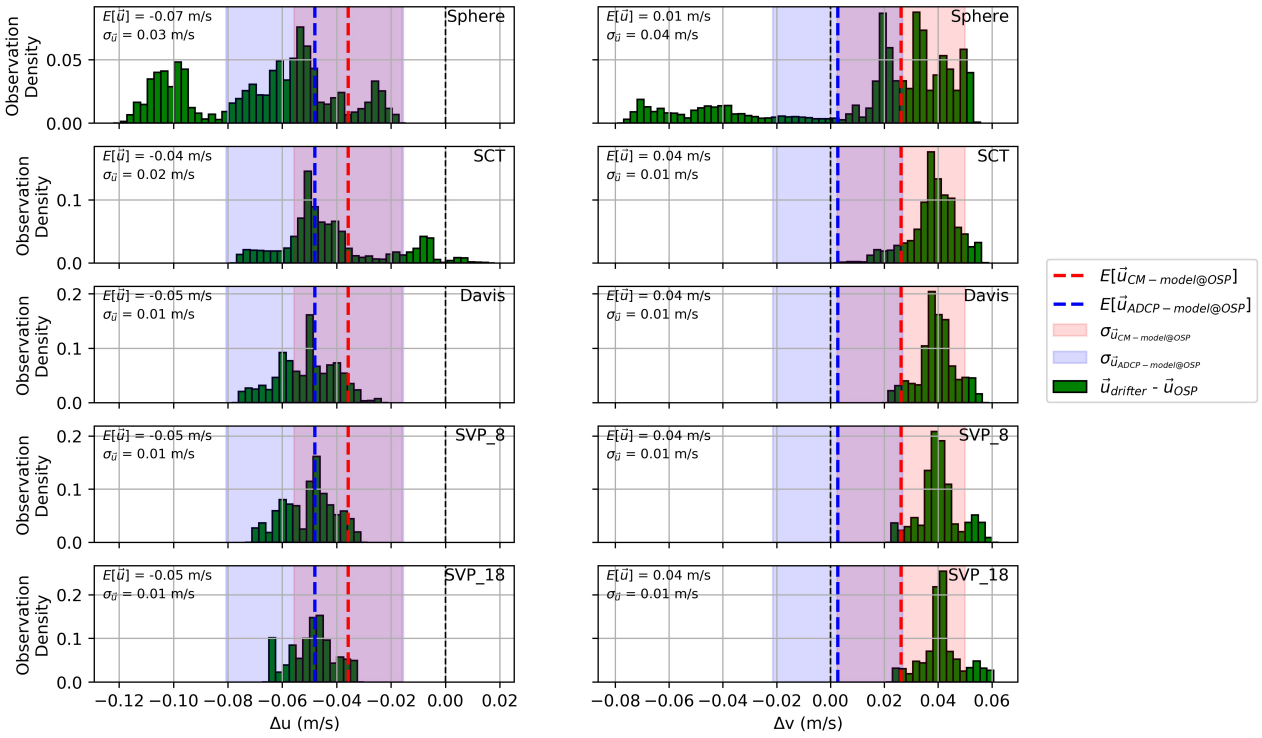


Figure 4.17: Differences in the daily-mean currents from GLORYS12 between OSP and the locations of the drifters. Each row represents values for a drifter type, with $u$-components (E-W, east positive) on the left and $v$ components (N-S, north positive) on the right. The differences between modelled and observed currents are shown for the observations from the current meter (red) and the ADCP (blue). Mean differences are shown by the dashed line while the shading represents one standard deviation.

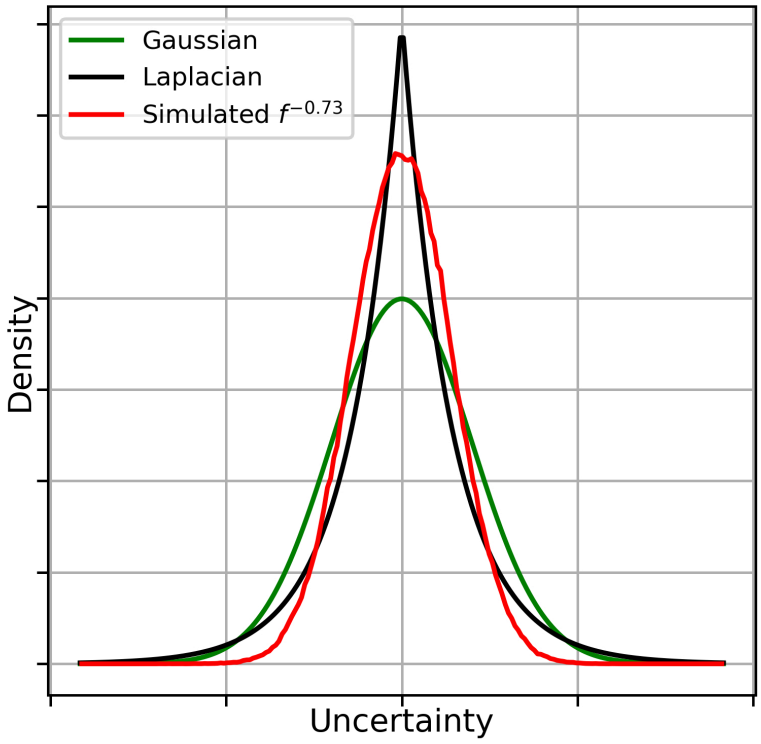


Figure 4.18: Generalized comparison of Gaussian and Laplace distributions with a simulated distribution from a red noise process with $f^{-\frac{3}{4}}$.

The fuzzified winds and currents exhibit uncertainties that at times exceed 5 m/s for winds and 0.2 m/s for currents at the lowest membership level, during times when winds and currents are strong. The influence of the relative instrument uncertainty on overall uncertainty is clearly seen in the reduced width of the fuzzy numbers when wind speeds are low, between midday on Feb 22 and midday on Feb 23 (Figure 4.19).

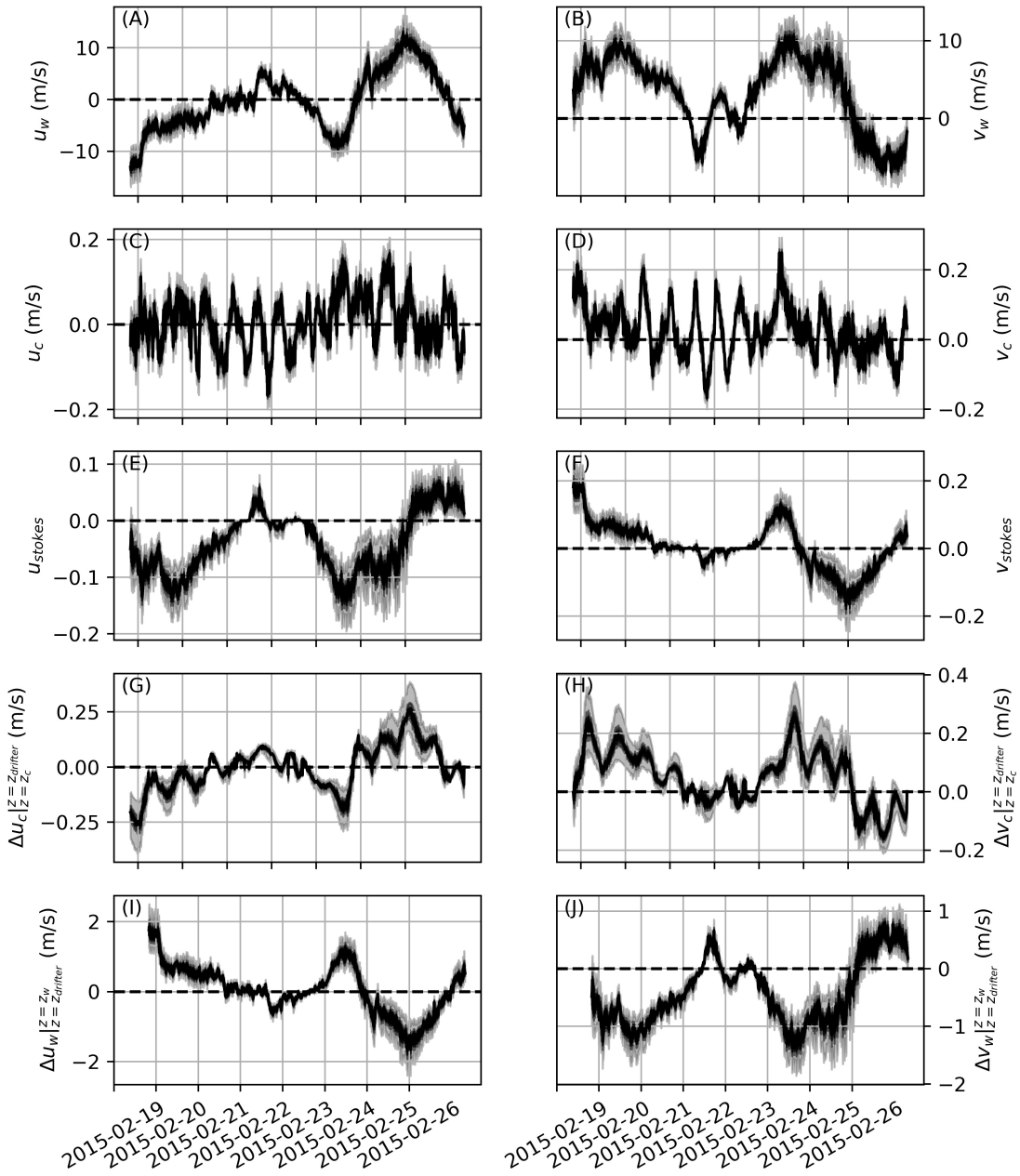


Figure 4.19: Components of fuzzy wind (panels A & B), current (panels C & D), Stokes drift (panels E & F), current shear (panels G & H), and wind shear (panels I & J) for the first 7 days of the drifter deployment. The crisp value (measurement) is shown in black and the three fuzzy membership levels are indicated in grayscale. For clarity in panels E - J only values for SCT drifters with the velocity profile applied by the 'centroid' scheme are shown.

Transforming the fuzzy winds according to (4.13) allows us to calculate fuzzy values of Stokes drift, current shear, and wind shear according to the procedures outlined in Sections 4.2.1 and 4.2.2. The time series of these fuzzy vectors are shown in panels E - J of Figure 4.19. Here only values for SCT drifters derived using the centroid method (4.5a) are shown for clarity. These variables were calculated for each drifter separately by each of the three methods of applying the shear profile given in (4.5). Stokes drift and current shear on the near-surface SCT drifters are strongly positively correlated with the wind velocity, though both diminish significantly when wind speeds are below ∼5 m/s. During times of strong winds both Stokes drift and current shear are comparable to, or larger than, the measured currents. These terms are therefore expected to have first order impacts on the drift trajectory prediction, consistent with the findings of [122]. During strong winds the magnitude of the uncertainty (i.e. the width of the lowest membership level) in the estimated Stokes drift and current shear is comparable to the magnitude of the estimate itself.

Wind shear appears to be a minor factor compared to Stokes drift and current shear. As expected, $\Delta\vec{u}_w|_{z=z_{drifterW}}^{z=z_w}$ is inversely correlated with wind velocity, i.e. it reduces the effective wind velocity in (4.16). The magnitude of wind shear is at most ∼20% of the wind speed, and subject to similar relative uncertainties as Stokes drift and current shear.

### 4.5.2 Optimal Forcing Combinations

Drift trajectories were computed for all combinations of forcing terms on the right-hand side of (4.16), methods of applying sheared forcing as per (4.5), and sources of current data (current meter/ADCP). Trajectories were then scored at each time step using (4.21) to determine the optimal combination of parameters by identifying the combination giving the highest skill score at each time step. We consider a forcing term to be part of the optimal solution if it is present in the solution with the highest skill score, for at least 50% of time steps. Wind shear is only considered during times when wind drag is an important forcing term, however current shear may be considered independently of measured currents. The optimal combination of forcing terms as a function of time is summarized for each drifter in Figure 4.20. The percentage of time each of the three methods of applying profile shear (4.5a - c) was most suitable is summarized in Figure 4.21. The optimal parameters for each drifter, and the trajectories predicted using these, are shown in Figures 4.22 - 4.26.

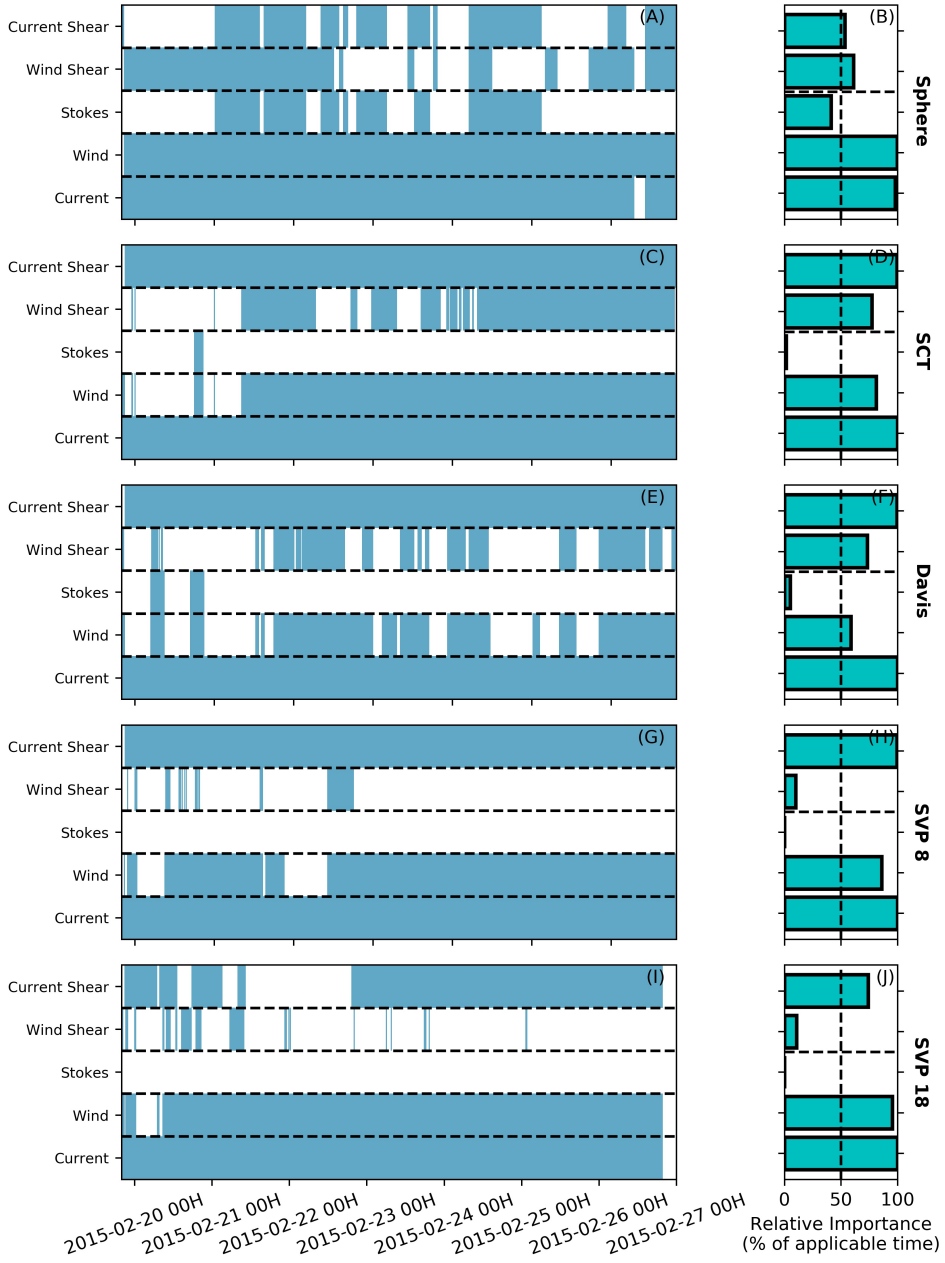


Figure 4.20: Summary of optimal forcing terms for each drifter as a function of time. In the left-hand panels (A, C, E, G, & I), timesteps where a forcing term contributed to the solution achieving the highest skill score are shaded in blue for each drifter type. In the right hand panels (B, D, F, H, & J) the percentage of applicable time each forcing term contributed to the optimal solution is summarized.

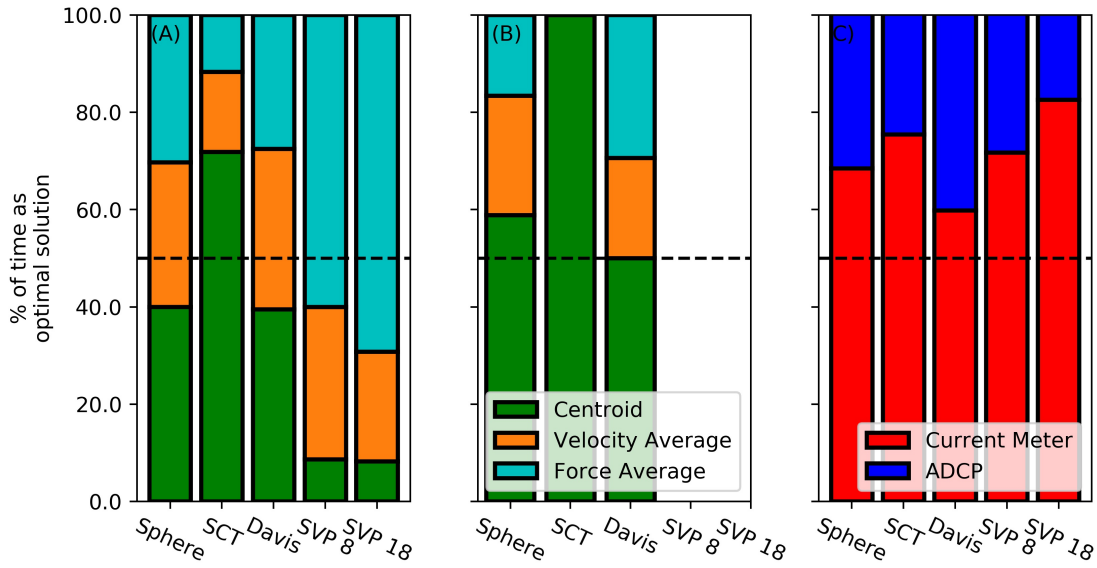


Figure 4.21: Percentage of time current shear (Panel A) and Stokes drift (Panel B) profiles applied via the centroid and area-weighted average of velocity or force produce the optimal solution for a the trajectories of each of the five drifters. No results are shown for Stokes drift on the SVP drifters, as the analysis indicates that Stokes drift does not contribute to the optimal trajectory prediction at any time. Panel C shows the percentage of time the currents measured by the current meter (red) and the ADCP (blue) contribute to the optimal solution.

Current, wind, and current shear all play an important role in predicting the trajectories of the drifters, and wind shear is an important term for the near-surface drifters (Sphere, SCT, and Davis) (Figure 4.20). This suggests that the importance of wind shear increases with the magnitude of the wind drag term. Since forcing from measured currents is consistently part of the optimal solution, we conclude that currents measured at OSP are relevant to the drifter location, despite the separation between these locations. In 60 - 80% of instances, forcing with the currents measured by the current meter produced a better solution than forcing with currents measured by the ADCP (panel C of Figure 4.21). This is likely due to the higher reporting frequency (every 10 min for current meter compared to every 30 min for ADCP), as well as the possible high frequency noise in the ADCP record discussed earlier. Stokes drift is not part of the optimal solution for any drifter type. It is more frequently of importance for the near-surface drifters however, and not important at any time for the SVP drifters. A possible explanation is that Stokes drift may lag behind wind

forcing as the wave field responds to changes in the wind. In general, the potential effects of Stokes drift should be further investigated by a sensitivity analysis to both different methods of calculating Stokes as well as time-lagged methods of applying this forcing. The optimal forcing terms are often simultaneously applicable in the latter half of the simulation period, however this is less likely to be the case during the first half of the simulation period. A possible explanation for this is that the lower winds during the first half of the simulation period result in more small scale current variability, such as inertial oscillations, which are not captured in the measurements at OSP.

The deterministically derived wind drag factor, $\alpha$, from (4.2) produces appropriate results. Values of $\alpha$ are 5.43%, 1.00%, 0.53%, 0.28%, and 0.27% for the sphere, SCT, Davis, and SVP 8 and 18 drifters, respectively. The values calculated for the SVP drifters agree well with the slippage estimate of 0.2% of the wind speed given by [112]. To assess the sensitivity of the results to the wind drag factor, the changes in mean skill score for trajectory predictions made using values ranging from $0.5\alpha$ to $1.5\alpha$ are given in Table 4.3. For the near-surface drifters any increase in the wind drag factor degrades the predictions, while decreases in the wind drag factor can improve the prediction. This improvement is minor (maximum 5.5%) for the SCT and Davis drifters, but can be significant (up to 78% for a 25% reduction of $\alpha$) for the Sphere drifters. This suggests that perhaps wind shear is underestimated here, which may be expected as winds are calculated at a 1 m reference height while the bulk of the exposed drifter parts are within 0.2 m of the water surface. It is also possible that some dynamics associated with objects that are primarily above the water surface are not captured here, as suggested by the significant differences in the Sphere drifter results. Altering the wind drag factor for the SVP drifter generally resulted in a negligible change to the results. Decreasing $\alpha$ generally resulted in a small (maximum 3.1%) decrease in skill while increasing $\alpha$ resulted in a similarly small (maximum 3.4%) increase in skill.

Table 4.3: Percent change in mean trajectory prediction skill for given adjustments to deterministic wind drag factor, $\alpha$.

| | **Prediction skill change from deterministic $\alpha$-value (%)** | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Drifter Type** | -50% | -25% | -10% | -5% | +5% | +10% | +25% | +50% |
| **Sphere** | +17.8 | +78.0 | +32.5 | +12.3 | -5.99 | -15.8 | -35.2 | -62.9 |
| **SCT** | -6.70 | +0.97 | +2.38 | +1.46 | -2.55 | -5.03 | -13.2 | -26.1 |
| **Davis** | +5.48 | +3.63 | +2.31 | +0.87 | -1.09 | -1.58 | -6.17 | -17.2 |
| **SVP 8** | -1.43 | -2.15 | -1.38 | -0.37 | +0.53 | +0.96 | +2.33 | +3.37 |
| **SVP 18** | -3.09 | -0.46 | +0.27 | -0.05 | +0.02 | +0.11 | -0.25 | +0.54 |

Optimal methods of applying sheared forcing vary by drifter type. For applying current shear, the centroid method produces the best results for near-surface drifters, while the area-weighted force method produces the best results for the SVP drifters (panel A of Figure 4.21). The grouping of these results is likely in part due to the 1 m vertical discretization used in solving for the current shear, which results in approximately linear velocity profiles over the draft of the near-surface drifters as these are almost entirely in the uppermost vertical level. For the longer SVP drifters, the effect of nonlinearities in the vertical velocity profile becomes both more relevant and better resolved. The centroid method provides the best results for applying Stokes drift for all applicable drifter types (panel B of Figure 4.21).

### 4.5.3 Trajectory Results

The optimal forcing combinations for each drifter lead to predictions that are skillful (i.e. the observed positions remain within the lowest membership level) for at least seven days. The observed and predicted trajectories of the Sphere, SCT, Davis, SVP 8, and SVP 18 drifters as well as the time series of model skill associated with the prediction are shown in Figures 4.22 - 4.26. The analogous results from the leeway method and associated skill scores are also shown.

The trajectories of the near-surface drifters (Sphere, SCT, and Davis) are generally well predicted, especially for the SCT drifters (Figures 4.22 - 4.24). In the first two days of the simulation, the length of all three trajectories is slightly underpredicted, however the model results converge with observations by the third day. This convergence coincides with a shift in the winds from consistent winds towards

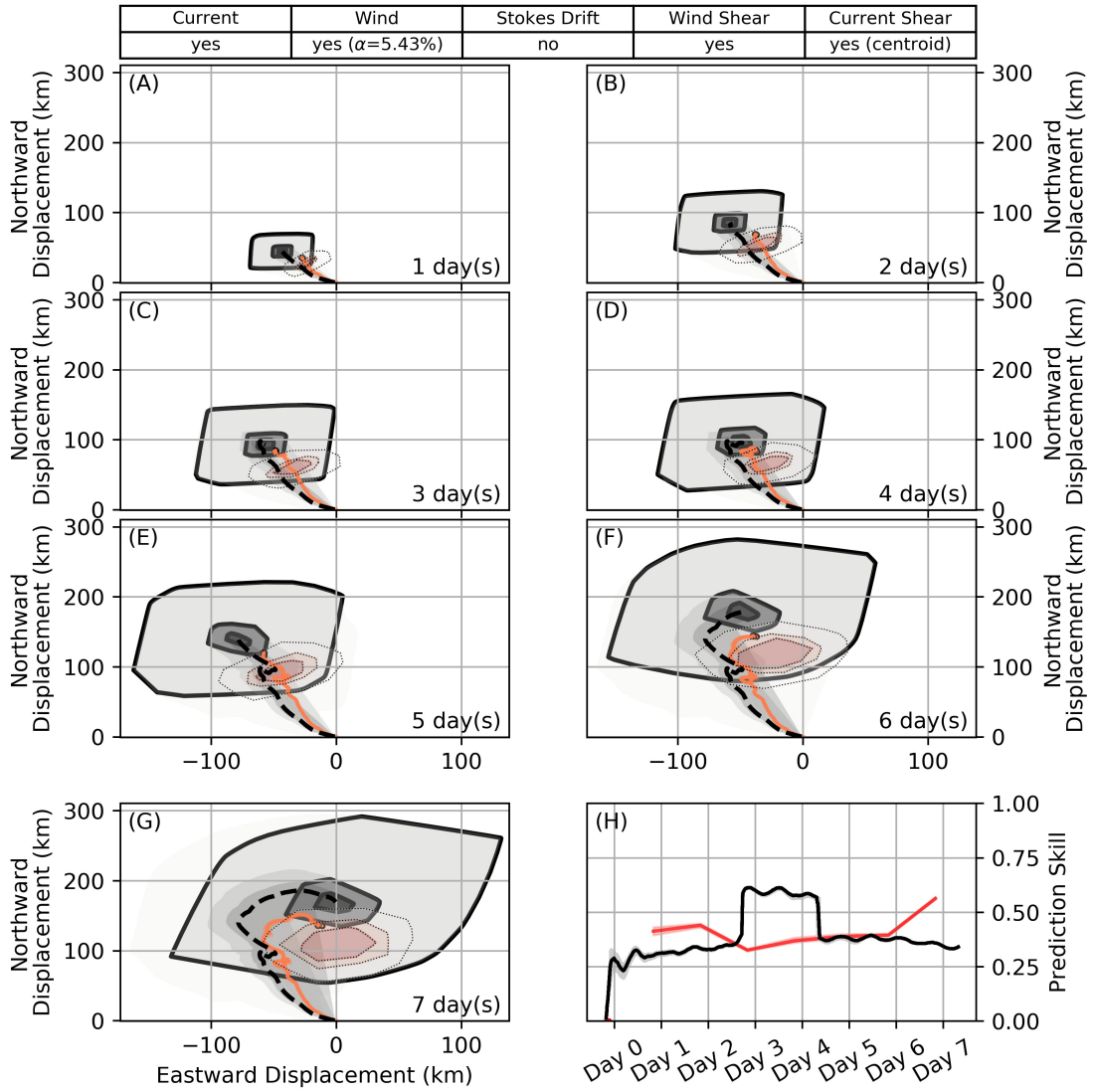


Figure 4.22: Daily predicted tracks of the sphere drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Membership levels of the possible displacement, $P_{\vec{x}}^{(j)}$ for $j = 0, 1, 2$, are indicated in grayscale shading, increasing in intensity towards the crisp solution ($j = 3$) which is indicated by the dashed black line. Equivalent results obtained using the leeway method are shown in red shading. Observed drifter tracks are superimposed in orange. Model skill is also shown in grayscale for the optimal forcing combination and in red for the leeway method. Shading indicates the maximum and minimum skill for the ensemble of drifters, while the solid line indicates the mean. Red markers on the x-axis of the skill plot indicate times when at least one observed drifter fell outside of the lowest membership level.

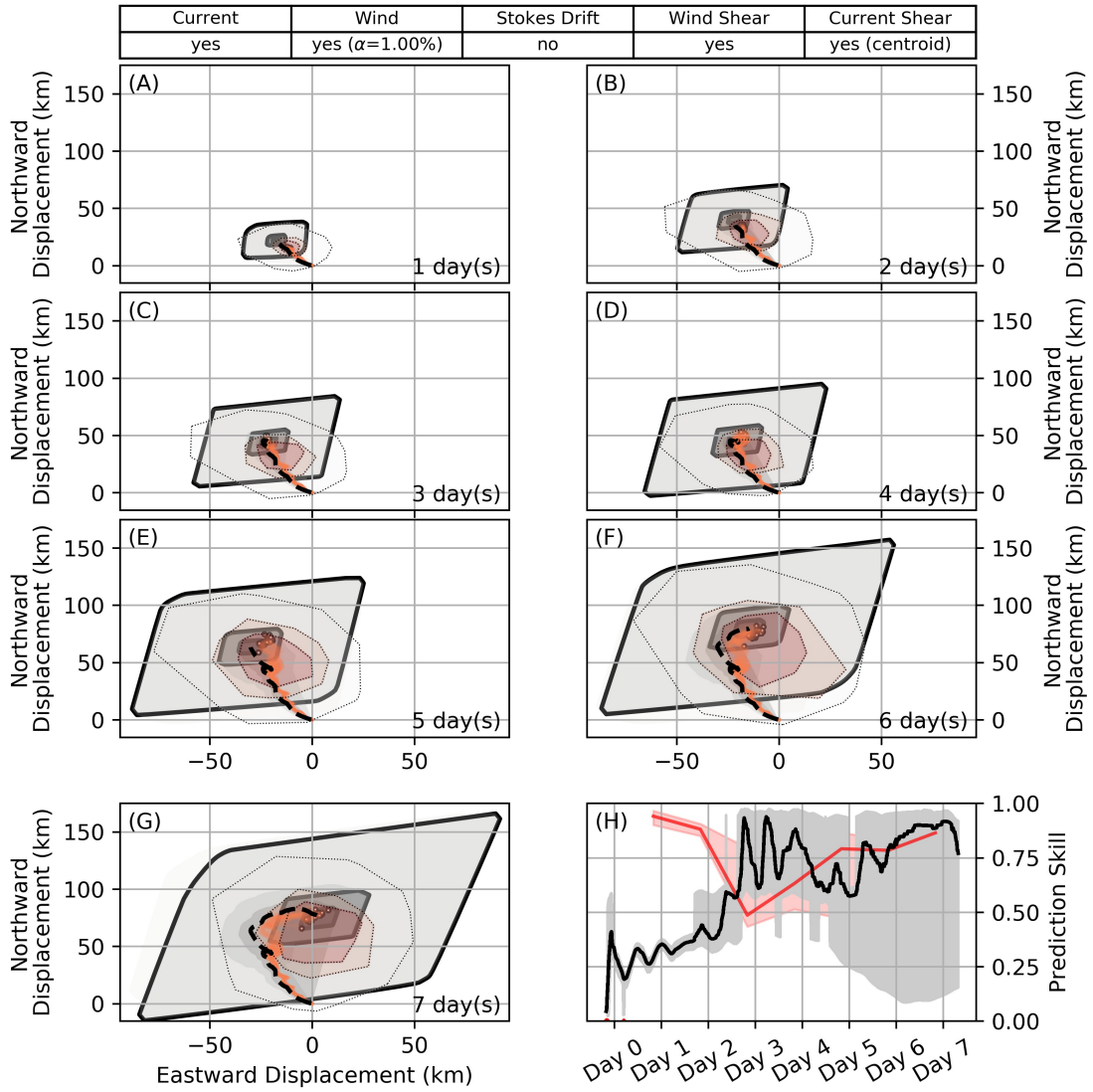


Figure 4.23: Daily predicted tracks of the SCT drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22.

the northwest to weaker and more variable winds. The predicted trajectories of the Sphere drifters consistently veer slightly to the west of the observations. On the fifth day of the simulations, the behaviour of the three near-surface drifter types diverges. From here, the northward extent of the Sphere drifter trajectories is over-predicted and the predicted trajectories of the Davis drifters begin to shift eastward of the observations, while the predicted tracks of the SCT drifters remain close to the observations. Modelled tracks of both the Sphere and Davis drifters continue

to propagate eastward at a higher rate than is observed on days 6 - 7, while the trajectories of the SCT drifters remain close to the observations. During this time the dispersion of the cluster of SCT drifters becomes comparable to the scale of the 0.67 membership level, which suggests that the size of these membership levels corresponds well with natural dispersion processes associated with unresolved high-frequency fluctuations in the forcing data.

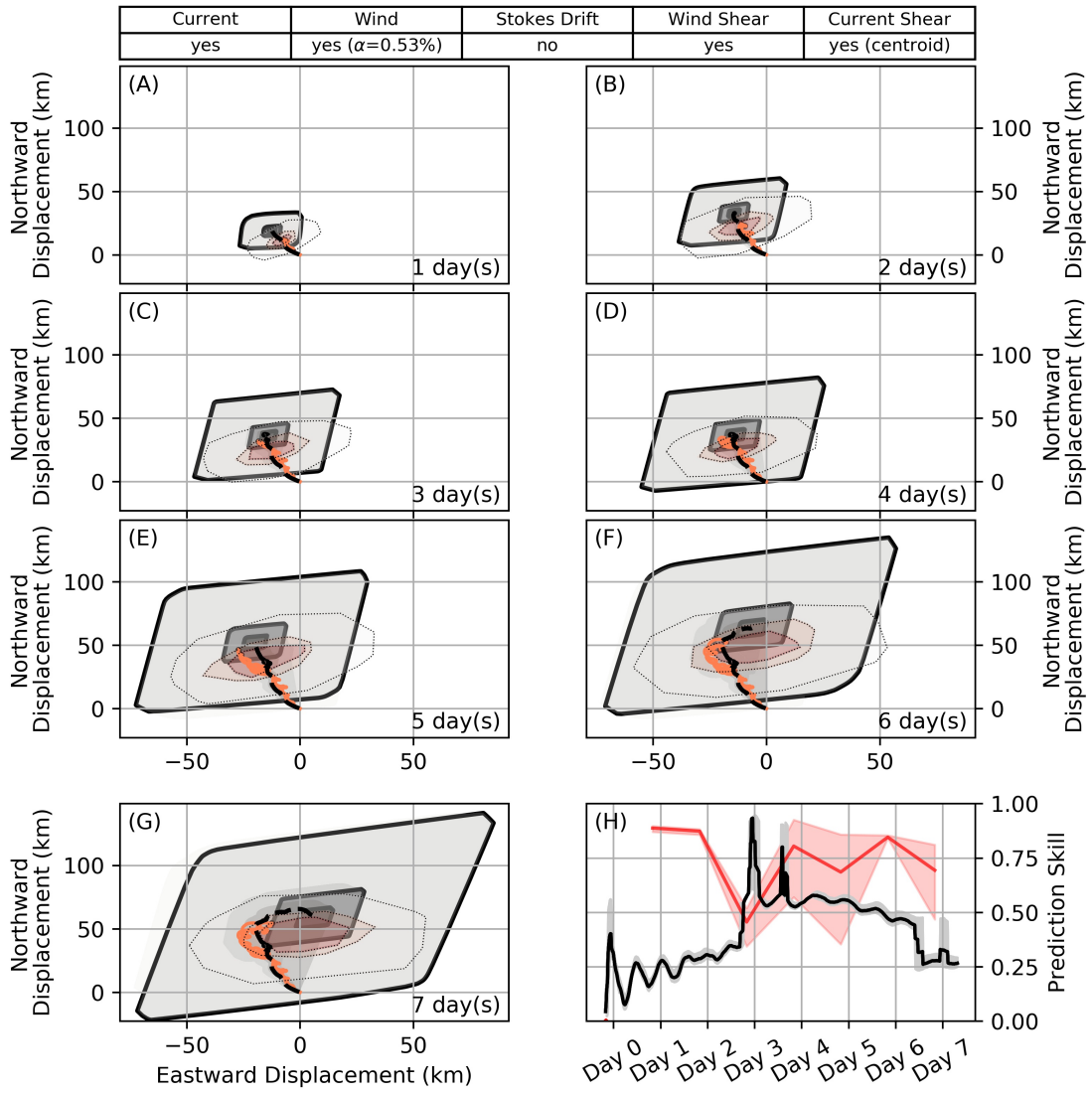


Figure 4.24: Daily predicted tracks of the Davis drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22.

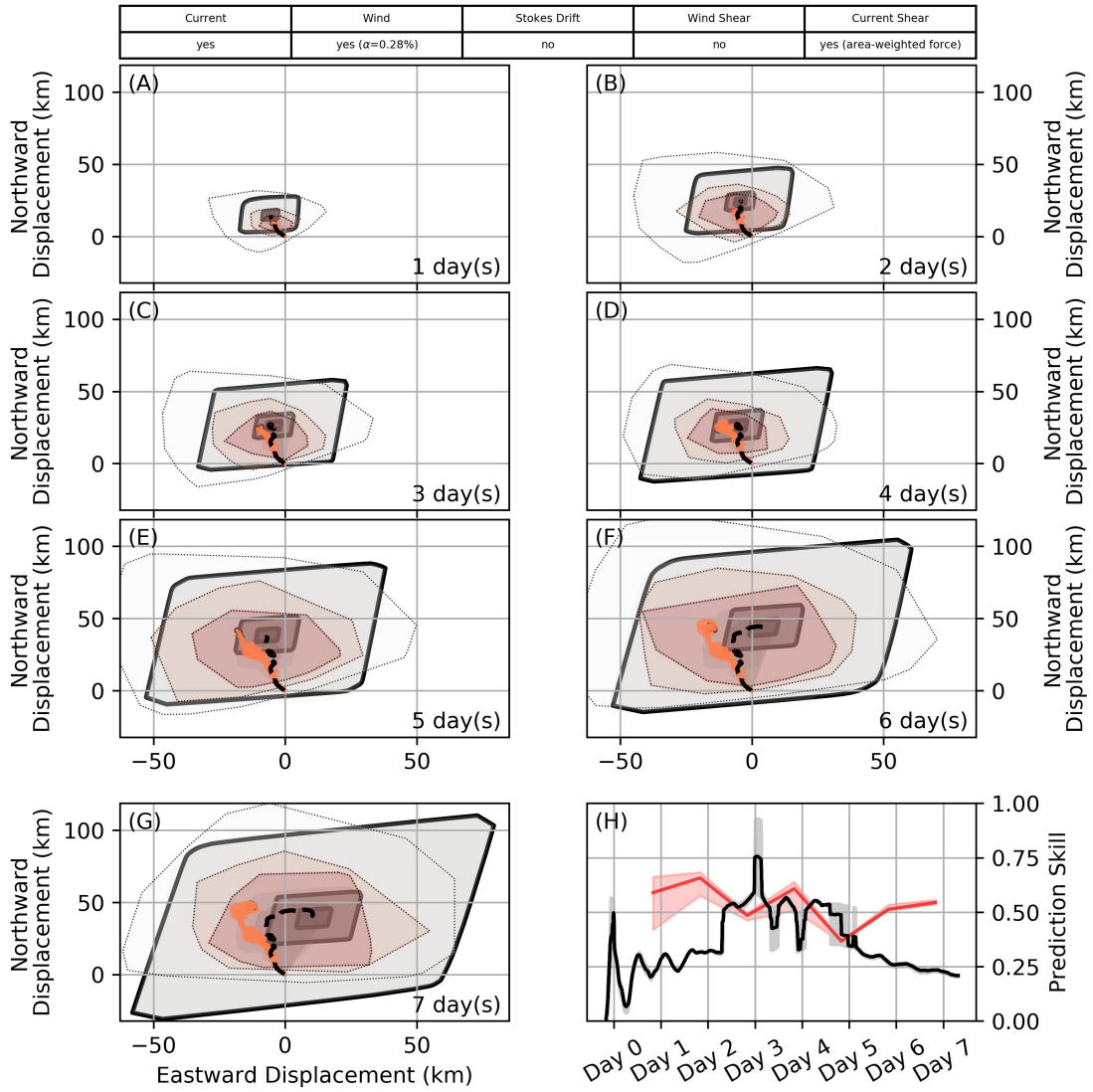


Figure 4.25: Daily predicted tracks of the SVP 8 drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22.

The near-surface drifter locations predicted by the leeway method tend to be slightly to the southeast of the locations predicted by the proposed fuzzy method, with the observed positions in between the two sets of model results. The lowest membership level of the predicted positions is larger for the proposed fuzzy method than for the leeway method, while the intermediate membership levels are larger for the leeway method. Skill scores for the two models are quite similar. Whenever a particular model captures one or more of the observed positions in an intermediate

membership level (0.33 or 0.67), its skill will rise above the other model temporarily. The skill of the leeway method is higher near the beginning and end of the simulation, however the proposed fuzzy method often has higher skill overall during the intermediate period due to the smaller intermediate membership levels.

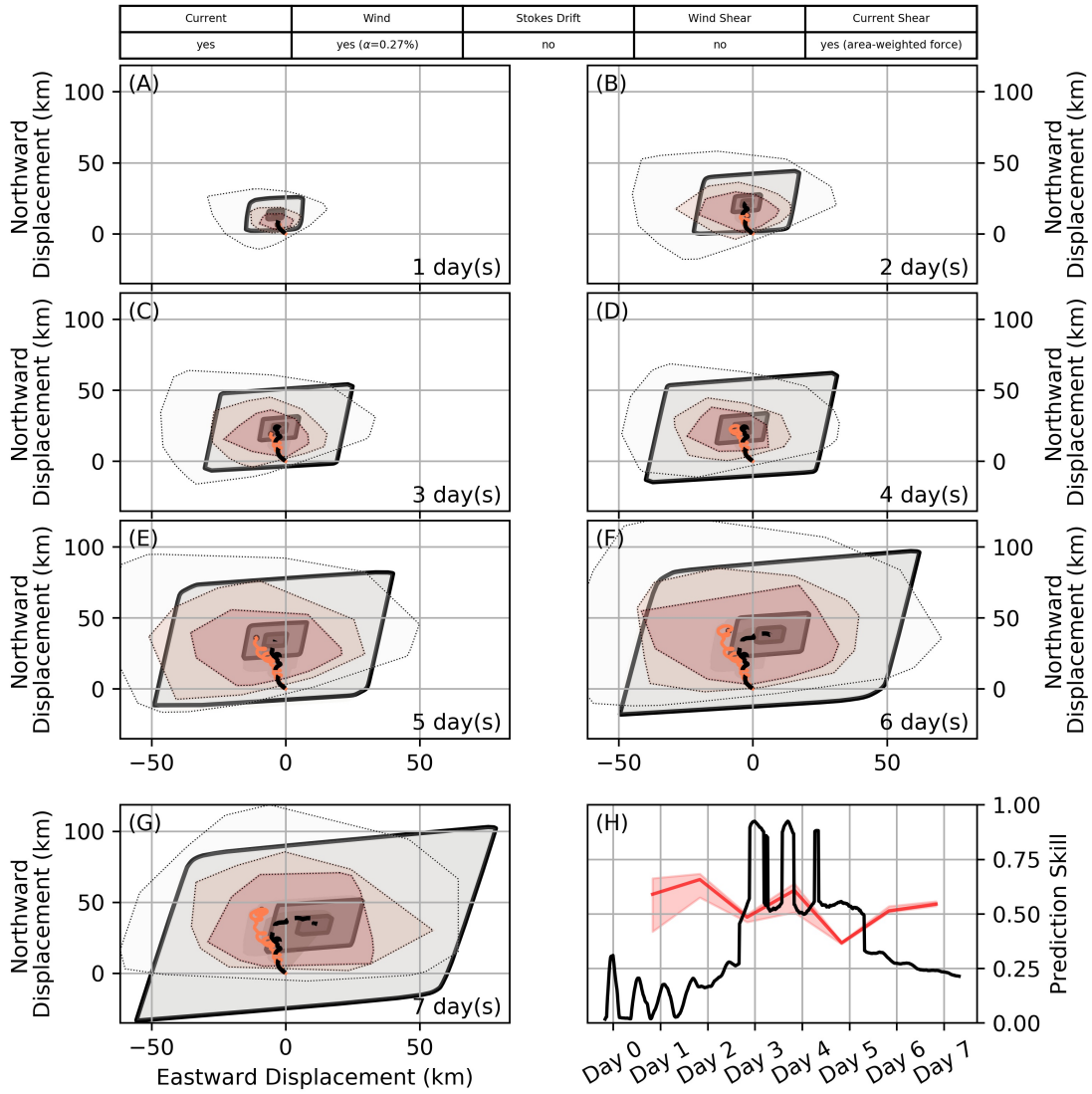


Figure 4.26: Daily predicted tracks of the SVP 18 drifters for 7 days after deployment using the indicated optimal forcing combination (panels A - G), and a time series of model skill during this period (panel H). Same layout as Figure 4.22.

For the more deeply drogued SVP drifters, the trajectory length is generally slightly under-predicted and the modelled trajectories also begin to veer eastward

of the observations around the fourth day of the simulation. After this point, the modelled trajectories strongly diverge eastward of the observations. The center of the predicted positions is similar between the proposed fuzzy method and the leeway method however, contrary to the near-surface drifters, the membership levels of positions predicted by the leeway method are generally larger than those from the fuzzy method here. Again the skill score of the two methods is similar, with the leeway method scoring higher near the beginning and end of the simulation while the fuzzy method scores higher in the intermediate period.

## 4.6 Discussion

The fuzzy number based scheme for describing and propagating uncertainty in trajectory predictions proposed in this paper appears to work well, given that the predicted sets of possible displacements covered the observed position of the drifters. The scheme is advantageous in that it does not require specification of empirical coefficients to describe the wind response of drifting objects, and can be applied anywhere concurrent measurements of wind and currents are available. Future work to extend the scheme to include forcing data from numerical models will further expand its utility. The work presented here can also be readily extended to arbitrary or unknown object geometries by describing the direct wind drag coefficient, $\alpha$, as a fuzzy number.

Regarding the representation of uncertainty, it is worthwhile to consider whether displacements in the lowest membership level should be reported, given that they are representative of the largest displacements thought to be possible, without making claims about the probability of such displacements (by definition of a possibility distribution). It is noted that the lowest membership level at times overestimated the uncertainty of predicted positions, which is to be expected as the probability density functions transformed to derive the fuzzy currents and winds were noted to overestimate the extreme values (i.e. the tails) of the represented uncertainty. The form of the input fuzzy numbers clearly affects the results of the trajectory model, similar to previous results from [69]. Increasing the number of membership levels in future work may help resolve this question by better resolving the skill score and the lowest membership level containing observed drifters. However increasing the number of membership levels comes with significant increases in computational demands, and the 'lowest useful' membership level will always vary as a function of the forcing data quality.

Direct wind drag is a dominant forcing factor on all considered drifter types, and it is shown that a deterministic, geometry-based wind drag coefficient leads to skillful predictions for at least seven days when the uncertainty in the forcing data is propagated through the prediction (Figures 4.22 - 4.26). However, the determined wind drag coefficients at times over-predicted the trajectory length for near-surface drifters and lower values of $\alpha$ would have resulted in higher model skill. Conversely, for the SVP drifters trajectory lengths were slightly under-predicted even without the addition of Stokes drift and wind shear terms, and higher values of $\alpha$ would have resulted in slightly increased model skill (see Table 4.3). Therefore some uncertainty remains around the processes relevant to the down-wind components of the motion, as direct wind drag, part of the current shear, and Stokes drift act co-linearly and their relative importance is less clear when the uncertainty in the wind forcing is accounted for. Due to this ambiguity in the co-linear forcing terms, a single set of leeway coefficients describing the down-wind motion was noted as beneficial in [12]. However within the framework proposed here, further sensitivity analysis of the parameters of the Stokes drift and current shear parameterizations in the proposed method may provide clarification of this ambiguity, though this is beyond the scope of the current paper.

The model proposed here avoids the need to specify empirically determined leeway coefficients while retaining results of a similar quality to the commonly used leeway method. This is a clear benefit, as leeway coefficients are not available for every object type and may not be representative of an object for which no coefficients have been determined. This is evident when comparing the correspondence of the leeway and fuzzy results for near-surface and SVP drifters. For near-surface drifters the leeway method results in a smaller range of predicted possible positions at the lowest membership level, though the range of intermediate membership levels is larger. This is to be expected as the lowest membership level represents all possible solutions, which may not be realized during a simulation with 5000 particles. For the SVP drifters however, the leeway method results in a larger area of possible displacements at all membership levels. For near-surface drifters closely aligned proxy objects are available, with leeway coefficients determined by the direct method (current measured during determination) [12]. For the SVP drifters, the closest available proxy object is a vertically oriented person in the water which in reality has a much shallower draft than the SVP drifters. The geometry of a vertical PIW is also quite variable, which is represented in the coefficients and results in a larger uncertainty than is predicted

by the fuzzy method.

Closer comparison of the modelled and observed trajectories also suggests that not all physics governing the evolution of a drift trajectory are captured by the trajectory model proposed here. The inclusion of uncertainty compensates for this by capturing the resulting discrepancies in lower membership levels, but the effect of the missing physics remains evident. For four of the five drifter types, a significant eastward displacement is predicted during the last two days of the simulation, in response to a change in the winds. This displacement is not seen to the same extent in the observations, which may point to a lagged response to wind forcing or spatial variability in the winds between OSP and the drifter locations. The modelled trajectories also more generally tend slightly to the east (left) of the observed trajectories, which may be due to absence of Stokes drift forcing or due to the chosen values of the constants in the current shear parameterization. A sensitivity analysis of these parameters may yield further insight here, but is beyond the scope of the current study. Alternatively, this eastward tendency may suggest that Coriolis-Stokes forcing plays a role, as its inclusion is expected to deflect the surface current further to the right of the wind [110, 159]. This may be expected as our estimates of Stokes drift have a similar magnitude to the measured currents, consistent with the results reported by [122]. Most common ocean models include parameterizations of Coriolis-Stokes forcing [122], and therefore extending the method proposed here to numerical modelling results may achieve these improvements and also meet the obvious need for representation of spatial variability in the forcing fields.

## 4.7 Conclusions

Uncertainty in surface drift trajectory predictions is of significant importance when responding to marine contaminant spills and performing search and rescue operations, along with a plethora of other applications. We have presented a method for modelling the trajectories of drifting objects that makes use of fuzzy numbers to characterize the uncertainty inherent in these predictions without relying on specification of empirical dispersion coefficients or random kicks. A comparison of results from this method to observations from five different types of surface drifters suggests that the method works well, and offers similar performance as the frequently used, empirical leeway method. Following a deployment near Ocean Station Papa in the northeast Pacific, 44 observed drifters of five different types remain within the range of modelled possible

displacements during the seven days following their deployment.

Moreover, our results show that the response of drifting objects to direct wind forcing can be reasonably well predicted based on the objects geometry provided that wind-induced Stokes drift and current shear are appropriately parameterized. Even with these parameterizations, we find direct wind drag to be a forcing term of first order importance for all drifters considered here. A novel skill metric for assessing model performance proved to be a useful tool for identifying the optimal combinations of parameterizations. The analysis presented here can be extended to objects of arbitrary geometry, and also unknown or uncertainty geometry when $\alpha$ is expressed as a fuzzy number. Further development and evaluation will expand its utility by incorporating forcing data from numerical models.

# Chapter 5

# Propagating particle tracking uncertainty defined by fuzzy numbers in spatially variable velocity fields

This chapter has been prepared as a submission to the *Journal of Marine Science and Engineering*.

## 5.1 Introduction

Accurate prediction of material drift in the ocean is critical for response to human misadventures in the marine environment, such as oil spill response and search and rescue operations [32], as well as scientific pursuits like mapping larval drift and ecosystem connectivity [120]. Drift is forced by the interaction of ocean currents, waves, and winds. These interactions range from simple additive or linearly scaled forcing [122] to more complex interactions giving rise to small scale motions, such as Coriolis-Stokes forcing [154] and leeway drift [10].

The information used to inform drift prediction is generally obtained from numerical models of currents, wind, and waves, as it is not feasible to measure these parameters at all times with high spatial fidelity. Models however are by definition an abstraction of reality, and are unlikely to faithfully reproduce all physical processes giving rise to currents, wind, and waves. This is especially true at small spatial and

temporal scales, where dynamics are not yet well understood [31]. Misrepresentation of physical processes may be due to the model code itself, or due to sub-optimal configuration of the model. Additionally, all instruments used to collect observational data for validating models are subject to instrument uncertainty and representativity errors arising from sub-optimal sampling locations.

Predicted drift trajectories that do not acknowledge the associated uncertainty, or include the uncertainty in an improper manner, can lead to a range of adverse consequences ranging from improper scientific conclusions to loss of life. It is important to note here that overestimation of uncertainty (i.e. too predicting too large a possible search area) can have impacts just as detrimental as underestimation.

An appropriate accounting of the uncertainty in drift prediction should aim to quantify uncertainty from each source as accurately as possible. This involves comparison of the forcing models' results to representative observations (with quantified instrument uncertainties), as well as estimating the energy in unresolved spatial and temporal scales of the currents and winds. If there are energetic time scales which are not resolved in the forcing models, then a phase-averaging ensemble approach will need to be employed for this component of the overall uncertainty. Steps towards this have been taken in previous work by [8], who employed fuzzy numbers [170, 171, 172] to propagate uncertainty through a drift trajectory model forced by time series measured at a single point. However this work did not consider spatially variable velocity fields.

Common approaches to uncertainty propagation in drift prediction treat uncertainty as a singular stochastic parameter. This stochastic parameter is often applied by simulating a large number of particles subject to a series of random velocity kicks drawn from a prescribed distribution [173, 36, 28]. More involved approaches have been built on this, such as the consideration of the autocorrelation in the velocity field through fractional Brownian motion [53]. In a similar vein, the leeway model for object drift [10] includes measures of the variability of object response to wind forcing, which can be applied in a randomized manner. While this approach relies on empirical, object-specific coefficients, it was found to perform similarly to the fuzzy-based analysis performed by [8].

Some efforts have also been made to reduce model uncertainty by applying hyper-ensemble techniques, i.e. combining results from several different ocean and atmospheric models [118, 119, 155]. These approaches show promise, but for operational application they are limited to areas where multiple model solutions, or capacity for

ensemble modelling with parameter perturbation, exist.

None of these approaches directly consider the misfit between model results and available observations, though both have the ability to indirectly consider this. The magnitude of the random kicks in drift prediction with a dispersion term may be informed by model - observation differences. Ensemble approaches may include models which assimilate observed ocean surface velocities, which has been shown to improve representation of modelled surface velocity structure [17]. However to the authors' knowledge these are not standard practices in operational drift prediction.

In the current paper, work is presented towards treating uncertainty in drift prediction on a per-source basis with consideration for the differences between the modelled current, wind, and wave fields that inform the drift prediction, and analogous observations of these parameters. Specifically, an algorithm for propagating such uncertainty in spatially variable flow fields is developed, as was suggested in [8]. This algorithm will be subjected to a sensitivity analysis of free parameters, in order to document its performance. This is a crucial step in developing a broader framework for using fuzzy numbers to describe and propagate uncertainty through a drift prediction system, as described by [8]. The current effort builds on work such as [93], who used interval analysis to tackle this problem, and [6], who describe spatial interpolation with fuzzy inputs. The results from the proposed scheme will be compared to analogous results obtained using common dispersion algorithms.

## 5.2 Materials and Methods

### 5.2.1 Forcing Field

Two analytical forcing fields are considered in this study (Figure 5.1). Both represent some variant of a counter-clockwise rotating monopole vortex. To facilitate the discussion of these circular velocity fields, both Cartesian and polar coordinates will be freely used.

The first field under consideration is the simplest one, representing the vortex in steady form with constant angular velocities. In mathematical form,

$$v_r(t, r, \theta) = 0$$
$$v_\theta(t, r, \theta) = sin(\frac{\pi r}{2})$$

Here $v_r$ is the radial velocity and $v_\theta$ is the angular velocity. All other symbols are as conventionally defined in polar coordinates.

The second field is designed to test performance in unsteady velocity fields, here a periodically reversing flow. The magnitude of angular velocity varies sinusoidally with time $t$.

$$v_r(t, r, \theta) = 0$$
$$v_\theta(t, r, \theta) = cos(t + \frac{1}{2})sin(\frac{\pi r}{2})$$

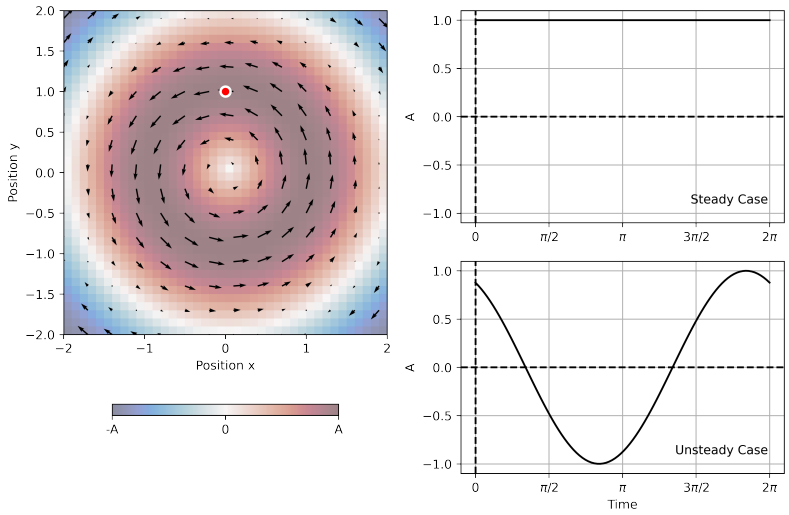


Figure 5.1: Analytical velocity fields used in this study. Both fields consider a counterclockwise rotating monopole vortex, with amplitude $A$ (left panel). In the steady case, $A$ is held constant at 1 (top right panel). In the unsteady case, $A$ varies sinusoidally with time, with a maximum amplitude of 1 (bottom right panel).

### 5.2.2 A Quick Primer on Fuzzy Numbers

Fuzzy numbers are used throughout this paper to describe uncertainty. They were first introduced by [171, 172] as a method to accurately describe imprecise quantities. A fuzzy number consists of a value that is believed to certainly be a possible value of the quantity being described (the *crisp* value, or kernel), as well as ranges of the quantity's value considered to be possible at lower degrees of belief. These ranges are described by the *membership function* as a function of their corresponding degree of belief. Degree of belief is given on the interval $(0, 1]$, with 1 being assigned to the crisp value. The lowest and highest values of the quantity thought to be possible form the *support* of the fuzzy number. The membership function increases monotonically from the lower bound of the support to the crisp value, and then decreases monotonically towards the upper bound of the support.

Fuzzy numbers are weakly related to probability theory, through the consistency principle which states that the degree of belief in a range of possible values of a quantity must be greater than or equal to the probability of this range of values [172]. This does not have a direct impact on the work presented here, but is mentioned to contextualize the work in the broader literature on uncertainty in particle tracking. In the present work, fuzzy numbers are used since their arithmetic is well defined which lends itself to straightforward and accurate aggregation of uncertainty from various sources and propagation of this uncertainty through complex mathematical models.

To perform arithmetic on fuzzy numbers, they are first discretized into a set of membership levels (also called $\alpha$-cuts in the literature) [117]. Membership levels are intervals describing the possible values of a fuzzy quantity at a fixed degree of belief. For mathematical models where no terms are repeated, arithmetic can be performed on each membership level separately using interval arithmetic [65]. If terms are repeated in the model to be solved, then interval arithmetic cannot be used as it overestimates the uncertainty in the result due to 'double-counting'. In these cases, the transformation method can be used to discretize the fuzzy number into array form and the resulting arrays is used to solve the equations using element-wise arithmetic [55]. The work presented in this paper does not directly require fuzzy arithmetic, however a brief summary is again presented for context. For further discussion in the context of ocean particle tracking, please refer to [8]. For general discussion of fuzzy mathematics, the reader is referred to the foundational papers [170, 171, 172] as well

as texts such as [65] and [55].

### 5.2.3 Uncertainty & Fractional Brownian Motion

The analytic velocity fields described in the previous section will be disturbed by synthetic random velocity fields representing uncertainty. For the purpose of this paper, describing the method for uncertainty propagation, these fields are arbitrarily defined. Rigorous definition of the uncertainty fields based on oceanographic principles is discussed in [8], and will be further discussed in future publications.

Broadly, uncertainty is defined as consisting of a global (or epistemic) component, $\sigma$, and a local (or aleatoric) component, $\sigma_l$. In reality, contributions to the global component of uncertainty in a particle tracking model come from the following sources: the deviations between the hydrodynamic and atmospheric models providing information on the current, wind, and wave fields (the input models) and analogous observations of these fields; unresolved spatial and temporal scales of motion in the input models and observations; instrument uncertainty in the equipment used to make observations of the input fields; and interpolation error due to the finite size of the model input grids.

The local component of uncertainty is related to the rate of change of the uncertainty from the above sources, which is given by the autocorrelation of the deviations between the input model fields and analogous observations. If the autocorrelation of these deviations decays exponentially in time then the local component of uncertainty is equivalent to Brownian motion with the same magnitude as the global uncertainty. Here we model the local uncertainty as a fractional Brownian function [79, 99, 125], which are autocorrelated by definition, and exploit the self-similar properties of these functions. In the literature, this self-similarity is given as follows:

$$x(t + \lambda\Delta t) - x(t) \stackrel{d}{=} \lambda^{\frac{1}{D}}[x(t + \Delta t) - x(t)]$$

Here $\stackrel{d}{=}$ implies equality of statistical properties. Therefore self-similarity suggests that statistically the changes in a time series $x(t)$ over an arbitrary interval $\Delta t$ are the same as those over an interval $\lambda\Delta t$ if they are scaled by a factor $\lambda^{\frac{1}{D}}$. $D$ is the fractal dimension of the process, which for a two-dimensional process lies on the interval $1 < D \leq 2$. If we take $x(t)$ to be the positional uncertainty of a particle in

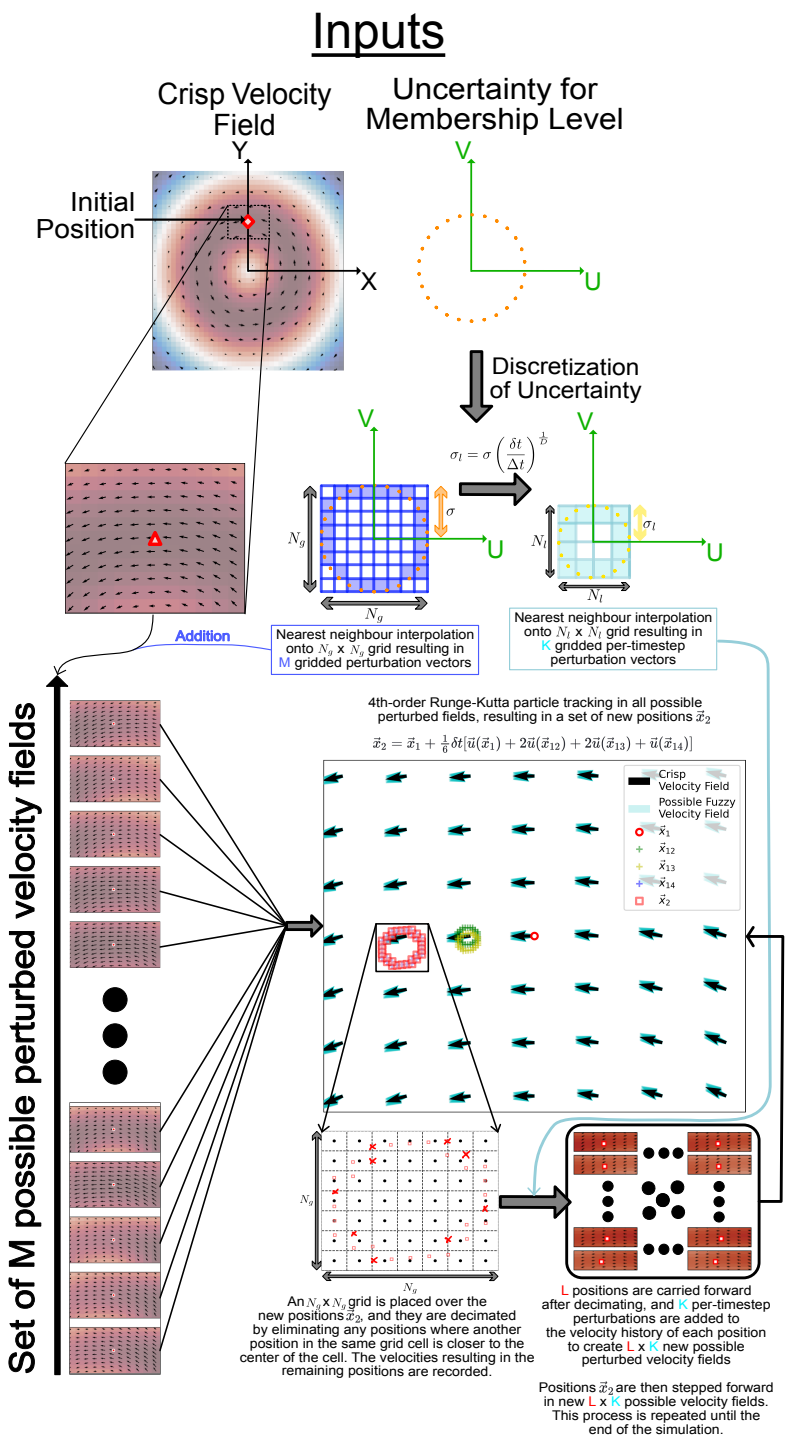


Figure 5.2: Schematic showing the fuzzy particle tracking scheme described in Section 5.2.4

a simulation with length $\Delta t$, and the time step of the simulation to be $\delta t = \lambda \Delta t$, then we can write the following power law for the local uncertainty, $\sigma_l$, in terms of the global uncertainty, $\sigma$.

$$\sigma_l = \sigma \left( \frac{\delta t}{\Delta t} \right)^{\frac{1}{D}}$$

As the fractal dimension of the uncertainty in drift velocity for a modelled particle is not known *a priori*, we will test values in the interval $(1, 2]$. Note that $D = 2$ is equivalent to traditional Brownian motion [125].

In this paper, we describe uncertainty as a fuzzy number [170, 171, 172] with five membership levels [117] corresponding to values of $\sigma = 0, 0.025, 0.05, 0.075, 0.1$ (which is equivalent to signal-to-noise ratios of $\infty, 40, 20, 13.\overline{3}, 10$).

### 5.2.4 Fuzzy Particle Tracking

In each of the velocity fields described in Section 5.2.1 we begin a particle tracking simulation from the point $(r, \theta) = (1, \pi/2)$, i.e. $\vec{x}_1 = (x, y) = (0, 1)$. At this point each of the simulated crisp velocity fields at time $t = 0$ has a value of $(u, v) = (-1, 0)$, where $u$ and $v$ are the velocity components in the $x$ and $y$ directions, respectively. The particle tracking problem is solved for each membership level separately ('Inputs' on Figure 5.2). Time steps are chosen as a fraction of the time it takes the crisp steady solution to propagate around the vortex, i.e. of the form $dt = \frac{2\pi}{T_i}$ with $T_i \in [30, 360]$. At the largest time step the solution therefore completes in 12 time steps, while 360 time steps are required for the smallest time step.

For each membership level, the corresponding value of $\sigma$ is mapped into $(u, v)$-space by taking the following in polar coordinates and converting to Cartesian, resulting in the membership function shown in Figure 4.2.

$$\begin{aligned} (v_r, v_\theta) &= (\sigma, \theta \in [\theta_1, \theta_2, ..., \theta_{N_g}]) \\ u &= u_r cos(u_\theta) \\ v &= u_r sin(u_\theta) \end{aligned}$$

The resulting values of $u$ and $v$ are mapped onto an $N_g$ x $N_g$ grid in $(u, v)$-space using

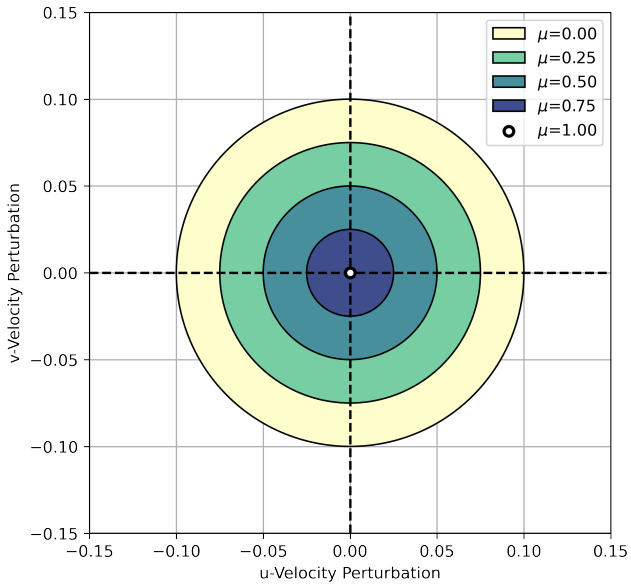


Figure 5.3: Membership, $\mu$, function of velocity perturbations.

nearest neighbour interpolation, and each of the discretized values is then individually added to the crisp velocity, resulting in $M \sim N_g^2$ possible perturbed velocity fields at the starting point of the simulation ('Discretization of Uncertainty' in Figure 5.2). In each possible unique velocity field, a particle at $\vec{x}_1$ is then advanced by one time-step using a $4^{th}$-order Runge-Kutta scheme with inverse distance weighted interpolation, resulting in $M$ new possible positions $\vec{x}_2$. These positions are mapped onto an $N_g$ x $N_g$ grid, and their number is decimated by eliminating any positions where another position in the same grid cell is closer to the center of the cell (see bottom middle panel of Figure 5.2). The perturbations leading to the remaining positions are recorded.

For further time stepping, discretized values of $\sigma_l$ (Equation 5.2.3) are added to each positions previous velocity. Similarly to $\sigma$, $\sigma_l$ is discretized in u-v space onto a grid with size $N_l$ x $N_l$ using nearest neighbour interpolation. The positions $\vec{x}_2$ are then advanced to the next time step within their unique velocity field consisting of the sum of the crisp velocity field, their previous perturbation, and the discretized values of $\sigma_l$. This process is repeated until the end of the simulation is reached (bottom half of Figure 5.2).

### 5.2.5 Skill Assessment

The algorithm presented in this section is evaluated by performing a sensitivity analysis to variations in $N_g$ (16 or 32) and $N_l$ (5, 7, or 9), as well as the particle tracking time step $dt$ and the resolution of the forcing velocity field $dx$. The objective here is to determine the combination of parameters that maximizes model skill while minimizing computational effort. Additionally, the choice of parameters should not diminish the representation of the uncertainty in the final result.

Simulations were performed with each possible combination of parameters, and scored using a cost function, written as follows:

$$C = \frac{1}{2} s \left[ \frac{T}{T_{max}} + \left( 1 - \frac{F}{F_{max}} \right) \right]$$

Here $F$ is the integral of the membership function of particle position, representing the amount of information on uncertainty contained in the solution. $F_{max}$ is the largest value of $F$ for a specified fractal dimension of aleatoric uncertainty, $D$. Trajectories are computed for $D \in [1.1, 1.3, 1.5, 1.7, 2]$, and the mean cost from all values of $D$ for a specific parameter combination is reported. $T$ is the time required for the simulation to complete on a consumer-grade laptop, with $T_{max}$ being the value of $T$ for the simulation with $F = F_{max}$. Finally, $s$ is the separation between the crisp numerical solution for the given $dt$ and $dx$ and the analytical solution.

## 5.3 Results

### 5.3.1 Steady Forcing Field

The proposed particle tracking scheme produced stable results in a steady velocity field for the investigated parameter space $N_g \in [16, 32]$ and $N_l \in [5, 7, 9]$ for model time steps ranging from $\delta t = \frac{2\pi}{360}$ to $\delta t = \frac{2\pi}{30}$. Time step is hereafter reported as the denominator of the given values, i.e. 360 for the smallest time step where the crisp solution is displaced by one degree per time step. The spatial resolution of the velocity field was varied from 1% to 15% of the vortex radius, at which point the crisp solution began to show signs of instability. At resolutions finer than 15% the vortex radius, the spatial resolution of the velocity field did not have a significant impact on the results. The results that follow were obtained using a spatial resolution of 10%

of the vortex radius.

Examining the required computational effort (runtime) and uncertainty content (integral of final position membership function) for simulations in the considered parameter space shows that both uncertainty content and computational expense increase with increased resolution ($N_g$, $N_l$) of the uncertainty space (Equation 5.2.4) and decreased time step (Figure 5.4). Regarding uncertainty content, the most evident change is with the increase in $N_l$ from 5 to 7 new positions per time step. The increase in represented uncertainty from $N_l$=7 to $N_l$=9 is modest in comparison, but associated with an approximate doubling of computational expense. Similarly, increasing $N_g$ from 16 to 32 positions results in a notable increase in uncertainty content for $N_l$=5, but more modest increases for $N_l \in [7, 9]$. In all cases this increase in $N_g$ is associated with an approximate tripling of computational expense. The increase in the range of required computational effort as a function of fractal dimension, $D$, indicated by the blue shading in Figure 5.4 is notable with increasing $N_g$ as broader dispersion at higher values of $D$ results in more positions being carried forward, i.e. not decimated, from one time step to the next (bottom panels of Figure 5.2).

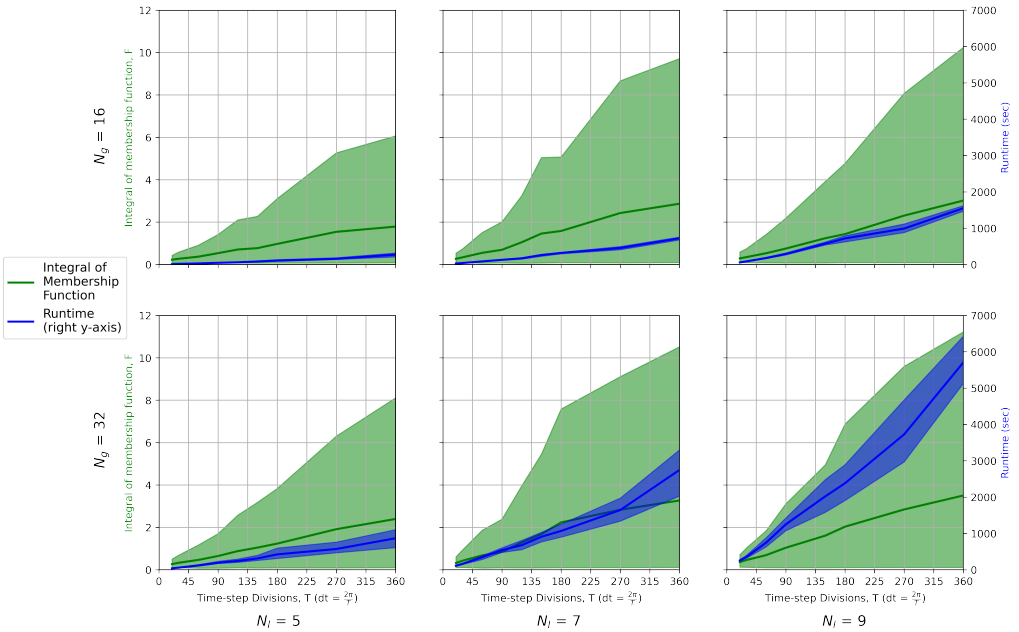


Figure 5.4: Mean represented uncertainty (green), given by the integral of the membership function of final positions, and computational runtime required (right y-axis, blue) for simulations in the steady velocity field. Shading indicates the range of values for the examined fractal dimensions, D.

In terms of the cost function (Equation 5.2.5) and its individual terms, it becomes evident that adequate representation of the actual uncertainty in the simulation can only be achieved with $N_g$=32, though depending on the users tolerance for unrepresented uncertainty using $N_g$=16 may yield lower total computational cost (Figure 5.5). Total cost is relatively constant at 0.3 - 0.4 for most cases, except for the case of $N_g$=32 and $N_l$=9 which has a mean cost of approximately 0.5, and the case of $N_G$=16 and $N_l$=7 at smaller time steps where the cost decreases to approximately 0.2. Interestingly the case with the highest cost is also the case with the most complete representation of uncertainty. This suggests that it may be prudent to set a threshold of acceptable unrepresented uncertainty for decreased computational cost. In Figure 5.5 this threshold is arbitrarily set at 0.2 (dashed line), i.e. a 20% under-representation of uncertainty is considered to be acceptable in order to optimize computational effort. Only the cases with $N_g$=32 and $N_l \in [7, 9]$ achieve this threshold, both at a time step division of $\sim$270 (i.e. the crisp solution being displaced 1.33 degrees per time step). The total cost for the case with $N_l$=7 is significantly lower at this time step. Therefore this may be considered the optimal combination of parameters for this case, given the assumptions noted above. Note that the crisp skill is nearly constant near 0.98, and therefore has minimal impact on the simulation cost.

The evolution of the membership function of particle position (shown for the case of $D$=1.5 in Figure 5.6) indicates that uncertainty evolves in a visually reasonable manner, and growth of the membership function is such that no membership level covers an excessive portion of the domain. This quality is maintained for all values of $D$ though the extent of the membership function does increase with $D$, approaching Brownian motion (position variance growing proportional to $t^{\frac{1}{2}}$) for $D$=2 (not shown). The membership function maintains convexity to within the limit of resolution provided by $N_g$ and $N_l$. Notable features of the membership function are: the lack of possible positions in the vortex kernel, most likely due to any overshoot in the particle tracking biasing positions towards the slower regions on the outside of the vortex; the elongated nature of membership levels in the direction of motion, again likely due to the rapid drop in crisp velocities away from the vortex peak amplifying the effects of uncertainty added to the peak crisp velocity; and the reversal of particle motion in the lowest two membership levels near the bottom of the domain, as perturbed particles are displaced into the reversing portion of the crisp velocity. These features suggest that the scheme behaves in a physically reasonable manner.

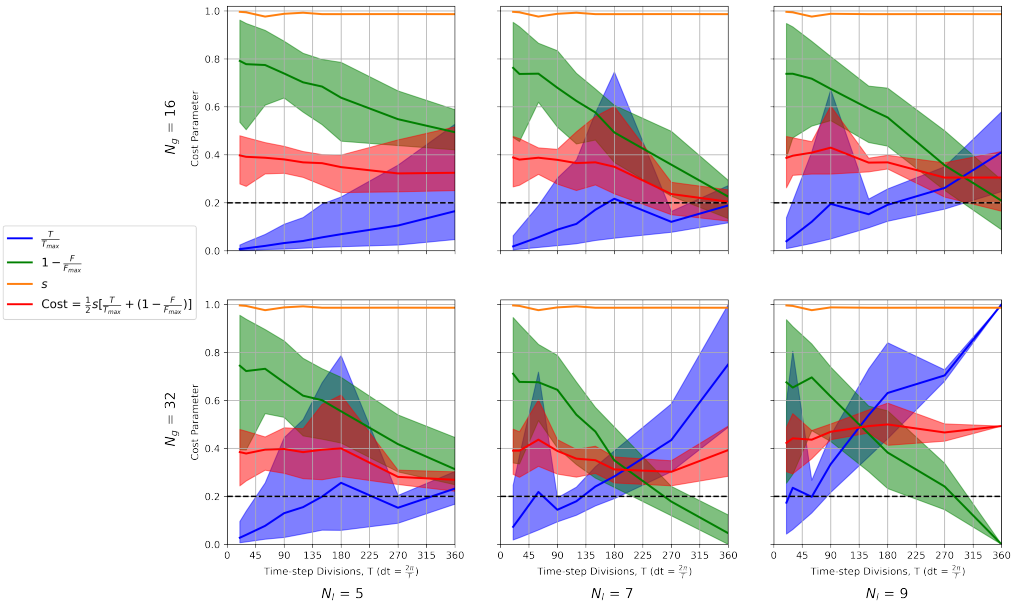


Figure 5.5: Performance summary for relevant combinations of free model parameters. Plots show mean cost (or components of cost) as a function of time step for the steady velocity case. Time cost ($T/T_{max}$) is shown in blue, uncertainty cost ($1-F/F_{max}$) in green, crisp skill ($s$) in orange, and total cost (Equation 5.2.5) in red. $N_g$ increases from first to second row, and $N_l$ increases from left to right. Shading indicates the range of values for the examined fractal dimensions, D.

### 5.3.2 Unsteady Forcing Field

In the unsteady test case, the proposed particle tracking scheme again yielded stable results for the investigated parameter space and the computational characteristics of which were similar to the steady case.

The required computational effort did not change significantly compared to the steady case. The uncertainty content is slightly increased overall, but the relationships observed within the parameter space are generally similar to the steady case (Figure 5.7). One exception of note is that for $N_g$=32, there is a more significant increase in uncertainty content from $N_l$=7 to $N_l$=9 than what was observed previously.

In terms of the cost function (Equation 5.2.5), a slight overall decrease in cost is observed due to a reduction in both time and uncertainty cost. As for the steady velocity field, only the cases with $N_g$=32 and $N_l \in [7, 9]$ reduce uncertainty cost below 0.2. On average, this occurs at a time step division of 270 for $N_l$=7 and ~200 for $N_l$=9. However the cost for the $N_l$=7 case is again appreciably lower, despite the smaller time step. Therefore this will again be considered to be the optimal case.

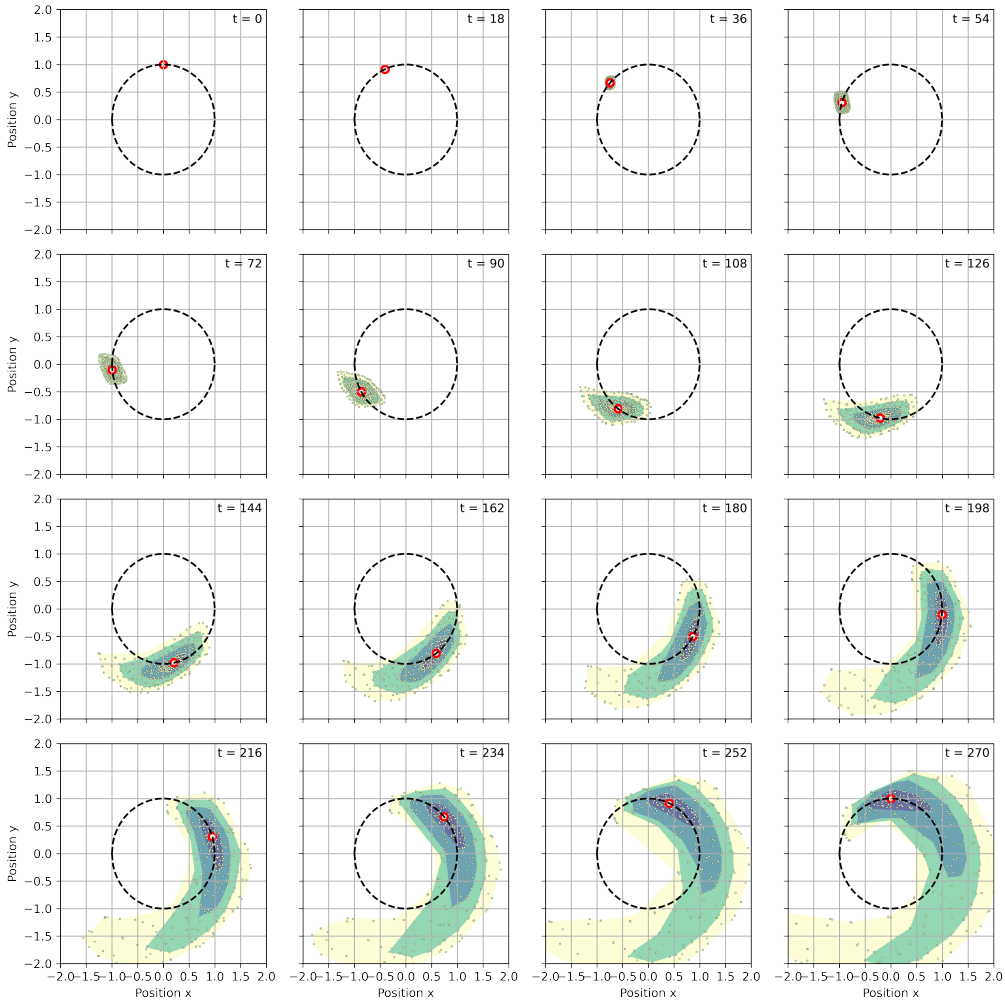


Figure 5.6: Evolution of particle trajectory for steady velocity case, with $\delta t = \frac{2\pi}{270}$, $N_g$=32, $N_l$=7, and $D$=1.5. The crisp position is shown by the red and white marker. The membership function of positions is shaded, and individual fuzzy positions are superimposed. $t$ (top-right corner of each panel) refers to the time index of the simulation.

For this scenario, the membership function of position (shown for $D$=1.5 in Figure 5.9) again indicates stable performance in a visually reasonable manner. Aside from the obvious difference in the crisp velocity field, the isotropy in the membership function is increased, with membership levels showing more significant lateral deviation from the crisp trajectory. This results in the membership function encompassing the center of the vortex by the end of the simulation, which was not seen in the steady velocity field. This is likely due to increased directional uncertainty at times when $|A| < 1$. A corollary to this is that the membership function at the final time step

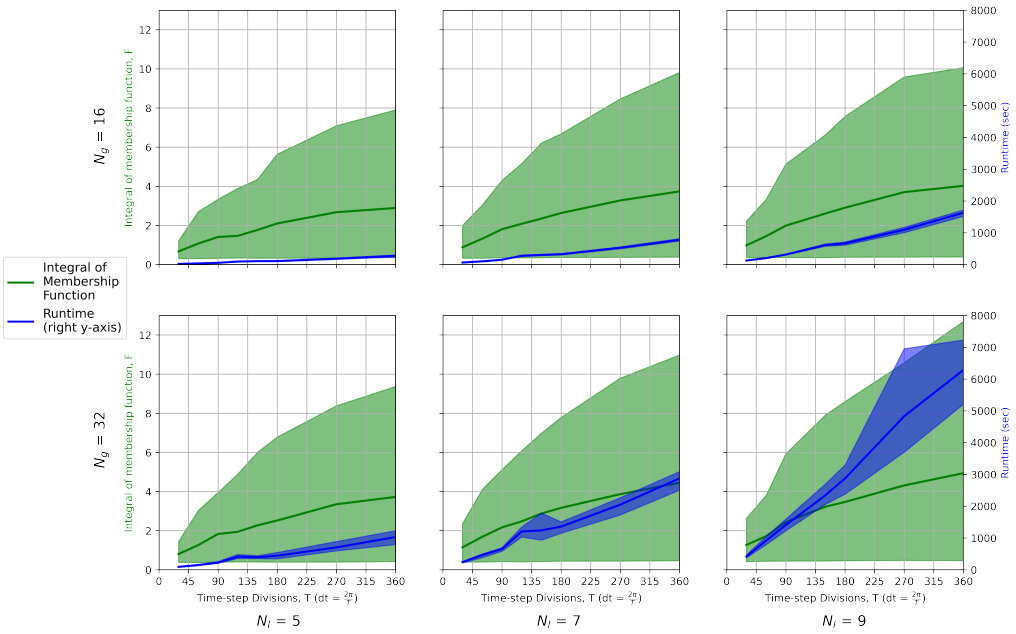


Figure 5.7: Mean represented uncertainty (green), given by the integral of the membership function of final positions, and computational runtime required (right y-axis, blue) for simulations in the unsteady velocity field. Shading indicates the range of values for the examined fractal dimensions, D.

exhibits shallower gradients of possibility, i.e. is more convex, than in the steady velocity field. As before, the results of the simulation are physically reasonable.

## 5.4 Discussion

The algorithm presented above offers the potential to account for uncertainty in the forcing data for a drift trajectory simulation in a systematic and deterministic manner, and to then propagate this uncertainty through the simulation. The accounting for uncertainty can be made using just a small number of data points, as a fuzzy number may be constructed from only three data points, and can readily accommodate anisotropic and biased uncertainty characteristics without requiring the assumption of a distribution. Uncertainty may be readily aggregated from various sources using the arithmetic of fuzzy numbers described in works such as [65] and [55]. Therefore the methodology is well suited for describing and propagating uncertainty in both observed and modelled current, wind, and wave fields, as in [8].

Depending on the characteristics of the aggregated uncertainty applied to the

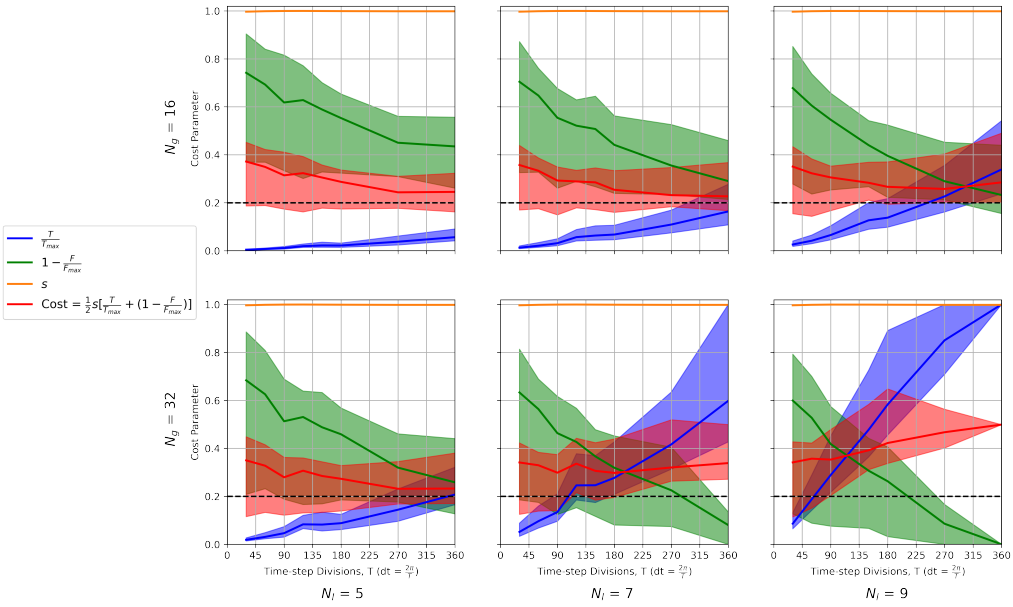


Figure 5.8: Performance summary for relevant combinations of free model parameters. Plots show mean cost (or components of cost) as a function of time step for the unsteady velocity case. Time cost ($T/T_{max}$) is shown in blue, uncertainty cost ($1 - F/F_{max}$) in green, crisp skill ($s$) in orange, and total cost (Equation 5.2.5) in red. $N_g$ increases from first to second row, and $N_l$ increases from left to right. Shading indicates the range of values for the examined fractal dimensions, D.

trajectory model, the character of the underlying Markov process may be varied from a slight deviation from the crisp solution ($D$ close to 1) to a random walk ($D$=2). The appropriate value of $D$ may be determined through fractal analysis of the time series leading to the formulated uncertainty, such as a time series of deviations between a forcing hydrodynamic model and analogous observations [cf. 78, 99, 7].

A remaining question is whether the parameter space investigated in Section 5.3 completely resolves the uncertainty contained in the formulated problem. It is feasible that, even though further improvements in cost are unlikely, further uncertainty may be resolved by considering increased values of $N_g$ and $N_l$. Results from such simulations with a time step division of 270 indicate that while further increases in resolved uncertainty are possible, the value of $F$ quickly begins to converge outside of the investigated parameter space (Figure 5.10). The increases in $F$ from $N_g$=16 to $N_g$=32 are appreciably larger than those from $N_g$=32 to $N_g$=64, whereby it can be concluded that further increases in $N_g$ would yield even smaller increases in $F$ while increasing computational expense exponentially. Through the local minimum of the

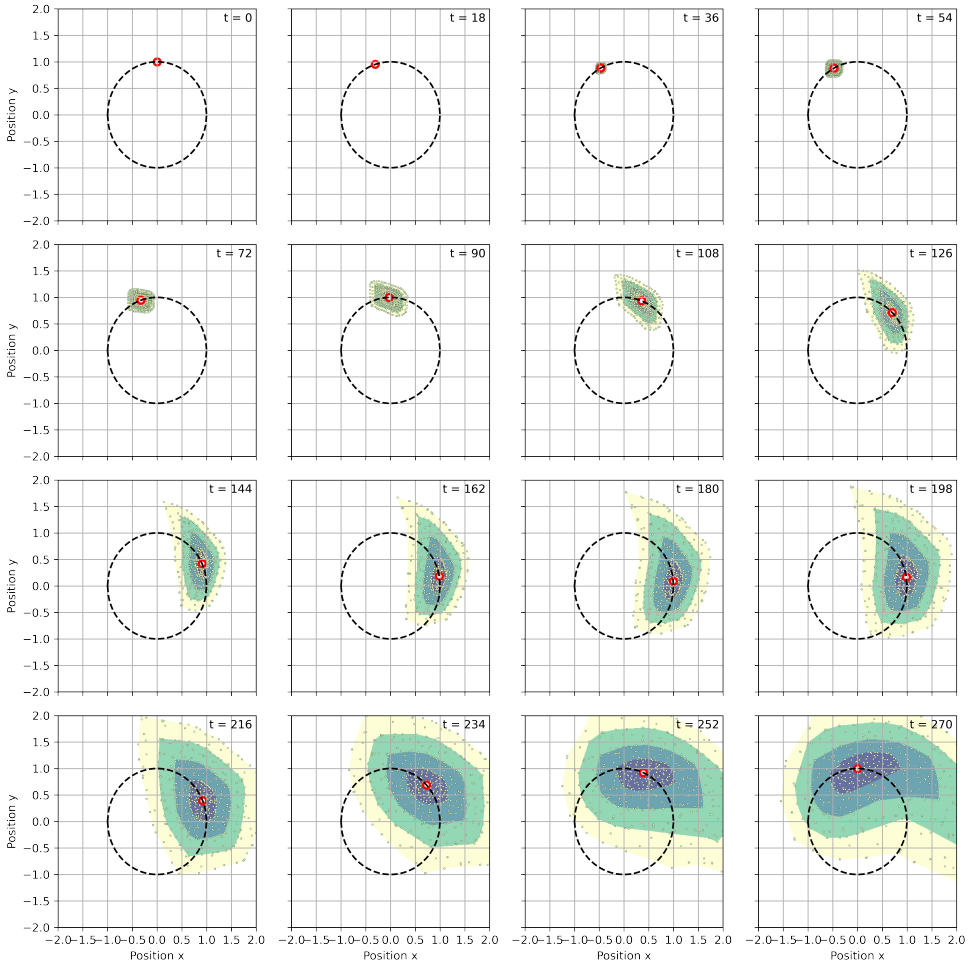


Figure 5.9: Evolution of particle trajectory for unsteady velocity case, with $\delta t = \frac{2\pi}{270}$, $N_g$=32, $N_l$=7, and $D$=1.5. The crisp position is shown by the red and white marker. The membership function of positions is shaded, and individual fuzzy positions are superimposed. $t$ (top-right corner of each panel) refers to the time index of the simulation.

cost function at $N_g$=32, it is evident that further increases of $N_g$ may not be necessary for practical application (top panel of Figure 5.10). The saturation of uncertainty is even more clearly evident for values of $N_l$ greater than 11. Here increases in $F$ are not discernible for $N_g$=16 and $N_g$=32, and very slight for $N_g$=64. The local minimum of the cost function here is at $N_l$=11, larger than the previously identified value of $N_l$=7.

The optimal combination of free parameters in the proposed trajectory model will likely be application-specific. Values given here are for a consumer-grade laptop with a 4.6 GHz processor, 24 Mb cache, and 32 Gb RAM. It is likely that code optimization

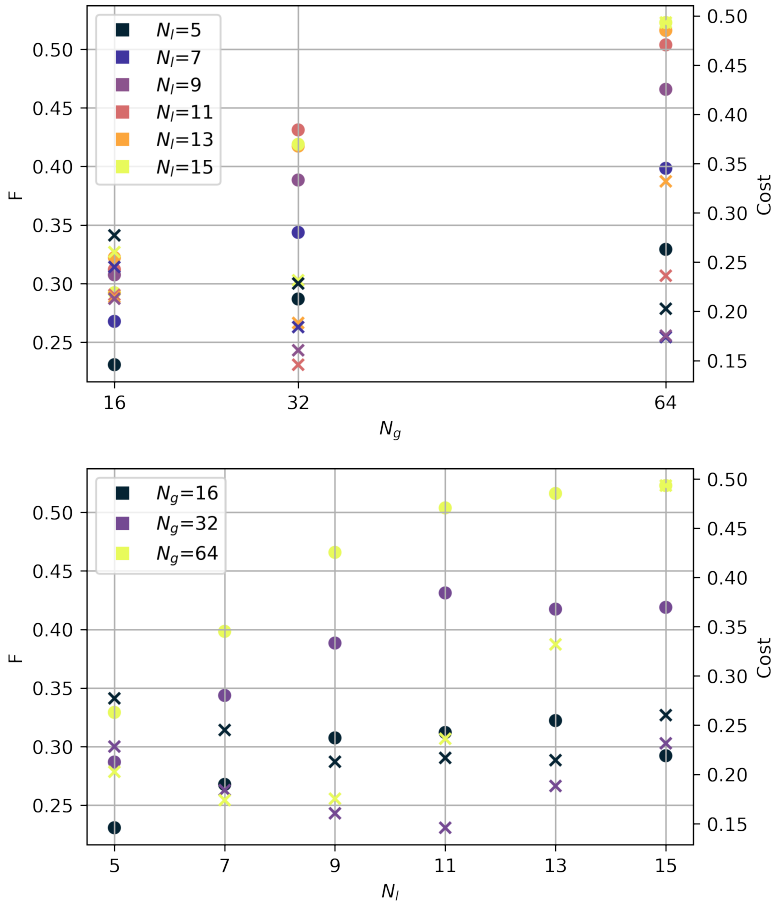


Figure 5.10: Uncertainty content ($F$, round markers) and cost (x markers) of additional simulations with $N_g$=64 and $N_l \in [5, 7, 9, 11, 13, 15]$ as a function of $N_g$ (top) and $N_l$ (bottom). Results are shown for the steady velocity case.

from the current serial formulation, to make use of parallel processing and/or GPUs, can result in significant improvements in computational speed. Therefore the metrics for computational effort and total given in this paper should be taken as guidelines, rather than firm conclusions.

## 5.5 Conclusions

A methodology for propagating uncertainty expressed as a fuzzy-valued vector through a particle tracking model in a spatiotemporally variable velocity field has been developed, and shown to produce realistic results with reasonable computational effort. This addresses a need for fuzzy-number based uncertainty propagation through mod-

els for the drift of objects floating at the ocean surface, as described in [8], as these models are generally forced with spatiotemporally variable output from hydrodynamic, atmospheric, and wave models. Uncertainty description based on fuzzy numbers allows for the consideration of bias and anisotropy with just a small number of input data points, and without requiring the assumption of a statistical distribution.

In future studies, this method will be applied to realistic scenarios with accurately described uncertainty to evaluate the performance against real-world observations and traditional models for uncertainty such as Brownian motion.

# Chapter 6

# Conclusions

Innovations to both the modelling of ocean surface drift and the interpretation of observed object drift have been presented in this thesis. In Chapter 3, the largest known observational study of surface drift in a coastal fjord system is described, analyzing data on unprecedented time scales of up to ten days. For the purposes of this study, a novel grid system was developed to reduce the problem of studying drift motion in a topographically constrained area to a single dimension. This methodology offers a framework for future studies in topographically complex regions, where no adequate ocean and atmospheric circulation models exist to inform drift studies. Based on this grid system, interpretation of observations made using 206 surface drifting buoys led to the formulation of a seasonally-resolved statistical model for their drift and dispersion. This model offered insight into the grounding behaviour of drifters in nearshore regions, and captured the mean time required for a floating object to ground ashore. Further, the derived statistics offer insight into long-term dispersion behaviour of drifting objects in the study area, which had not previously been described in existing literature but is well contextualized within it. The proposed statistical model was found to skillfully reproduce historical observations of oil sheening after a spill in the study region in 2006. It can therefore be considered to be a suitable tool for environmental risk assessment and response planning.

Further work in Chapter 4 presented a novel method for assessing and propagating uncertainty in a drift trajectory model, based on fuzzy numbers. Within this method, the effect of geometry on the wind response of the drifting object is accounted for deterministically and found to produce very similar results to the commonly used leeway method which relies on object-specific empirical coefficients to describe wind response. The geometrically derived coefficients used in this method appear to be well

optimized, when considered in conjunction with explicit accounting for uncertainty and wind-related drift forcing through upper ocean current shear and Stokes drift. The use of fuzzy numbers allows for a comprehensive accounting of uncertainty from varied sources and enables separate treatment of real turbulence in the forcing field for drift and uncertainty from other sources. In this chapter, forcing is taken from an observation platform in the northeast Pacific and considered to be spatially constant. Using a novel skill metric to compare drift predictions made using the proposed fuzzy number-based method to the tracks of surface drifting buoys with varying geometry deployed near the forcing observation site enabled a sensitivity analysis of the forcing terms in the drift trajectory model. Through this sensitivity analysis contributions to existing literature are made by presenting a comprehensive analysis of available methods for applying vertically sheared forcing over the draft of a drifting object, and by establishing that reducing wind speeds for model application by considering boundary layer effects between the 4m measurement height and the ocean surface does not improve trajectory predictions.

In Chapter 5 the method from Chapter 4 is expanded upon to include forcing input from spatially variable velocity fields. In order to do this, a $4^{th}$-order Runge-Kutta solver with inverse distance weighted interpolation was expanded to include possibility as an additional dimension. Positions considered as possible solutions were allowed to evolve with autocorrelation according to a specified fractal dimension, which resulted in the expected behaviour of the uncertainty content in the solution decreasing as the fractal dimension approached one. This method was tested on two analytical velocity fields, one steady in time and one unsteady. The optimal values for free parameters in the model were established for these velocity fields through a sensitivity analysis to minimize a cost function, along with a methodology to repeat this analysis for novel input velocity fields. The results presented in this chapter offer a method to directly propagate the differences between numerical models providing forcing data for drift prediction and their analogous real-world observations through a trajectory model, in addition to uncertainty from other sources.

In future work, the methods presented in Chapters 4 and 5 will be used to assess uncertainty in drift prediction systems forced by high-resolution numerical ocean and atmospheric models and to propagate the determined uncertainty through the drift prediction. In conjunction with the recent push to establish high-resolution ocean forecasting systems along many of the worlds coastlines, this has the potential to significantly improve our ability to effectively respond to marine emergencies.

# Appendix A

# Additional Information

## A.1 Fuzzy Numbers - A Hypothetical Example

It is of interest to predict the drift speed of a ship that has lost power at sea during a winter storm, with the only available information being forecast wind speeds. From the literature, one might reasonably conclude that the ship will drift at 3% of the wind speed and this value is therefore assigned a state of 'certainly possible' and a membership of 1. However it is also known that the forecast wind speeds are generally within 10% of the true value. Therefore we can say that ship drift at a rate between 2.7% and 3.3% of the wind speed is 'likely possible' and assign this range a membership of 0.7, or a similar value below 1 that is representative of the likelihood of error due to inaccurate wind forecasts. Further, it is known that strong surface stratification may increase the ship's drift speed by up to 2% of the wind speed. The ship's location is relatively far from significant freshwater inflows and given the winter season diurnal heating is considered to be unlikely to produce sufficient stratification to increase drift speed. However no information is available to rule out this scenario, and therefore we consider the range of ship drift between 2.7% and 5.3% of the wind speed to be 'maybe possible' and assign it a membership of 0.4. Finally, we are confident that none of our assumptions are in error by more than 20% and therefore we set the range 2.16% to 6.36% as our lowest membership level, the largest interval with non-zero membership. Values outside this range are considered to be 'impossible'. The resulting membership function of the fuzzy number describing ship drift as a percentage of wind speed is shown in Figure A.1.

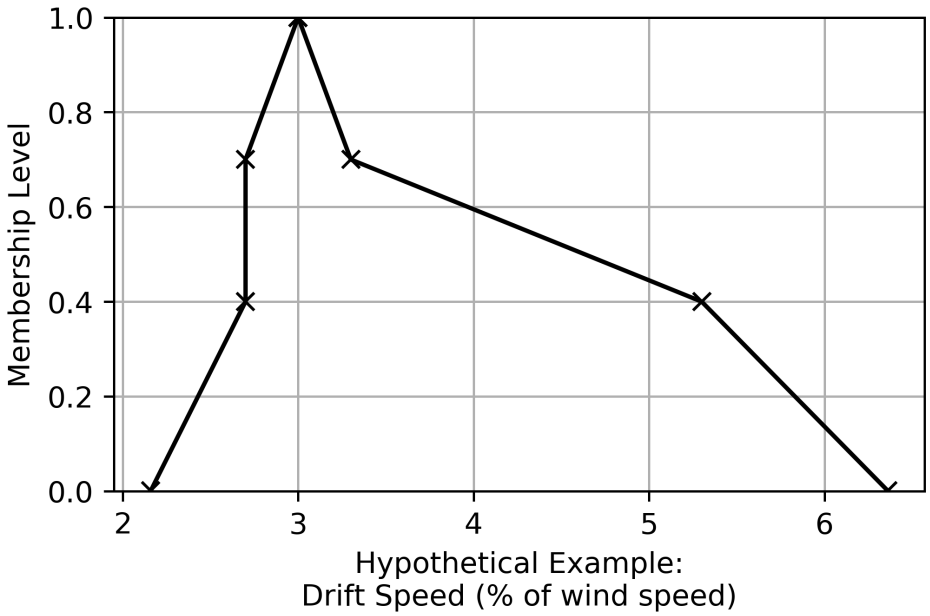


Figure A.1: Membership function derived as hypothetical example in the appendix, for illustrative purposes only.

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